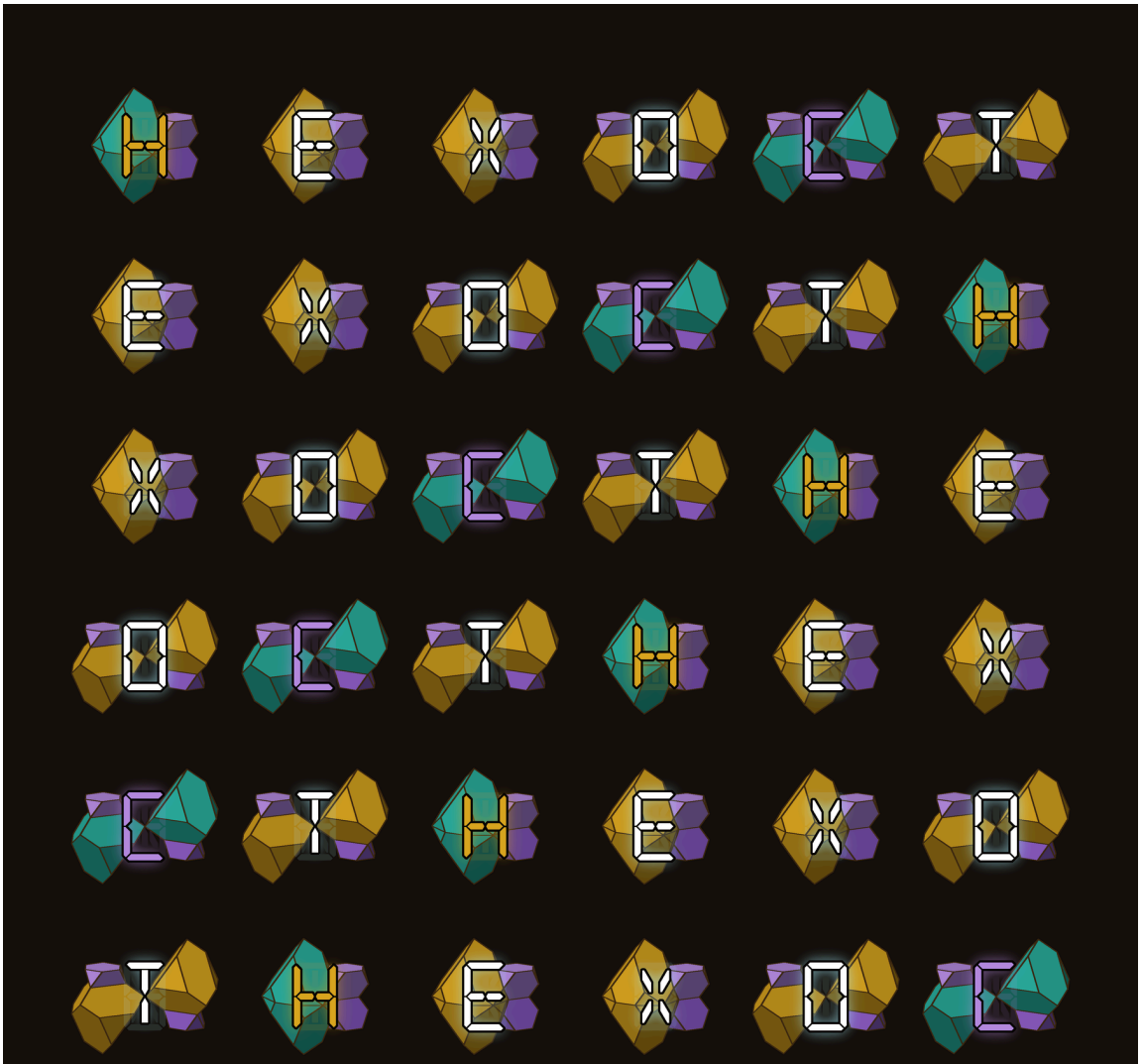


HexOct

The Hexagonality Law

with the Window Law, the Rule of Halves & the Swap Rule

DICTO



HexOct

The Hexagonality Law

with the Window Law, the Rule of Halves & the Swap Rule

The Law gave the pieces,
The pieces prove the law.

DICTO

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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).
Titled HexOct: The Hexagonality Law from version 5 (DICTO, 30 October 2026); versions 2 to 4 were titled HexOct: The Laws of Stacking, and the first edition The Laws of Stacking (28 October 2026), all the same record.
Version 7 adds Part IV, The Window Law (Chapter 15), with the subtitle "with the Window Law, the Rule of Halves & the Swap Rule" (DICTO, 1 November 2026), and reads the whole book within it: the Hexagonality Law is the Window Law's window of one layer. Version 6 added thirteen full-page plates after Chapter 14: the two tiles, HexOct-K, seven stackings built from the tiles, and Chucky, Fang and the Triskelion Tetrahedron.
Cover: the HexOct square (DICTO): HEXOCT across the top and down the side, each letter once in every row and column; H, E, X on the h-tile and O, C, T on the c-tile; where the tile matches its letter, H on the h-tile and C on the c-tile, the tile is teal.

Opening

Stack oranges in a crate and each new layer sits in the hollows of the one below. There are two ways to place it: straight above the layer two steps down, or in the third position, the one neither of the last two layers used. Repeat the first choice and the stack is hexagonal; repeat the second and it is cubic. Every mixture of the two is also a perfect packing, and nature uses many of them: silicon carbide, cobalt, ice, the oxides of batteries and the magnets of new electronics all come in such stackings.

This book follows three laws found in this one setting, in the order they were found, the two pieces they give, and the theorem that holds the three laws together. The **Rule of Halves** (Chapter 1): the gaps between the balls of any stacking are built from the gaps of the two pure stackings, whole or in halves. The **Swap Rule** (Chapters 3 and 4): when the layers are squashed or stretched, the two mixed pieces always add up to the two pure ones. The **Hexagonality Law** (Chapters 5 to 7): every count and every volume of the pieces is a straight line in one number, the share h of hexagonal layers. The second part tests the last law on real materials, from silicon carbide to spin ice, and measures for each how far it stands from the straight line. The third part, HexOct, turns back from the laws to the pieces: the space of every layer is one tile, and there are only two tiles, one cubic and one hexagonal, that stack in any order (Chapter 14). The fourth part, the **Window Law** (Chapter 15), holds the three laws in one theorem: whatever a stacking fixes through a window of g neighbouring layers averages to a linear function of the shares of its c/h words of length g. The Hexagonality Law is the window of one layer, and the Swap Rule is exactly what brings a count of two layers down to it.

The Hexagonality Law, defined

DICTO's Hexagonality Law is a law of structural crystallography: in every close-packed crystal structure, however its layers are stacked, every count of the structure (its pieces, holes, bonds and swaps) is a straight line in the structure's hexagonality, the crystallographers' share of hexagonal layers.

Its reach, stated exactly (Chapter 15). The counts of the law are totals per layer of what the layers hold: every count read from one layer, and every count of two neighbouring layers that obeys the Swap Rule. The pieces, holes, bonds and swaps of this book are of that kind, since each is borrowed, whole or in half, from one layer's pure stacking (the Rule of Halves) or changes only by swapping (the Swap Rule). A count that sorts sites by what two neighbouring layers do together, and breaks the Swap Rule, such as the octahedra of a perovskite with exactly 0, 1 or 2 shared faces, is not of this kind: the Window Law gives it exactly, as a plane in h and the share p_{hh} of neighbouring hexagonal pairs, with the Swap defect as the coefficient of p_{hh}.

How the rules fit together.

The Window Law always holds, for every window g: a count read from g neighbouring layers is a linear function of the shares of the c/h words of length g. **The Rule of Halves** and **the Swap Rule** decide which window a count needs. When a count needs only one layer, or two layers and obeys the Swap Rule, the Window Law reduces to **the Hexagonality Law** (g = 1), the straight line in hexagonality. Pair counts that break the Swap Rule are planes in (h, p_{hh}).

rule	place in the hierarchy	where we prove it
Window Law (#84)	always holds, for every window g	Chapter 15, Main Theorem
Rule of Halves (#57)	decides the window: halves are read from one layer, whole pieces from two	Chapter 1, Theorem (the rule of halves)
Swap Rule (#61, #64)	decides the window: Swap defect Δ = 0 brings two-layer counts down to one	Chapters 3 and 4, Theorem of §3; Chapter 15, Corollary 2
Hexagonality Law	g = 1: a straight line in h	Chapter 15, Corollary 1; Chapter 5, Theorem (Hexagonality Law)
beyond the Swap Rule	Δ ≠ 0: a plane in (h, p_{hh}), coefficient Δ	Chapter 15, Main Theorem (d), Corollary 2

We prove this in Chapter 15 (the Window Law, #84), with the Rule of Halves in Chapter 1 and the Swap Rule in Chapters 3 and 4.

What this book will show

By the end of the book seven things will have been demonstrated, each with proof or with the published data that tests it.

(1) **Finitely many pieces.** However equal balls are stacked in close-packed layers, the space between and around them is shared out into pieces of finitely many kinds, every one borrowed, whole or in half, from the two pure stackings, cubic and hexagonal (the Rule of Halves, Chapter 1, and its pyrochlore form, Chapter 2).

(2) **Change by swapping.** When the layers are squashed, stretched or restacked, the pieces change only by swapping: the two mixed pieces always add up to the two pure ones (the Swap Rule, Chapters 3 and 4, proved in four classical crystal families).

(3) **One straight line.** For every piece fixed by its two neighbouring layers, the Swap Rule is equivalent to the Hexagonality Law: every count and every volume of the pieces is a straight line in the share h of hexagonal layers, for random stackings, faulted ones and even the never-repeating Fibonacci stacking (Chapters 5 to 7).

(4) **Exact in real materials.** The counts that decide real chemistry and physics obey the law exactly in real crystals: where hydrogen can sit in a metal, how many faces the octahedra of an oxide share, how many bonds of diamond, ice or a semiconductor are eclipsed, how many kagome planes pass through each atom of a magnet (Chapters 10 to 13).

(5) **A measure of every material.** Where a property is not a pure count, its distance from the straight line is one number for each material, the share of interactions that reach beyond the next layer. It sorts the materials from tin, on the line, to silicon carbide, far from it, and it marks where the law stops (Chapters 8 to 13).

(6) **Two pieces, HexOct.** Cut the space of a stacking layer by layer and every layer's share is one of only two tiles, the c-tile and the h-tile, placed by one motion; any tile sits on any tile, so every stacking, periodic or not, is a string of the two, and every string is a stacking (Chapter 14). With keys on their faces the tiles can carry out any computation.

(7) **One frame, the Window Law.** Every count read from g neighbouring layers is a linear function of the shares of the c/h words of length g, which have exactly 2^{g-1} free values. At one layer this is the Hexagonality Law; at two

layers exactly one more number enters, the share p_{hh} of neighbouring hexagonal pairs, with the Swap defect as its coefficient, so a count of two layers is a straight line in h exactly when it obeys the Swap Rule. It is tested with exact counts on perovskites, Frank–Kasper phases and the approximants of quasicrystals, where the approximants carry zero defect on the chamber that holds the quasicrystal (Chapter 15).

The work began with building, in PET bottles and printed pieces, and grew through three open apps (Kaleidohedra, Rhombiverse and Polyhedraverse) into the papers gathered here. Each chapter is one paper, kept word for word in its theorems and proofs; repeated introductions are cut, and each chapter names the paper and DICTO DOI it comes from. The sources checked for every chapter are listed together at the end.

Notation

A, B, C	the three positions of a close-packed layer
c, h layer	a layer whose neighbours above and below sit on different positions (cubic) or the same position (hexagonal)
h	hexagonality: the share of hexagonal layers; 0 for cubic (3C), 1 for hexagonal (2H)
1 – h	cubicity, used for ice
3C, 2H, 4H, 6H, 9R …	stackings named by the number of layers in the repeat and its symmetry (cubic, hexagonal, rhombohedral)
J_1, J_2, J_3	couplings between layers one, two and three apart (the ANNNI layer model, convention of Chapter 8); the Hexagonality Law holds exactly when $J_2 = J_3 = 0$; J_2 is the Swap defect ($\Delta = -4J_2$, Chapter 8) and J_3 needs a window of three layers
p_{hh}	the share of neighbouring hexagonal pairs (hh) among the pairs of neighbouring layers; between max(0, 2h – 1) and h
window g	the number of neighbouring layers a count is read from (Chapter 15)
Δ	the Swap defect of a count of two layers, F(c, c) + F(h, h) – F(c, h) – F(h, c); zero is the Swap Rule

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Prologue. From the Octet to Every Stacking: the Web of Bonds

From paper #58, DICTO DOI 10.5281/zenodo.23665944

How cutting gaps in the right place, and watching which pieces touch, led to one rule for every way of stacking balls

In short. Pile up oranges in a shop and there are small gaps between them. Those gaps come in two shapes, and the way the layers of oranges are stacked decides how the gaps are arranged. Over ten papers we asked a simple question again and again: if the gaps and the oranges' own spaces are shared out as solid pieces, what shapes do the pieces take, and which pieces touch which? The answer ends in one rule, the rule of halves: however the layers are stacked, every piece is borrowed, whole or in half, from just two ways of stacking.

The core statement

Plato built his elements from solids, Democritus from indivisible atoms, and some of today's physics pictures space itself as made of small polyhedral grains. The old question behind them is how far space can be shared out into pieces of a few fixed kinds. For stacked balls there is now an exact answer:

> **However layers of equal balls are stacked, the space between and around them is shared out into pieces of only finitely many kinds, and every one of them is borrowed, whole or in half, from just two pure orders, the cubic and the hexagonal. Endless variety of order, from a finite set of quanta.**

This is geometry, proved exactly in paper #57; the link to atoms and to physics is one of lineage and analogy, not a claim about nature.

1. Pieces that fill space

DICTO came to the division of space through his hands, cutting and rebuilding PET-bottle caps into DICTOspheres. He narrowed the range to three spheres, each building its own crystal: the dodecahedron builds icosahedra and tetrahedra, the cuboctahedron builds cubes and rhombic dodecahedra, and the rhombic dodecahedron builds truncated octahedra and the octet. In Deltaverse he let deltahedra find their own lattice, where Rhombiverse had laid the lattice down first; in Polyhedraverse he cut rhombic dodecahedra in half and joined the halves into new solids, among them a slightly irregular truncated octahedron that still filled space, a forerunner of the golden split, and turned those cuts into the RHOMBIS puzzle. The apps became his playground, a place to pick up and turn solids freely; there he filled the gaps of dodecahedral and icosahedral structures with bidiminished and tridiminished icosahedra. He even planned spheres made in two halves, to bridge five-fold and four-fold symmetry. Cutting into halves and filling what is left run through every paper in this series; the rule of halves is where they meet.

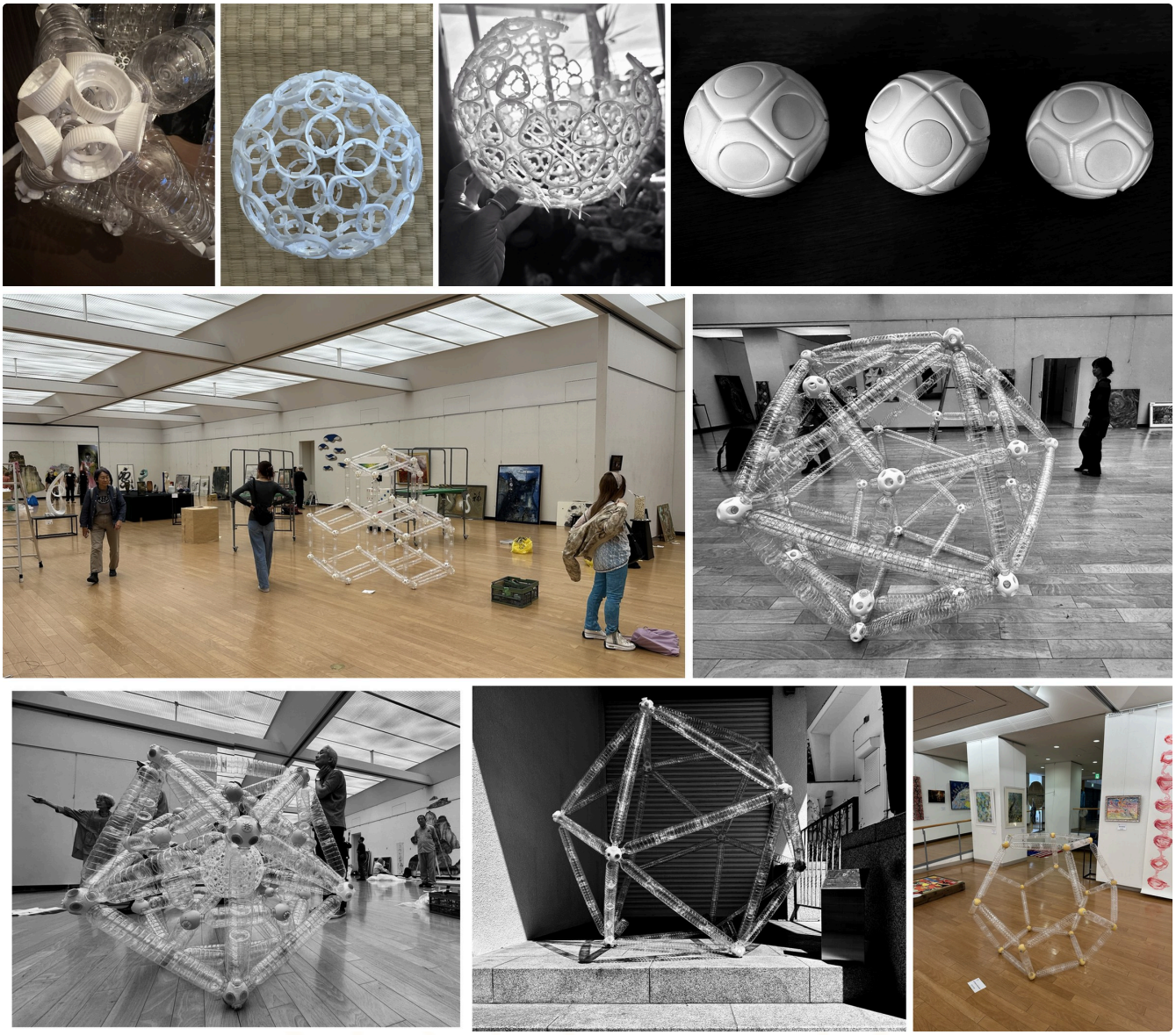


Figure P.1. From the hand to the room. Top: PET-bottle caps; DICTO's OG sphere in cap rings; a Goldberg arrangement mid-build; the three DICTOspheres (cuboctahedron, rhombic dodecahedron, dodecahedron). Middle and bottom: large PET-bottle structures by DICTO; middle left, a tetrapod arrangement at the Kyoto Municipal Museum of Art, 26 October 2026; middle right, 2025; bottom left, 2024; bottom centre, the icosahedron outdoors at the Insho Domoto Museum of Fine Arts, 2025; bottom right, the dodecahedron, 2022. Photographs: DICTO.

Some shapes fill space on their own, like bricks or cubes. Most do not. DICTO's first question was whether any mix of the five Platonic solids (the tetrahedron, cube, octahedron, dodecahedron and icosahedron) can be made to fill space. Paper #47 showed that every mix can, all 31 of them, as long as a small set of seven connector pieces is allowed to plug the gaps.

The icosahedron, the twenty-sided die, is the most stubborn. Put icosahedra in a square grid and two kinds of gap are left over: a squashed icosahedron in the middle, which DICTO called the Unborn, and thin wedges between them, the AXE. The AXE sticks out wherever it is glued. In paper #48, the Golden Split, we cut each AXE into four small pyramids from one inside point and handed each pyramid to the neighbour it touches. At exactly one point, the golden point, nothing sticks out any more: the icosahedra and the Unborns both become neat, convex solids that fill space together. DICTO named them the Icosaxes and the Unborn Axes.

Paper #51, the Delian Bond, did the same with the oldest mixed filling of all: the octet, where tetrahedra and octahedra fill space together. Slide one flat cut through every place where a tetrahedron meets an octahedron, and give each side to its own piece. As the cut slides, one kind grows and the other shrinks, but the two always fill space. At one end the pieces are Kepler's rhombic dodecahedra, at the other end cubes. Halfway, at one exact

place, the two pieces weigh the same: the small one has doubled and the large one halved. Doubling is the ancient Delian problem, doubling a cube, and DICTO named this equal pair Samson and Delilah.

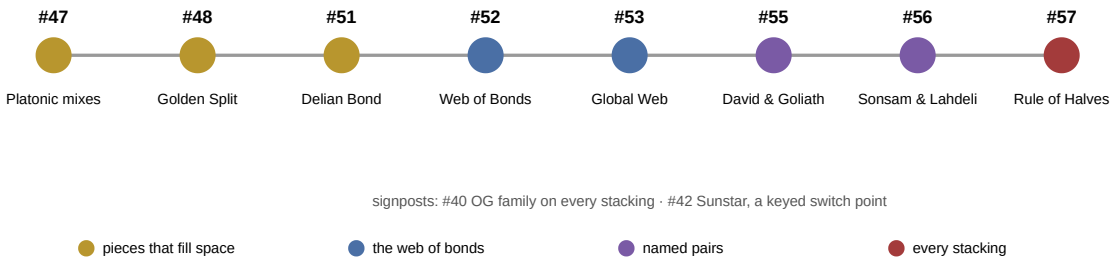


Figure P.2. The series as one route, from the octet to every stacking.

2. The web of bonds

Next we stopped looking only at the pieces and looked at who touches whom. Draw a line, a bond, between the centres of every two pieces that share a face. The lines make a web. Paper #52 showed that both of DICTO's families above are really one fixed crowd of points of two kinds, with one number slid between the kinds, like turning a dial. Each piece is simply the region that belongs to its point. As the dial turns, the pieces change shape, but the web changes in only two ways. The bond between a piece of one kind and a piece of the other never breaks. The bonds between pieces of the same kind hand over (one pair of pieces stops touching as another pair starts): in the golden family slowly and smoothly, in the Delian family all at once, at the octet.

The dial has a physical meaning: it is the size of a ball. The golden family is the arrangement of atoms in caesium chloride, the Delian family that of fluorite, and the range where the balls can touch ends exactly at the limits Linus Pauling gave for these crystals.

DICTO guessed that these two ways of handing over would turn up elsewhere. Paper #53 turned the same dial on ten more crystal arrangements and recorded every change exactly. The guess holds: the all-at-once hand-over appears in the square grid with its centres, in the honeycomb and in rock salt; the smooth hand-over appears in zinc blende, in the gaps of the hexagonal stacking, and in two famous metal alloy structures, A15 and C15.

3. Every stacking: the rule of halves

Real crystals rarely stack in only one pure way. They mix: ABAC, ABCACB, and endless others. An early signpost was paper #40, where two short rules placed DICTO's Ogdoecontahedron and its family into the stack of tetrahedra and octahedra, layered in any order, even in a pattern that never repeats.

Paper #57 turned the dial on every stacking at once and found a rule simple enough to say in two sentences. A four-faced (tetrahedral) gap always takes the piece it would have in the pure stacking of the layer it sits closest to. An eight-faced (octahedral) gap sits halfway between two layers, so its piece is two halves: the top half borrowed from the layer above, the bottom half from the layer below. That is all. So every stacking, however long or strange, is built from the pieces of the two pure stackings, and there are only finitely many kinds of piece. Where a cubic layer meets a hexagonal one, pieces appear that are half of one and half of the other, which DICTO named the Janus pieces (after the Roman god who looks both ways), Chucky and Fang.

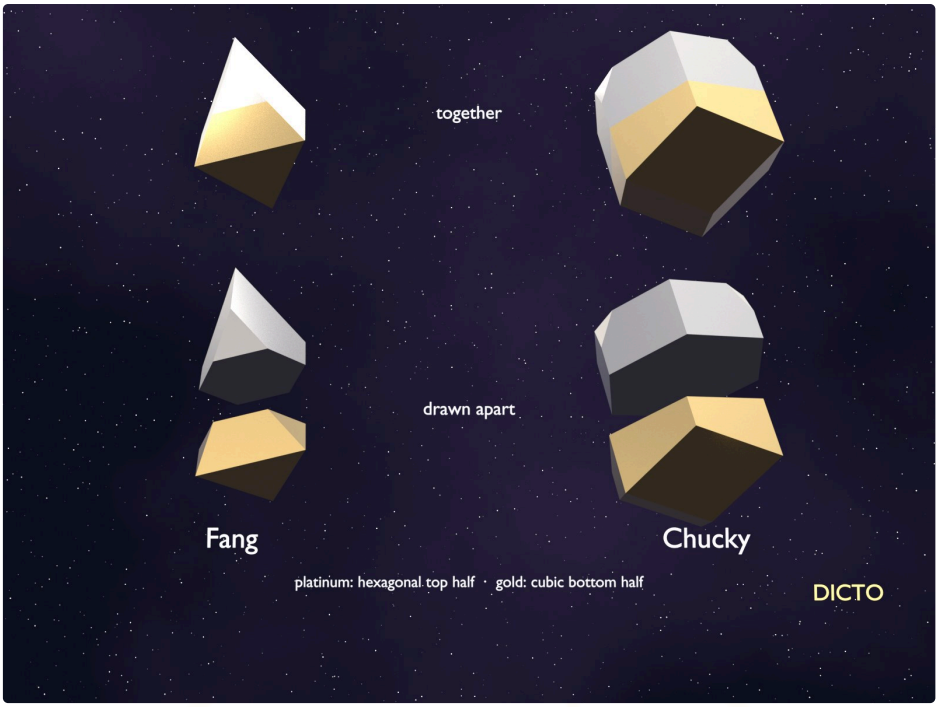


Figure P.3. The Janus c/h pieces Fang and Chucky, whole and drawn apart: a hexagonal top half (platinum) on a cubic bottom half (gold).

Part I. The laws



Chapter 1. The Rule of Halves: the Holes of Every Close-Packed Stacking

From paper #57, DICTO DOI 10.5281/zenodo.23647898

Theorem: in every close-packed stacking, along the whole weight walk, each tetrahedral hole's piece is the piece of the cubic or the hexagonal stacking named by its nearest layer, and each octahedral hole's piece is a top half and a bottom half, each the half of the cubic or the hexagonal piece named by the layer on that side; so the pieces of all stackings are finite in kind and an octahedral piece's volume is always the mean of its two parents' volumes.

In short. Stack layers of balls and the gaps between them are small octahedra and tetrahedra. Stack them in the cubic way (ABCABC…) or the hexagonal way (ABAB…) and these gaps are arranged differently; crystals also use countless mixed stackings, such as ABAC or ABCACB. In papers #51, #53 and #56 we slid one number, a weight, between the two kinds of gap and followed how space is shared out among them: the web of bonds. Here we follow it on every stacking at once, and find a simple rule. A tetrahedral gap always takes the piece it would have in the pure stacking of the layer it sits closest to. An octahedral gap sits half-way between two layers, and its piece is two halves: the top half is the top half of the cubic or the hexagonal piece, as the layer above decides, and the bottom half likewise from the layer below. So every stacking, of any length or pattern, is built from the two pure ones, and the kinds of piece are finite: we list them all, with exact volumes. Where a cubic layer meets a hexagonal one, octahedral pieces appear that are half cubic and half hexagonal, which DICTO calls the **Janus c/h pieces**; two of them he named **Chucky** and **Fang**; and its volume is exactly the mean of its two parents'. The proof is exact, with one bound done by hand. The rule was not found in the sources checked; its classical counterpart for the balls themselves, that each ball's surroundings are set by its own layer, is the reason the h/c notation of crystallography exists.

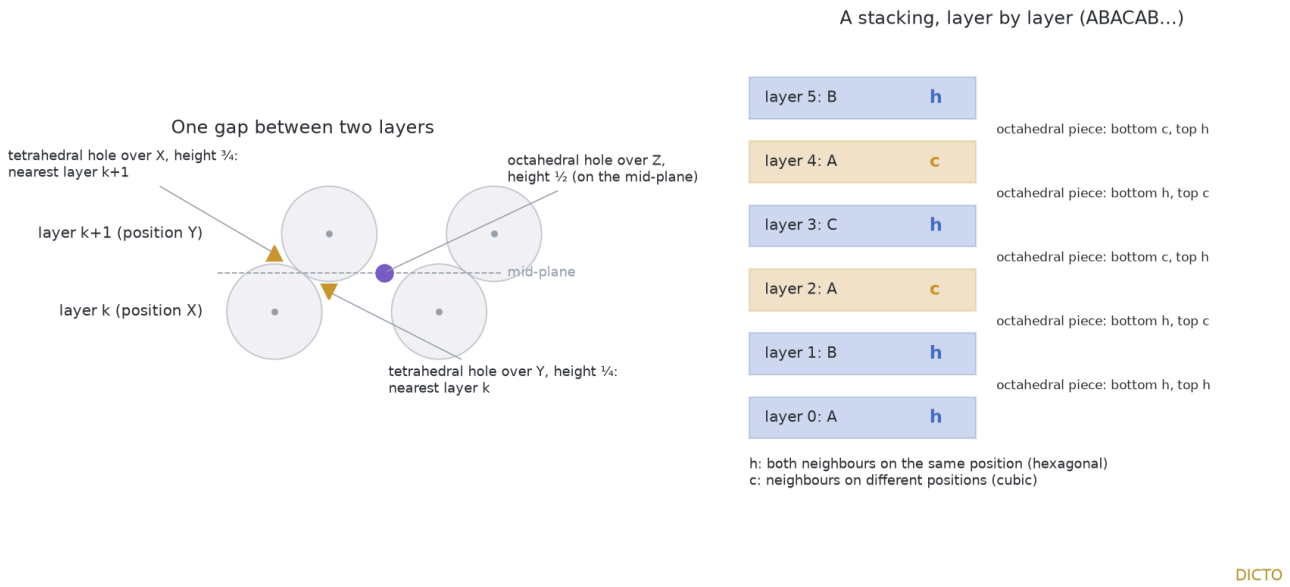


Figure 1.1. Left: one gap and its three holes. Right: a stacking layer by layer; a layer is h when both neighbours sit on the same position, c otherwise; each octahedral piece takes its bottom half from the layer below and its top half from the layer above.

1. Where it started

Paper #56 [1] followed the weight walk of #53 [2] on the hexagonal stacking and found DICTO's Sonsam and Lahdeli, Ian Dury and Blockhead, the hexagonal twins of the cubic pieces of #51 [3]. DICTO then proposed following the web of bonds onto the other close-packed stackings. On the double-hexagonal stacking ABAC and the six-layer ABCACB the walk mixed the cubic and hexagonal pieces and added two pieces neither had, which DICTO

named Chucky and Fang; the nine-layer samarium stacking ABACACBCB added nothing new. That pointed to a local rule, which we prove here.

2. The set-up

> *Each gap gets a share of space; one weight decides how big a share the tetrahedral gaps get; slide it and the shares change shape.*

Unit balls sit in close-packed layers (triangular lattices of spacing 1) at heights kd, $d = \sqrt{(2/3)}$; layer k sits on one of the three positions A, B, C, and no two consecutive layers share a position. Layer k is **h** if layers k – 1 and k + 1 sit on the same position and **c** otherwise (Jagodzinski's notation [4]). In the gap between layer k (position X) and layer k + 1 (position Y) there are three holes (Figure 1.1): an octahedral hole over the third position Z at height (k + ½)d, a tetrahedral hole over X at height (k + ¾)d and one over Y at height (k + ¼)d. The holes are the sites of a power diagram [5]: each site p, with weight w_p, owns the points x for which $|x - p|^2 - w_p$ is smallest; octahedral sites have weight 0 and tetrahedral sites weight λ. The pieces are convex and fill space; two pieces sharing a face are bonded. The walk is λ sliding over $-11/24 \le \lambda \le 3/8$, the range in which, in at least one of the two pure stackings, both kinds of piece are present (the hexagonal tetrahedral pieces shrink to points at –11/24, the cubic ones at –3/8, the octahedral pieces of both at 3/8). On each open stretch of λ between the finitely many **stops**, where some piece changes its faces, every corner of every piece moves on a straight line (Lemma 1). The pure stackings are the cubic ABC… (all layers c; the octet of #51) and the hexagonal AB… (all layers h; the twinned octet of #56).

Definitions. The *walk*: for each λ in the range we set $w_{tet} = \lambda$, $w_{oct} = 0$ and take the power diagram. A *piece*: the cell of one hole. A *stop*: a λ at which some piece gains or loses a face; a *stretch*: an open interval of λ between consecutive stops. A *window*: the six layers k – 2, …, k + 3 round the gap of layers k and k + 1, up to relabelling A, B, C; it is fixed by the h/c types of layers k – 1, …, k + 2, so there are 16. The *top* and *bottom halves* of an octahedral piece: its parts above and below the horizontal plane through its hole.

3. Main theorem

> **Theorem (the rule of halves).** For every close-packed stacking and every λ with $-11/24 \le \lambda \le 3/8$:
>
> 1. the piece of a tetrahedral hole is congruent to a tetrahedral piece of the pure stacking (cubic if c, hexagonal if h) of the type of its nearest layer: the hole over X, at height (k + ¾)d, is nearest to layer k + 1, the hole over Y to layer k;
> 2. the piece of an octahedral hole, cut by the horizontal plane through the hole, has a top half congruent to the top half of an octahedral piece of the pure stacking of the type of layer k + 1, and a bottom half congruent to the bottom half of an octahedral piece of the pure stacking of the type of layer k.
>
> **Corollaries.** (i) Up to congruence there are finitely many pieces in all close-packed stackings together: 21 combinatorial kinds, listed in the Appendix with exact volumes. (ii) The volume of an octahedral piece is the mean of the volumes of its cubic and hexagonal parents (or equal to the parent's, when both layers have the same type).

> *The two pure stackings are the only ingredients; every other stacking just decides, layer by layer, which ingredient each half comes from.*

4. Proof

Lemma 1 (corners move on straight lines). A corner of a piece is the common point of three dividing planes $2(q_i - p)\cdot x = |q_i|^2 - |p|^2 - w_{q_i} + w_p$. Their normals do not depend on λ and their right-hand sides are affine in λ; on a stretch the same three planes define the corner, and their normals are linearly independent (otherwise they would not meet in a point), so the corner is $M^{-1}b(\lambda)$, with M the fixed matrix of the three normals and b(λ) affine: affine in λ. At a stop the set of planes through a corner may change, which is why the stretches are treated one by one. □

Step 1: six layers decide. The pieces of the holes of gap k depend only on the layers k – 2, …, k + 3. (i) *Exact computation.* Up to relabelling A, B, C, these six layers are fixed by the types of the four inner layers k – 1, …, k + 2: 16 windows. For each we built a periodic stacking containing the six layers and computed its power diagram exactly, in rational arithmetic, over the whole walk (the census engine of #53 [2], which also checks on every stretch that its finite list of sites suffices and that the pieces' volumes add up to the cell's). In every window, on every stretch, every corner of every piece of gap k lies within squared distance ½ of its hole (176 cases; squared distance along a straight path is convex in λ, so the stretch ends suffice). (ii) *By hand.* A hole three or more gaps away is at least 2.5d higher or lower than every hole of gap k, so $|q - p| \geq 2.5d$, and $|w_p - w_q| \leq 11/24$. Its dividing plane lies at distance $(|q - p|^2 + w_p - w_q)/(2|q - p|) \geq ((2.5d)^2 - 11/24)/(5d) = (89/24)/(5d)$ from p (the function $t \mapsto (t^2 - c)/(2t)$ increases), whose square is $(7921/576)/(50/3) = 23763/28800 > ½$. So for every λ no such plane meets the ball of squared radius ½ about p. (iii) *Convexity.* Let C be the piece of p computed from the holes of gaps k – 2 to k + 2 only. In any stacking with the same six layers the piece of p is C cut by the planes of farther holes; these miss the ball, and by (i) the piece in the test stacking lies inside the ball; a convex body inside the ball whose boundary there is made of C's planes only is C itself. So all such stackings give the same piece. This holds for every stacking, periodic or not, since any stacking shows one of the 16 windows round each of its gaps. □

Step 2: the 16 windows. For each window, stretch and hole of gap k we searched the 24 rigid motions that fix the hole's vertical axis and map the triangular layers to themselves (the 12 symmetries of the triangular lattice about the axis, with or without the reflection $z \mapsto -z$; they serve only to exhibit congruence). For tetrahedral holes, one such motion carries the corner functions (Lemma 1) of the piece exactly onto those of the stated pure piece: all cases. For octahedral holes, one motion carries the corner functions above, on and below the mid-plane, and the points where edges cross it (exact rational functions of λ), onto those of the stated pure halves: all 160 halves. Equal corner functions give equal pieces for every λ in the stretch. Together: 176 of 176 pieces, 0 failures; no stretch needed to be split where a corner crosses the mid-plane.

Step 3: stops. At a stop each piece is the limit of the pieces on the two adjacent stretches (the engine checks that the limit is the piece computed directly at the stop), so the congruences hold there too. □

The corollaries. (i) follows because the windows are finitely many and so are their stretches. (ii): each pure octahedral piece has a centre of symmetry at its hole (the octet's octahedral holes are centres of inversion, and so are the octahedral holes of hexagonal close packing), and inversion through the hole swaps the two halves, so each half has exactly half the parent's volume; by the theorem the top half of a mixed piece is congruent to the top half of one parent and its bottom half to the bottom half of the other, so its volume is half the one plus half the other. In particular, exactly: Fang = $5\sqrt{2}/324$, the mean of the cubic octahedron $\sqrt{2}/81$ and the hexagonal rhombohedron $\sqrt{2}/54$; Chucky = $383\sqrt{2}/768$, the mean of the cubic rhombic dodecahedron $\sqrt{2}/2$ and the hexagonal piece $191\sqrt{2}/384$.

Limits. The theorem covers $-11/24 \leq \lambda \leq 3/8$. At the ends of the range, and at $-3/8$ for the cubic tetrahedral pieces, some pieces shrink to points or edges; there the statement holds with those pieces read as points, as limits of the stretches. Steps 1(i) and 2 are exact computations; Step 1(ii) and (iii) and Lemma 1 are by hand; no floating-point arithmetic enters the proof.

Computational methods. Python with exact rationals (fractions, sympy), using the census engine of #53 [2]: for a periodic stacking it lists the holes within a fixed distance, computes each piece exactly at a rational λ, finds the stretch round it exactly, and expresses every corner as an affine function of λ, with checks that the site list suffices and that the volumes add up to the cell. All scripts and outputs are in the study folder [S]; a fresh copy of the repository on a second computer reproduced every result of Steps 1 and 2 and the exact volumes identically.

Use of AI. The computations, proofs and text were prepared with Claude (Anthropic) under DICTO's direction. An open model run locally (gpt-oss-120b) was used as an independent referee of the proof and of this chapter (it found no mathematical error and asked for clarifications now included) and drafted the order of sections; it wrote none of the final text.

5. The pieces

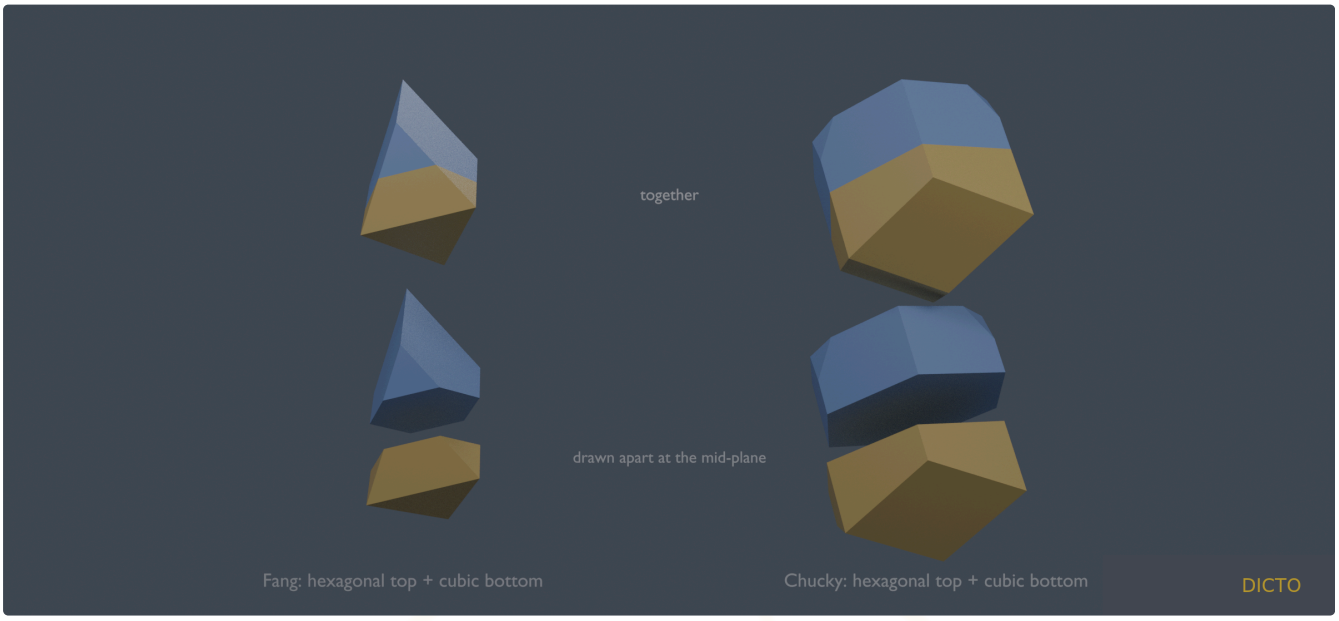


Figure 1.2. Fang (λ = 5/24) and Chucky (λ = –3/8) in the double-hexagonal stacking ABAC: hexagonal top half (blue), cubic bottom half (gold); whole above, drawn apart at the mid-plane below.

Chucky (DICTO's name): the octahedral piece at λ = –3/8 where a hexagonal layer lies above and a cubic layer below; 19 corners, with 3 triangles, 3 squares, 6 pentagons and one nonagon; volume 383√2/768. **Fang** (DICTO's name): the octahedral piece at λ = 5/24 in the same position; 7 corners, with 4 triangles and 3 squares; volume 5√2/324. Both belong to longer families of mixed pieces along their stretches (Appendix), and the stacking with the cubic layer above and the hexagonal one below gives their mirror images. DICTO calls all the octahedral pieces with one cubic and one hexagonal half the **Janus c/h pieces** (Janus, the two-faced god of doorways: each looks into one stacking above and the other below); Chucky and Fang are its named members, and two further families, of 9 and of 27 corners, complete it. **Triskelion Tetrahedron** (TT, DICTO's name): the tetrahedral piece of ABAC at λ = 5/24 with 16 corners (4 triangles, 6 hexagons). By the theorem it is the cubic piece there, a member of the sliding family of Samson in #51 [3]; the name marks that member; volume 79√2/324. The other pieces of the mixed stackings are the cubic and hexagonal pieces of #51 and #56; the two further Janus c/h families live on whole stretches and meet no stop.

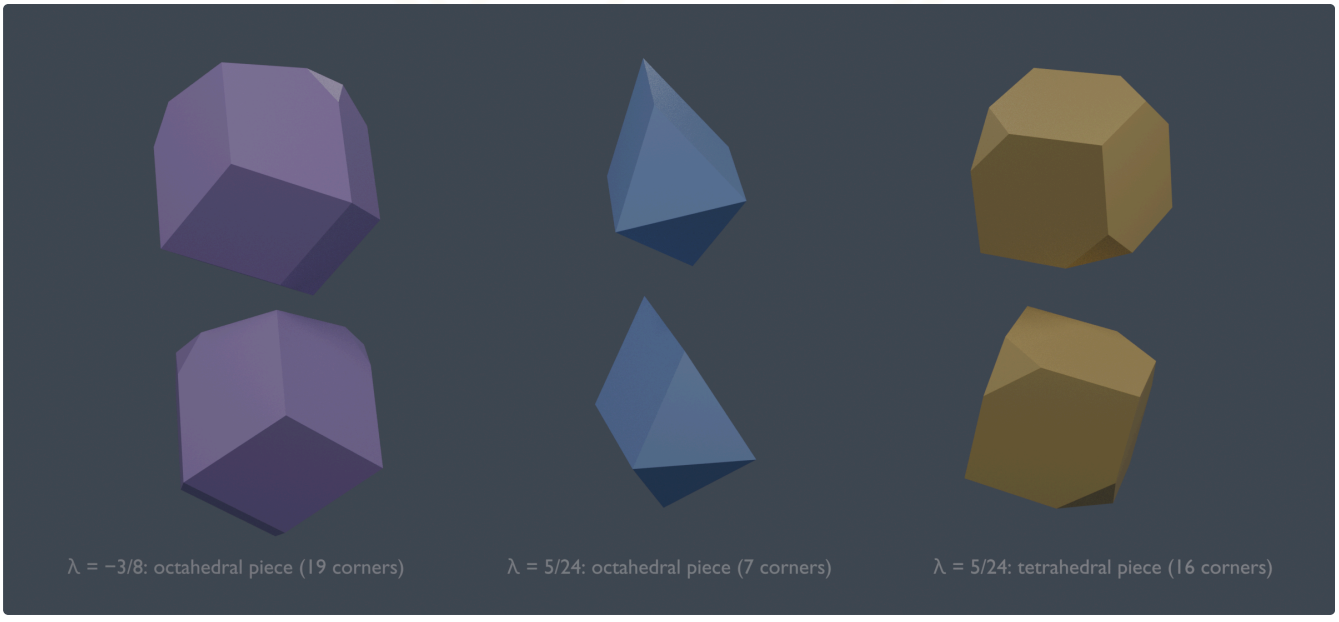


Figure 1.3. The three pieces of ABAC first met in this study, each from two sides: Chucky (19 corners), Fang (7 corners) and TT (16 corners, the cubic piece).

6. What is known, and what was not found

Known and credited. Close-packed stackings and their polytypes, and the h/c notation, whose purpose is that each ball's own surroundings are set by its layer's type [4]; our theorem is the counterpart for the holes, along the whole walk. Power diagrams [5]. The pure octet and twinned octet pieces [1, 3] and the walk [2]. Hales's proof of the Kepler conjecture divides the cubic and hexagonal packings into pieces built from their near-regular tetrahedra and octahedra ("decomposition stars"), for those two packings only [6]. Binary crystals of hard tetrahedra and octahedra: the cubic and hexagonal forms c-OT2 and h-OT2 (in h-OT2 octahedra sit on octahedra), which "can be combined in infinitely many different stacking variants", and small amounts of stacking faults observed in simulation [7]; that study has no cells round the holes and no rule relating mixed stackings to the pure forms. Radical (Laguerre) tessellations of binary hard-sphere crystals [8]. **Not found in the sources checked:** the pieces of the holes of mixed stackings, their finite catalogue, and the rule of halves. Searched: DICTO's reference library (eight queries on polytypes, hole cells, local environments, Delaunay and Voronoi cells of close packings, parameter families of power diagrams), five web searches, and Hales's three overview papers read for the purpose by a local model, which found no statement on mixed stackings or halves.

7. Open directions

- Other weights for the two kinds of tetrahedral hole (a three-kind walk, a map instead of a line), where the rule may split further.
- Keyed forms of Chucky and Fang, and a self-keyed Blockhead (#56).
- A world in the app to choose any stacking, slide λ and see every piece coloured by its parents.

Appendix: every piece of every close-packed stacking

From appendix.json [S]. Columns: the kind of hole; the stretch of λ (open interval between two stops); what the piece is made from by the theorem (a cubic or hexagonal tetrahedral piece, or the types of an octahedral piece's top and bottom halves); the number of corners and the faces of the piece on that stretch; its exact volume as a function of λ, for balls of diameter 1. At a stop a piece is the limit of the adjacent stretches' pieces.

hole	stretch of λ	made from	corners	faces	volume
tetrahedral	(-11/24, -3/8)	hexagonal piece	4	4 triangles	$\sqrt{2}\cdot(24\cdot\lambda + 11)^3/6144$
tetrahedral	(-3/8, -1/8)	cubic piece	4	4 triangles	$\sqrt{2}\cdot(8\cdot\lambda + 3)^3/96$
tetrahedral	(-3/8, -1/8)	hexagonal piece	4	4 triangles	$\sqrt{2}\cdot(24\cdot\lambda + 11)^3/6144$
tetrahedral	(-1/8, 5/24)	cubic piece	16	4 triangles, 6 hexagons	$\sqrt{2}\cdot(512\cdot\lambda^3 - 576\cdot\lambda^2 + 216\cdot\lambda + 69)/384$
tetrahedral	(-1/8, 5/24)	hexagonal piece	19	6 triangles, 3 pentagons, 3 hexagons, 1 nonagon	$-\sqrt{2}\cdot(1536\cdot\lambda^3 + 5184\cdot\lambda^2 - 3384\cdot\lambda - 1013)/6144$
tetrahedral	(5/24, 3/8)	cubic piece	16	4 triangles, 6 hexagons	$\sqrt{2}\cdot(512\cdot\lambda^3 - 576\cdot\lambda^2 + 216\cdot\lambda + 69)/384$
tetrahedral	(5/24, 3/8)	hexagonal piece	19	9 squares, 4 hexagons	$\sqrt{2}\cdot(8\cdot\lambda + 1)\cdot(64\cdot\lambda^2 - 80\cdot\lambda + 37)/256$
octahedral	(-11/24, -3/8)	cubic top + cubic bottom	14	12 squares	$\sqrt{2}/2$
octahedral	(-11/24, -3/8)	cubic top + hexagonal bottom	19	3 triangles, 3 squares, 6 pentagons, 1 nonagon	$-\sqrt{2}\cdot(13824\cdot\lambda^3 + 19008\cdot\lambda^2 + 8712\cdot\lambda - 1741)/6144$
octahedral	(-11/24, -3/8)	hexagonal top + cubic bottom	19	3 triangles, 3 squares, 6 pentagons, 1 nonagon	$-\sqrt{2}\cdot(13824\cdot\lambda^3 + 19008\cdot\lambda^2 + 8712\cdot\lambda - 1741)/6144$
octahedral	(-11/24, -3/8)	hexagonal top + hexagonal bottom	24	6 triangles, 6 hexagons, 2 nonagons	$-\sqrt{2}\cdot(13824\cdot\lambda^3 + 19008\cdot\lambda^2 + 8712\cdot\lambda - 205)/3072$
octahedral	(-3/8, -1/8)	cubic top + cubic bottom	30	8 triangles, 12 hexagons	$-\sqrt{2}\cdot(512\cdot\lambda^3 + 576\cdot\lambda^2 + 216\cdot\lambda + 3)/48$
octahedral	(-3/8, -1/8)	cubic top + hexagonal bottom	27	7 triangles, 9 hexagons, 1 nonagon	$-\sqrt{2}\cdot(46592\cdot\lambda^3 + 55872\cdot\lambda^2 + 22536\cdot\lambda - 13)/6144$
octahedral	(-3/8, -1/8)	hexagonal top + cubic bottom	27	7 triangles, 9 hexagons, 1 nonagon	$-\sqrt{2}\cdot(46592\cdot\lambda^3 + 55872\cdot\lambda^2 + 22536\cdot\lambda - 13)/6144$
octahedral	(-3/8, -1/8)	hexagonal top + hexagonal bottom	24	6 triangles, 6 hexagons, 2 nonagons	$-\sqrt{2}\cdot(13824\cdot\lambda^3 + 19008\cdot\lambda^2 + 8712\cdot\lambda - 205)/3072$
octahedral	(-1/8, 5/24)	cubic top + cubic bottom	6	8 triangles	$-\sqrt{2}\cdot(8\cdot\lambda - 3)^3/192$
octahedral	(-1/8, 5/24)	cubic top + hexagonal bottom	9	5 triangles, 3 pentagons	$-\sqrt{2}\cdot(6656\cdot\lambda^3 - 14400\cdot\lambda^2 + 6840\cdot\lambda - 955)/6144$
octahedral	(-1/8, 5/24)	hexagonal top + cubic bottom	9	5 triangles, 3 pentagons	$-\sqrt{2}\cdot(6656\cdot\lambda^3 - 14400\cdot\lambda^2 + 6840\cdot\lambda - 955)/6144$
octahedral	(-1/8, 5/24)	hexagonal top + hexagonal bottom	12	2 triangles, 6 pentagons	$\sqrt{2}\cdot(1536\cdot\lambda^3 + 5184\cdot\lambda^2 - 3384\cdot\lambda + 523)/3072$
octahedral	(5/24, 3/8)	cubic top + cubic bottom	6	8 triangles	$-\sqrt{2}\cdot(8\cdot\lambda - 3)^3/192$
octahedral	(5/24, 3/8)	cubic top + hexagonal bottom	7	4 triangles, 3 squares	$-5\cdot\sqrt{2}\cdot(8\cdot\lambda - 3)^3/768$
octahedral	(5/24, 3/8)	hexagonal top + cubic bottom	7	4 triangles, 3 squares	$-5\cdot\sqrt{2}\cdot(8\cdot\lambda - 3)^3/768$
octahedral	(5/24, 3/8)	hexagonal top + hexagonal bottom	8	6 squares	$-\sqrt{2}\cdot(8\cdot\lambda - 3)^3/128$

Chapter 2. DICTO Janus c/h Pyrochlore

From paper #60, DICTO DOI 10.5281/zenodo.23678750

The rule of halves of #57 carried to the pyrochlore network, its layers stacked in any order

> **In short.** The pyrochlore network is made of small tetrahedra that touch only at their corners. It hides inside every diamond-like crystal: take the middle of every bond, and the four middles round each atom make one of these tetrahedra. Diamond-like crystals can stack their layers in many orders, so the pyrochlore network can too. We share out all of space round it by one weight, as in #57, and prove the same rule: every piece is borrowed, whole or in half, from the pure cubic or the pure hexagonal stacking.

1. The idea

DICTO's idea (27 October 2026): "close packing halves rule pyrochlore network" — the rule of halves of #57 (DICTO DOI 10.5281/zenodo.23647898) on the pyrochlore network, its layers (kagome and triangular, seen down a three-fold axis) stacked in any order. The series paper #58 (DICTO DOI 10.5281/zenodo.23665944) tells the road to #57.

2. Main theorem

Theorem. In every close-packed stacking, and for every weight λ from –11/36 to 1/4, divide space by weight among the balls and diamond atoms (weight λ) and the two empty holes of each gap (weight 0). Then every piece is a piece of a pure stacking: a ball's piece is the ball piece of its own layer's pure stacking (cubic if the layer is c, hexagonal if h); a diamond atom's piece that of its nearest layer; and each hole piece, cut at its gap's mid-height, is a bottom half from the layer below and a top half from the layer above.

Where a cubic layer meets a hexagonal one, the hole pieces are therefore **Janus c/h pieces**, as in #57: a cubic half glued to a hexagonal half (in 4H, octahedral pieces of volume 0.346 and 0.325, beside 0.339 in the cubic and 0.332 in the hexagonal stacking). The tetrahedra of the network — the pieces of the balls and diamond atoms — are always whole, each from one layer; the Janus pieces are the network's open spaces.

2a. Proof

The proof follows #57 and uses its exact engine unchanged (rational arithmetic throughout).

Locality. Near any gap, a stacking is fixed by the cubic or hexagonal type of the four layers round it: sixteen windows. Every corner of every piece of the central gap lies within squared distance 11/32 of its site (exact, at the ends of every stretch; convexity). A site whose letter the window does not fix is at least 2¼ layer spacings away in height, and the weights differ by at most 11/36, so its dividing plane stays at squared distance at least $((27/8 - 11/36)/(2\cdot\sqrt{(27/8)}))^2 \approx 0.698$ from the site: it cannot touch the piece. So each piece is the same in every stacking that shares its window.

Pieces along every stretch. On each stretch of the walk (stops –11/36, –1/4, –3/20, 0, 1/8, 1/6, 1/4) every corner of every piece moves on a straight line in λ. For each of the sixteen windows and each stretch, one symmetry of the layer carries the corners of each whole piece (ball, diamond atom) onto those of its pure piece, and the corners above, on and below the mid-plane together with the cut points (exact rational functions of λ) of each half (octahedral and Y holes) onto those of its pure half: 384 of 384. As an independent check, the pieces are congruent to their pure pieces at twelve exact weights covering every stretch: 768 of 768.

3. Construction

Balls of spacing 1 in close-packed layers $\sqrt{(2/3)}$ apart, stacked by a word in A, B, C. The diamond-type crystal (as silicon carbide) is the balls plus #57's tetrahedral hole "over X at 3/4" above each ball. The middles of its bonds

form the pyrochlore network: round each atom four middles make a regular tetrahedron of edge 1/2, and every corner belongs to exactly two tetrahedra (checked for 3C, 2H, 4H and 6H). Each gap holds four sites: the ball S, the diamond atom C, the octahedral hole O and the other tetrahedral hole Y; the two empty holes are exactly the network's open spaces.

4. The weight walk

Computed exactly with #57's census engine, unchanged (rational arithmetic; every built-in check passed).

Cubic (3C). Stops at –3/8 (the atoms' pieces vanish), **–1/4** (atoms regular tetrahedra, holes truncated tetrahedra of volume $23\sqrt{2}/96$: the classical tetrahedral–truncated tetrahedral honeycomb), **+1/4** (the roles swap: atoms truncated tetrahedra, holes tetrahedra), and +3/8 (the holes' pieces vanish).

Hexagonal (2H). Stops at –11/36, –1/4, –3/20, 0, 1/8, 1/6 and 1/4. At –1/4 the diamond atoms' pieces are still regular tetrahedra but the balls' pieces have ten corners: the regular pyrochlore tetrahedra are exact pieces only on cubic layers.

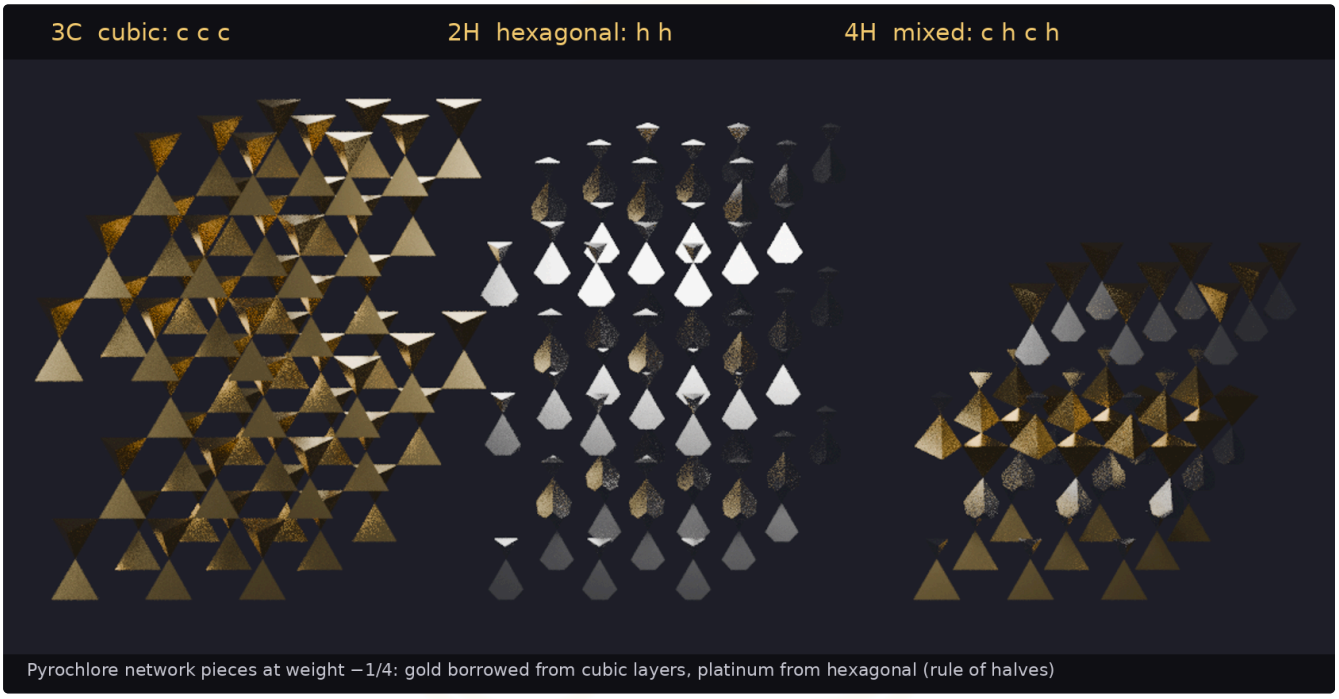


Figure 2.1. The network's pieces at λ = –1/4 in the cubic (gold), hexagonal (platinum) and mixed 4H stackings; open spaces left out.

5. Prior art and credit

The pyrochlore lattice as the bond-middle lattice of diamond, and as alternating kagome and triangular layers along a three-fold axis, is standard in the physics of frustrated magnets; the cubic and hexagonal stackings of diamond-type crystals (diamond and lonsdaleite, zincblende and wurtzite) are standard; the tetrahedral–truncated tetrahedral honeycomb is classical; the rule of halves is #57 (DICTO). Not found in the sources searched (a library of 302 references, and the web, 27 October 2026): the rule of halves for pyrochlore stackings, or this weighted division of them.

6. Methods and data

Study folder k-research studies/2026-10-27-pyrochlore-halves: pyro.py (construction), cells.py and halves.py (float division and halves), pyro_walk.py (exact walks with #57's engine), pyro_exact.py and pyro_stretch.py (the proof), figures.

7. Name

Title by DICTO: DICTO Janus c/h Pyrochlore — the pyrochlore network whose pieces, like #57's Janus c/h pieces, are borrowed from the cubic (c) and hexagonal (h) stackings.

Chapter 3. The Swap Rule

From paper #61, DICTO DOI 10.5281/zenodo.23682596

Ganesha, the Janus tetrahedra of squashed stackings, and the border at true close packing

Abstract. Stack layers of balls, cubic or hexagonal in any order, and share the space between the balls' holes by weight: each hole gets a piece. Paper #57 showed that at true close packing an octahedral piece is two halves, each borrowed from a pure stacking, while a tetrahedral piece is borrowed whole from its nearest layer. Here the layers are squashed or stretched. We prove that for every layer spacing from half the close-packed squash upwards, every hole piece, tetrahedral ones included, is a top half from the pure stacking of the layer above and a bottom half from the pure stacking of the layer below. Two consequences follow. The **Swap Rule**: the two mixed pieces always add up to the two pure ones. And a sharp **border**: the tetrahedral pieces are whole exactly when the layers are at least as far apart as in true close packing; squash them even slightly and each mixed tetrahedral piece gets its head from one side and its body from the other. DICTO calls these pieces **Ganesha**. Below half the squash the halves stop being pure, yet the Swap Rule for volumes still holds, down to a quarter of the squash exactly. The rule was found by DICTO's idea of looking for simple rules in pictures of computed values; a map of eleven spacings shows where each rule holds.

1. The idea

DICTO's idea (27 October 2026): the rule of halves of #57 (DICTO DOI 10.5281/zenodo.23647898) was spotted in a picture of computed values; other such pictures may hide other simple rules (halves, thirds, ...). A rule finder was built to look for them (Splash Rules, study 2026-10-27-splash-rules). Its first run, on the #57 walk with the layer spacing changed, found a rule nobody had looked for: when the layers are squashed, the mixed tetrahedral pieces stop being whole, yet their volumes still add up exactly. DICTO named the rule the Swap Rule and the new pieces Ganesha, after the god whose head was swapped.

2. Set-up

Balls of spacing 1 lie in triangular layers at positions A, B, C, stacked by a word with no letter twice in a row; the layer spacing is d, with $d^2 = s \cdot ⅔$ (s = 1 is true close packing). Each gap between a layer X and the next layer Y holds three holes per cell: an octahedral hole over the third position at mid-gap, a tetrahedral hole over X at ¾ of the gap and one under Y at ¼. As in #57 the octahedral holes get weight 0 and the tetrahedral holes weight λ, and space is divided by power distance (a Laguerre or power diagram); each hole's share is its *piece*. A layer is **c** (cubic) if its two neighbours differ, **h** (hexagonal) if they are equal; the pure stackings are ABC (all c) and AB (all h).

3. Main theorem

> **Theorem.** Let $s \geq ½$. Put $\lambda_-(c) = -¼ - s/8$, $\lambda_-(h) = -⅓ - s/8$ and $U(s) = \min(⅓ + s/24, 3s/8)$. In every close-packed stacking, periodic or not, take any hole piece of a gap whose lower layer has type t and upper layer type t′, at any λ with $\max(\lambda_-(t), \lambda_-(t')) \leq \lambda \leq U(s)$. Cut it by the horizontal mid-plane of its gap. The part below is congruent to the lower half of the piece of the same kind in the pure stacking of type t, and the part above to the upper half of that piece in the pure stacking of type t′. This holds for all three kinds of hole piece.

The weight range contains the whole walk of the cubic stacking for every $s \geq ½$, and the whole walk of every stacking for $s \geq 1$.

Corollary 1 (the Swap Rule). Write V(t|t′) for the volume of a piece over a t layer and under a t′ layer. Then, for all three kinds of piece,
V(c|h) + V(h|c) = V(c|c) + V(h|h).

Proof: V(t|t′) = L(t) + R(t′) with L and R the pure lower and upper half volumes; both sides equal L(c) + L(h) + R(c) + R(h).

Corollary 2 (the border at true close packing). For s ≥ 1 the far half of a tetrahedral piece is the same over a c layer as over an h layer, so the piece is borrowed whole from its nearest layer, as in #57. For every s with ½ ≤ s < 1 there is a weight in the cubic walk where the far half over a c layer is a proper part of the far half over an h layer, with strictly smaller volume: the mixed tetrahedral pieces are then Janus c/h pieces in their own right. These are the **Ganesha** pieces.

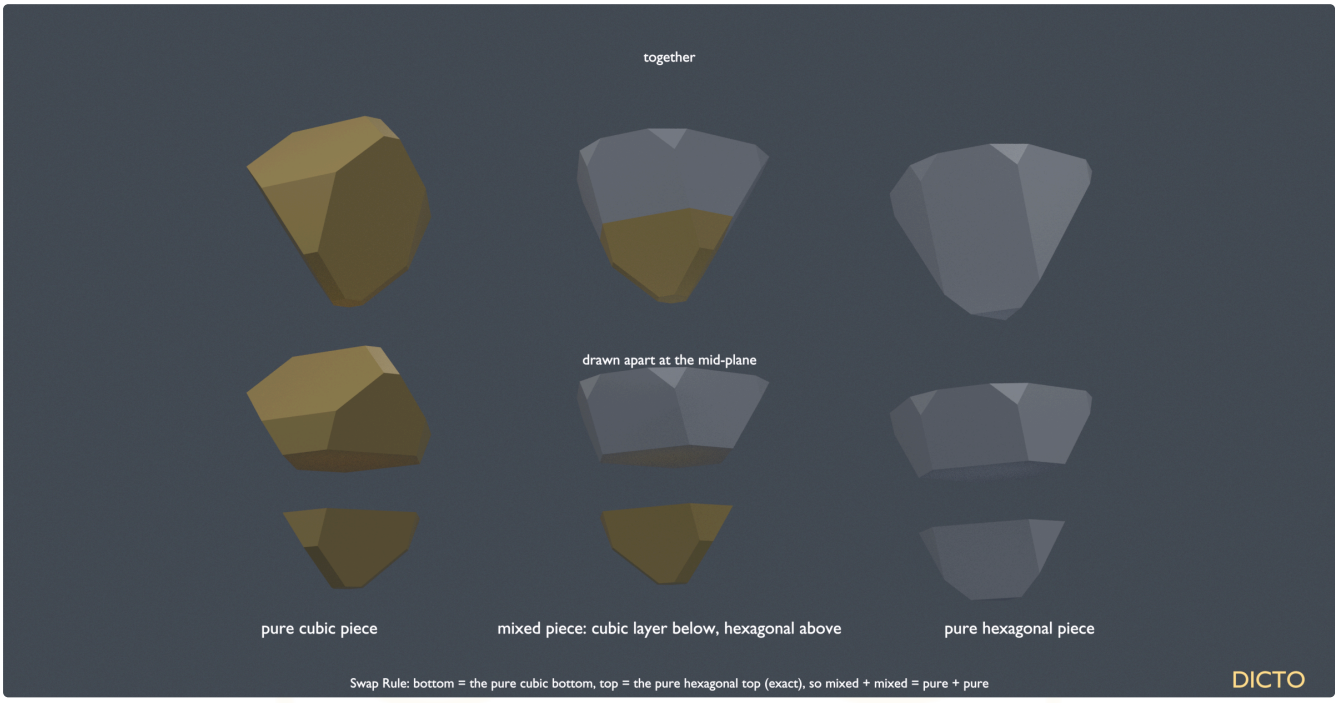


Figure 3.1. Ganesha. Left, the pure cubic tetrahedral piece (gold); right, the pure hexagonal one (platinum); middle, a mixed piece of the stacking ACBC (cubic layer below, hexagonal above), whole (top) and drawn apart at the mid-plane (bottom): its bottom is the cubic bottom, its top the hexagonal top. Layer spacing² = ¾·⅔, λ = 1/10; volumes of the parts 0.058005 and 0.196269.

Corollary 3 (4 : 5 : 6). At true close packing, for 5/24 < λ < 3/8 the three octahedral pieces (cubic, Janus, hexagonal) have volumes in the exact ratio 4 : 5 : 6 and vanish together at λ = 3/8. Near the end of the walk each octahedral piece is cut out by its tetrahedral neighbours only. In the cubic stacking it has 8 of them, one on each face, and is a regular octahedron |x| + |y| + |z| ≤ 1 (volume 4/3 at that scale); in the hexagonal stacking the octahedral holes sit face to face along the stacking axis, only 6 tetrahedral neighbours remain, and the piece is the rhombohedron |x + y − z|, |x − y + z|, |−x + y + z| ≤ 1 (volume 2). Hence 4 : 6, and the Janus piece, half of each, gives 5.

4. The map

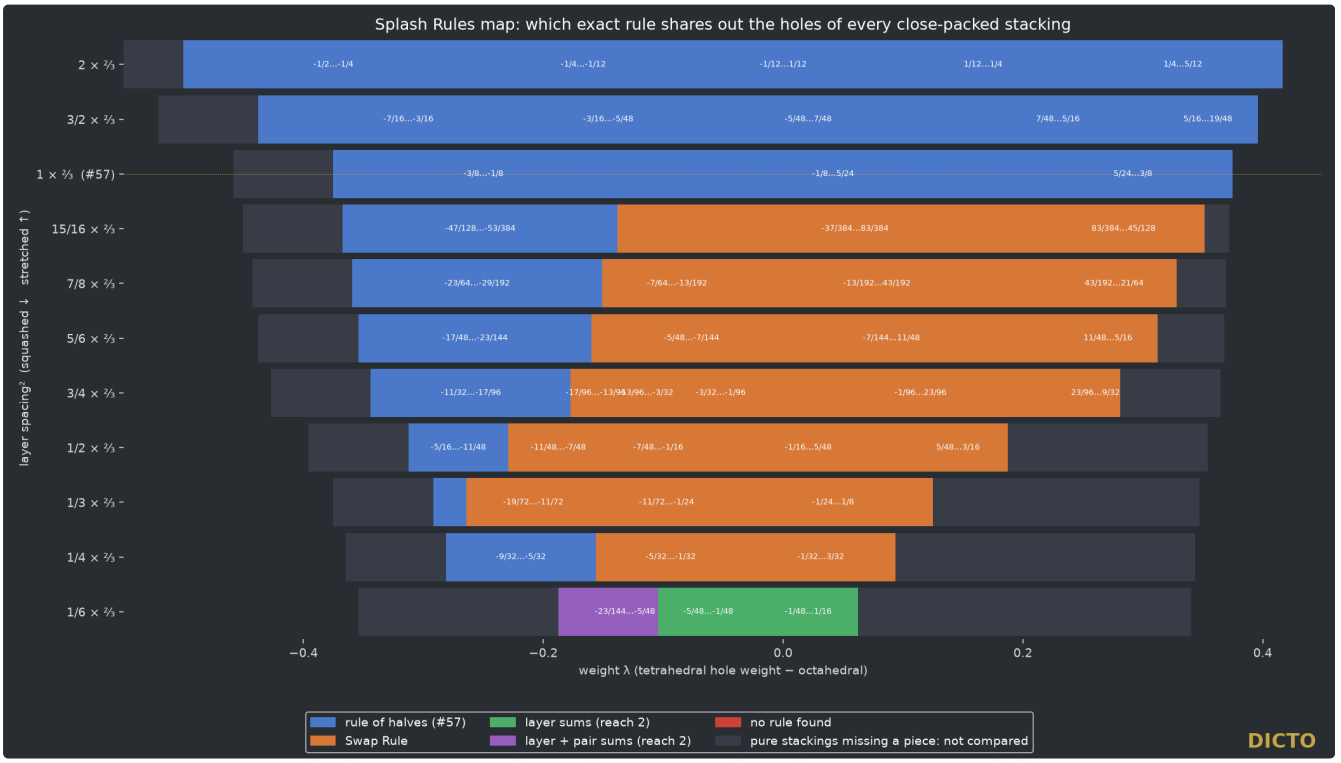


Figure 3.2. Which exact rule shares out the holes, for eleven layer spacings (rows; squashed below, stretched above; #57 dotted) and every window of the weight walk (columns). Blue: rule of halves only. Orange: Swap Rule with Ganesha pieces. Green and purple (s = 1/6): pieces reach two gaps; each volume is a sum of one share per layer reached (green), plus one share per neighbouring pair of layers (purple). Grey: a pure stacking lacks the piece, nothing to compare. Every window has a rule. Stackings with up to six layers; all values exact.

The map was made by the rule finder: each piece is labelled by the layer pattern around it, the shortest label that fixes its exact volume is found, and each volume is fitted exactly as a combination of the pure ones. At every spacing tested from 15/16 down to ¼ the Swap Rule appears, so it switches on as soon as the layers come closer than true close packing; the theorem proves this for $s \geq ½$.

5. Proof

One polyhedron for everything. Use lateral coordinates x, y, the height ζ in layer spacings and $q = d^2 = 2s/3$, and put $u = q \cdot \zeta$. Then every power-distance comparison between two holes becomes a linear inequality in the five variables (q, λ, x, y, u). So a piece, its cut at the mid-plane ($u \leq q/2$) and the whole range of spacings and weights ($q \geq ⅓$, $\lambda_- \leq \lambda \leq U$, all linear in q) form one convex polyhedron in five dimensions.

Locality, for all spacings at once. The lower half of a piece is fixed by the holes of its own gap and of the gap below, which depend only on the type t of the lower layer. That no other hole of any stacking cuts into it is a linear statement on the polyhedron, proved by one exact Farkas certificate per outside hole: 1070 certificates, all checked in exact rational arithmetic, covering every $s \geq ½$ ($s \to \infty$ included) and every λ in range. Lateral and vertical domination arguments reduce the outside holes to finite lists. The upper half follows by the mirror $\zeta \to 1 - \zeta$.

Range of the walk. Summing bisector inequalities against neighbours whose weighted offsets cancel shows when a piece must be empty; this gives the bounds λ_- and U (four cases, symbolic in q and λ).

Border. For $q \geq ⅔$ ($s \geq 1$), 38 certificates show that the holes of the gap below never reach a tetrahedral piece's far half, so it is the same for c and h. For $½ \leq s < 1$, six holes present only below a c layer cut the h far half (the cutting function is 1/12 or 1/6 at $s = ½$, 0 at $s = 1$, and concave), twelve holes present only below an h layer do not cut the c far half, and interior points are exhibited: the c far half is a proper part of the h far half.

Sharpness. At s = ½ exactly, the octahedral hole of the next gap first reaches past the mid-plane (the cutting function is 0 there and positive just below). At s = 2/5 the exact engine finds halves that are not the pure ones, and above U(s) (s = ¾, λ = 5/16) the halves rule and the Swap Rule both fail. As an independent check, the engine of #57, rebuilding whole stackings from scratch, agrees with the theorem in 4992 of 4992 exact comparisons (six stackings, s = ½, 5/8, ¾, 1, 3/2, 2).

6. Below half the squash

Below s = ½ the halves are no longer pure, yet the Swap Rule for volumes still holds.

> **Theorem 2.** For ¼ ≤ s < ½ and every λ up to 3s/8, in every close-packed stacking, each hole piece has volume V(t|t′) = A(t) + B(t′): one term fixed by the lower layer's type, one by the upper layer's. Hence the Swap Rule V(c|h) + V(h|c) = V(c|c) + V(h|h) holds for every s ≥ ¼. The bound ¼ is exact.

Why. Below s = ½ the octahedral hole of the next gap reaches across the mid-plane and carves a fixed piece D out of the lower part of this gap's octahedral and tetrahedral pieces, but only when the layer it shares is cubic. D lies against the mid-plane, out of reach of every hole of the gap below that differs between a cubic and a hexagonal layer, so it is the same piece whatever the far layer is. The lower part of a piece is therefore the pure lower half minus D when the upper layer is c; by the mirror, the upper part is the pure upper half minus a second fixed piece when the lower layer is c. Each correction is one constant per cubic layer, so the volume stays a sum of one share per layer. The halves fail only because D sits on one side of the mid-plane: in the pure cubic stacking every piece loses it, in a mixed one only some do. In exact numbers, at s = 2/5 and λ = –29/100 the octahedral pieces move δ = 2/3125 across the mid-plane inside the same piece, and the two mixed tetrahedral pieces carry +7/37500 and –7/37500, which cancel in the sum.

Proof. The same five-variable polyhedron as in §5, now for q from 1/6 to ⅓: 1706 exact Farkas certificates show that no other hole cuts a lower part and that, on each carved region, every hole that differs between c and h is inactive. At s = ¼ exactly, every hole two gaps away stops cutting (13 critical cases; the cutting function is 0 at q = 1/6 and changes sign there). At s = 1/5 the exact engine shows octahedral volumes depending on all four layers round the gap and tetrahedral ones on three, so the two-type rule fails. At s = 1/6 the pieces reach two gaps and the layer-sum rules of Figure 3.2 take over.

7. Prior art and credit

Power diagrams are affine in the weights, and absorbing the layer spacing into the weights (u = q·ζ) is that standard fact used once more; Farkas certificates are textbook linear programming. The rule of halves and the Janus c/h pieces Chucky and Fang are DICTO's #57. No source checked states a halves rule or a Swap Rule for hole pieces, or treats the layer spacing as a parameter of the hole power diagram.

8. Conclusion

True close packing is a border. At or above it, the holes of any stacking are shared out by the rule of halves of #57, and the tetrahedral pieces are whole. Below it, down to half the squash, every hole piece is a top from one pure stacking and a bottom from the other; the mixed tetrahedra become Ganesha pieces, and the two mixed pieces always add up to the two pure ones. Further down, to a quarter of the squash, the halves give way but the sums do not. However the layers are stacked, the space is still built from the halves of two pure orders. Chapter 15 gives the rule its exact weight: a count fixed by two neighbouring layers is a straight line in the hexagonality exactly when it obeys the Swap Rule, and where it does not, the failure F(c, c) + F(h, h) – F(c, h) – F(h, c), the Swap defect, is the coefficient of the one further number the count needs.

Chapter 4. The Swap Rule in Crystals

From paper #64, DICTO DOI 10.5281/zenodo.23709753

Rock salt, cadmium iodide, zinc blende and the Laves phases on every stacking

Abstract. Many crystals come in a cubic and a hexagonal form that differ only in how their layers are stacked, and in many mixed stackings between them. Paper #57 found that at close packing the space of every stacking is shared out into pieces borrowed from the two pure stackings, whole or in halves (the rule of halves), and #61 proved the Swap Rule: the two mixed pieces always add up to the two pure ones. Here we take both rules to four classical crystal families and prove, exactly, that **the Swap Rule holds in all four**, in all 299 cases where it can be tested: rock salt and nickel arsenide (NaCl, NiAs), the half-filled layers of cadmium chloride and cadmium iodide ($CdCl_2$, CdI_2), zinc blende and wurtzite with two different atoms (ZnS), and the Laves phases ($MgCu_2$, $MgZn_2$, $MgNi_2$). The rule of halves holds in full for rock salt and cadmium iodide: each octahedral piece is exactly half cubic and half hexagonal, each ball and tetrahedral piece is borrowed whole from the layer it rests on. In zinc blende the atoms keep the rule of halves, while the empty holes stop being fixed mixtures and still swap exactly. In the Laves phases, near the real alloys, every stacking is cut into exactly the same pieces.

1. The idea

DICTO's rule of halves (#57) and Swap Rule (#61) were found on the plain close-packed stacking of equal balls. DICTO asked whether they belong to that one structure or to stacking itself. The test is to fill the holes of the stacking with atoms as real crystals do, and to ask the same two questions of every resulting piece: is it borrowed from the pure cubic and hexagonal stackings, and do the mixed pieces always add up to the pure ones?

2. Set-up

A close-packed stacking is a word in the layer positions A, B, C; each layer is cubic (c) when its two neighbours differ and hexagonal (h) when they are the same. Every gap between two layers X and Y has a ball site at its foot, two tetrahedral sites (one over Y at ¼ of the gap, one over X at ¾) and one octahedral site (over the third position at ½). A crystal family fills some of these sites with atoms. Its space is shared out by weight (a power diagram, as in #57): atoms of one kind have weight λ, empty holes weight 0, and λ runs over the whole range in which every site keeps a piece. The four families:

Family	Atoms (weight λ)	Empty (weight 0)	Pure cubic / hexagonal
Rock salt / NiAs	balls, octahedral sites	tetrahedral sites	NaCl / NiAs
$CdCl_2$ / CdI_2	balls, octahedral sites in every other gap	tetrahedral sites, the other octahedral sites	$CdCl_2$ / CdI_2
Zinc blende / wurtzite	balls (λ), one tetrahedral site per gap (λ + μ, a second atom; μ = ±1/20, ±1/10)	octahedral and the other tetrahedral sites	ZnS sphalerite / wurtzite
Laves phases	big atoms on the diamond-type sites (λ)	— (small atoms, weight 0: a kagome of three and one more per gap)	$MgCu_2$ (C15) / $MgZn_2$ (C14); $MgNi_2$ (C36) mixed

The Laves network is built on every stacking by one rule (§5) and checked exactly: every small atom has 6 small and 6 big neighbours (an icosahedron), every big atom 4 big and 12 small, at squared distances ¼, 11/32 and 3/8 (ball spacing 1), the ideal Laves geometry. On the cubic stacking it is exactly $MgCu_2$.

3. Main theorem

Theorem. In each of the four families, on every close-packed stacking of up to six layers (all of them, up to symmetry; odd ones doubled for $CdCl_2$/CdI_2) and at every weight λ of its range, every piece has its volume fixed

by the layers round its gap (four layers; six for the Laves foot atom), and

(1) **Swap Rule.** For every kind of piece whose volume is fixed by the types of its two neighbouring layers, V(c under h) + V(h under c) = V(c under c) + V(h under h): 299 of 299 cases, none failing. (Not testable: 40 zinc blende cases where a piece vanishes in one pure stacking, and the Laves foot atom above λ = 17/96, which reads three layer types.)

(2) **Rule of halves, rock salt and cadmium iodide.** Every octahedral piece, filled or empty, is exactly half the cubic piece and half the hexagonal piece. Every ball takes the pure piece of its own layer's type, and every tetrahedral piece that of the layer holding its base.

(3) **Zinc blende.** The balls and the second atoms keep the rule of halves (the balls in 40 and the second atoms in 34 of the 43 windows where both pure pieces exist; the balls are undetermined in the other 3), the empty octahedral holes are a fixed cubic–hexagonal mixture in only 4 of the 43, yet every piece swaps exactly.

(4) **Laves phases.** For −11/32 < λ < 17/96, which contains the plain Voronoi cells of the real alloys (λ = 0), every Laves stacking is cut into exactly the same pieces: the cubic and hexagonal pieces are equal. Above 17/96 five of the six kinds of piece keep the rule of halves and swap exactly; the big atom at the foot of each gap reads three layer types, not two.

4. The results family by family

Family	Weight range	Windows	Swap Rule	Rule of halves
Rock salt / NiAs	−3/8 to 3/8	5	20 of 20	in full: octahedral ½c + ½h, balls and tetrahedral whole
$CdCl_2$ / CdI_2	−3/8 to 3/8	10	80 of 80	in full, filled and empty octahedral sites alike
Zinc blende / wurtzite, four values of μ	within −3/8 to 3/8	53	172 of 172	atoms yes; empty holes mostly not fixed mixtures
Laves phases	−11/32 to 11/32	5	27 of 27	identical pieces below 17/96; above, 5 of 6 kinds

The rock salt and cadmium iodide pattern is exactly #57's: three balls from each side make an octahedral piece half of each, a tetrahedral piece with its base in one layer belongs to that layer. Zinc blende repeats what #61 found below half the squash: the Swap Rule outlasts the rule of halves, because each piece's volume stays a sum of one share from each neighbouring layer even when the shape is no longer a fixed mixture.

5. The Laves network on every stacking

In each gap X, Y (third position Z) the big atoms sit over X at 0 and ¾ of the gap, as the diamond-type atoms of #60; the small atoms form a kagome over X shifted by half a ball spacing, (½, 0), (0, ½), (½, ½), at ⅜, and one more over Z at ⅞. On the cubic stacking this rule gives the same small atoms as the classical construction of $MgCu_2$ (each empty hole of the big-atom diamond surrounded by an inverted tetrahedron of small atoms), checked exactly; on the hexagonal stacking it gives $MgZn_2$, and on every stacking up to six layers it gives the ideal Frank–Kasper neighbourhoods, with the next neighbours clearly farther (squared gap 31/96 or more).

6. Proof

For each family and each stacking word we run the exact weight walk of #57 (rational arithmetic throughout): it finds every λ at which some piece changes its combinatorial type and, between these stops, every piece's volume as an exact polynomial in λ, with the engine's own checks (pieces tile the cell, faces matched, distance bounds) — no check failed. Locality: two stackings that agree on the four layers round a gap (six for the Laves foot atom) give the same volume, every time (0 clashes). Each piece's volume is then compared, as a polynomial, with a·V(cubic) + b·V(hexagonal) for constant rational a, b (the rule of halves), and the swap sum is checked as a polynomial identity.

Scripts and logs: study 2026-10-27-splash-rules, run2/ (rocksalt_mix.py, cdi2_mix.py, zincblende_mix.py, laves_net.py, laves_mix.py, laves_mix6.py).

7. Prior art and credit

Polytypes are classical: Ramsdell's notation (3C, 2H, 4H …) and the c/h notation of Jagodzinski; the cubic and hexagonal forms of ZnS, SiC, CdI_2 and the NaCl–NiAs pair (Verma and Krishna, *Polymorphism and Polytypism in Crystals*, 1966). The Laves phases C15, C14 and C36 are polytypes of one common slab, with c, h and ch stackings, and are built from Z12 and Z16 Frank–Kasper clusters. A study of block-polymer Laves phases found the free energy of C36 to be almost exactly the average of C14 and C15, a statement of the same shape as the Swap Rule, but for energies and not proved exactly. The weight walk, the rule of halves and the Swap Rule are #57 and #61. We found no earlier statement that the pieces of these crystals obey the rule of halves or the Swap Rule exactly.

8. Conclusion

The Swap Rule is not a property of one packing. In rock salt, cadmium iodide, zinc blende and the Laves phases, on every stacking, the two mixed pieces add up to the two pure ones, exactly, in every one of the 299 cases where both pure pieces exist and two layer types fix the piece. Where atoms sit in octahedral sites the pieces are DICTO's Janus halves, as at close packing; where the holes stay empty the shapes may wander but the volumes still swap; and in the Laves phases the stacking, near the real alloys, makes no difference to the pieces at all. One rule runs through all four families. By Chapters 5 and 15, every count of these swapping pieces therefore totals a straight line in the hexagonality.

Chapter 5. The Focus

From paper #66, DICTO DOI 10.5281/zenodo.23717687

One study of space in pieces, and the Hexagonality Law

Abstract. DICTO concluded that “everything is the same study now: all aspects of everything we have been looking at have come to a focus”. This chapter sets out the new theorem that ties the flat stackings together. Stack equal balls in layers, each layer cubic or hexagonal by its neighbours, and share space out among the balls and the holes between them by weight, as real crystals do. **The Hexagonality Law:** in every stacking, each kind of piece totals exactly what its layers would hold in their own pure stackings, so the total is a straight line in the number of hexagonal layers, whatever their order. We prove it from the Swap Rule (#61) wherever a piece is fixed by its two neighbouring layers, show that it is equivalent to the Swap Rule there, and check it exactly in 1064 cases across pyrochlore, rock salt, cadmium iodide, zinc blende and the Laves phases, with no failure; in cadmium iodide, whose filled and empty layers alternate, it holds for each kind of layer separately. It holds even for the one Laves piece that reads three layers, where it is not yet proved. In the physics of stacking variants, straight-line laws in the hexagonality have been measured for sixty years; this is an exact geometric law of that form.

1. The focus

Over the last papers the threads of this work converged: the rule of halves for stackings of balls (#57), its pyrochlore and crystal forms (#60, #64) and the Swap Rule (#61). DICTO concluded that “everything is the same study now: all aspects of everything we have been looking at have come to a focus”. The common thread is that space in the flat stackings is shared out into a few kinds of pieces built from the tetrahedron and the octahedron, whole or in halves, and that when a stacking changes, pieces change by swapping. This chapter proves the theorem that makes “change by swapping” exact for the stackings: the Hexagonality Law (§3).

2. Stackings, layers and pieces

Close-packed layers of equal balls sit in three positions A, B, C, and each new layer sits on the one below in one of the two other positions, so a stacking is a word such as ABC (cubic, 3C), AB (hexagonal, 2H) or ABCB (4H). Each layer is *cubic* (c) when its two neighbours differ and *hexagonal* (h) when they are the same: Jagodzinski's notation. In a stacking with period n, n_c layers are cubic and n_h hexagonal; the hexagonality is n_h/n. Every gap between two layers holds a ball site at its foot, two tetrahedral sites and one octahedral site. A crystal family fills some of these sites with atoms, and space is shared out among all sites by weight, in a power diagram, as in #57 and #64: atoms of one kind have weight λ (zinc blende's second atom λ + μ), empty holes weight 0. Each site's cell is a *piece*, and its *role* is its kind of site (ball, octahedral, one of the tetrahedral sites; for the Laves phases the six sites of their net, #64). Everything is exact in rational arithmetic, with volumes as polynomials in λ.

3. The Hexagonality Law

Theorem (Hexagonality Law). Let a role have, in the gap between layers of types x (below) and y (above), a piece of volume F(x, y), the same in every stacking. Then the following are equivalent:
(i) the Swap Rule $F(c, h) + F(h, c) = F(c, c) + F(h, h)$;
(ii) in every stacking, the total volume of the role over one period is $T = n_c \cdot F(c, c) + n_h \cdot F(h, h)$.
So each layer holds, in every stacking, exactly what it holds in the pure stacking of its own type, and the total per layer is the straight line $F(c, c) + (n_h/n) \cdot (F(h, h) - F(c, c))$ in the hexagonality.

Proof. (i) ⇒ (ii). Put $g(c) = 0$, $g(h) = F(c, h) - F(c, c)$ and $f(x) = F(x, c)$. Then $F(x, y) = f(x) + g(y)$ for all four pairs: for (x, c) by definition, for (c, h) by definition, and for (h, h) by the Swap Rule, since $f(h) + g(h) = F(h, c) + F(c, h) - F(c,$

c) = F(h, h). Summing over the n gaps of one period, gap k between layers k and k + 1, gives $T = \Sigma_k f(t_k) + \Sigma_k g(t_{k+1}) = \Sigma_k (f + g)(t_k)$, because the period closes up. As (f + g)(c) = F(c, c) and (f + g)(h) = F(h, h), $T = n_c F(c, c) + n_h F(h, h)$. (ii) ⇒ (i). The stacking ABCB (4H) has layer types alternating h, c, so its four gaps are (h, c), (c, h), (h, c), (c, h) and T = 2F(h, c) + 2F(c, h); (ii) gives T = 2F(c, c) + 2F(h, h). ∎

Two kinds of layer. In cadmium chloride and cadmium iodide the octahedral sites are filled in every other gap only, so the layers come in two kinds, alternating. The same proof, with f and g allowed to depend on the kind, gives $T = f(c)n_c' + f(h)n_h' + g(c)n_c'' + g(h)n_h''$, where ′ and ″ count the layers of each kind: the law holds for the hexagonality of each kind of layer separately. A single hexagonality is not enough there: it fails exactly for the stackings ABCB and ABCBAB, the two in which the two kinds of layer have different hexagonalities.

Exact check. For every family of the crystal runs of #60 and #64, every close-packed stacking up to six layers (AB, ABC, ABCB, ABACB, ABCBAB, ABCACB, up to symmetry), every window of the weight λ and every role, we took the total volume over one period from the exact weight walks and tested it against $n_c \cdot T(ABC)/3 + n_h \cdot T(AB)/2$, as an identity of polynomials in λ.

Family	Exact tests	Linear in the hexagonality
Pyrochlore (#60)	96	96
Rock salt / nickel arsenide (#64)	80	80
Zinc blende / wurtzite, μ = ±1/20, ±1/10 (#64)	688	688
Laves phases (#64)	120	120
Cadmium chloride / iodide (#64), by kind of layer	80	80 (four unknowns from six stackings, consistent)
Total	**1064**	**1064**

(In cadmium iodide a single hexagonality was also tested, in 320 cases: 74 fail, all in ABCB and ABCBAB, as the two-kinds statement predicts.) The theorem's hypothesis, that a piece is fixed by its two neighbouring layers, was checked in #64 for every role except the Laves foot atom above λ = 17/96, which reads three layer types. For that role the law holds in every test above, but the proof does not apply: it is an open problem whether it holds for every stacking.

4. Physics

In the physics of stacking variants (polytypes), a property of a crystal is written as a sum over its layers, with terms for each layer alone and for pairs of layers further apart: the axial next-nearest-neighbour Ising (ANNNI) picture of Heine and coworkers for silicon carbide (Cheng, Needs and Heine, 1988). In that language the Hexagonality Law says that for the volume of every piece the pair terms are exactly zero: each layer contributes alone. Straight-line laws in the hexagonality are classical in measurement. The indirect band gap of the silicon carbide polytypes rises with the hexagonality, roughly in a straight line (Choyke, Hamilton and Patrick, 1964), and basal lattice constants of metallic polytypes correlate linearly with it; later work finds these laws approximate. The Hexagonality Law is an exact geometric law of the same form, for the space each atom and hole occupies. We make no claim that it explains the measured laws; it is a candidate for an exact part of them. One further echo, of form only: causal dynamical triangulations build spacetime in slabs between slices, from simplices typed by how they meet the slice above and the slice below, as our pieces are typed by the layers above and below their gap.

5. Conclusion

Space in flat stackings is shared out into a few kinds of pieces built from the tetrahedron and the octahedron, whole or in halves, and when a stacking changes, the pieces change by swapping, never by appearing from nothing. For the stackings the last clause is now a theorem: the Swap Rule is exactly the statement that every layer keeps, in every stacking, what it holds in its own pure stacking, so every kind of piece totals a straight line in the hexagonality. It holds in every crystal family we have computed, and in cadmium iodide for each kind of layer. Chapter 15 sets this theorem in a wider frame, the Window Law: whatever a stacking fixes through g neighbouring layers averages

to a linear function of the shares of its c/h words of length g, and the Swap Rule is exactly what brings a two-layer count down to one layer.

6. Prior art and credit

Jagodzinski's c/h (h/k) notation, the hexagonality and the polytypes are classical (Verma and Krishna, *Polymorphism and Polytypism in Crystals*, 1966). The ANNNI description of polytype energies is Heine's (Cheng, Needs and Heine, 1988). Linear trends of band gaps and lattice constants in the hexagonality are measured results (Choyke, Hamilton and Patrick, 1964), and an Ising-type additivity model has been used to estimate the band gaps of long polytypes (arXiv 2306.17756). We found no statement that the volumes of the weighted pieces of a close-packed stacking are exactly linear in the hexagonality, or that this is equivalent to the Swap Rule. The rule of halves and the Swap Rule are #57, #61 and #64.

Chapter 6. Random Stackings and Stacking Faults

From paper #67, DICTO DOI 10.5281/zenodo.23718636

The Hexagonality Law at work: colloids, ice and the faults of crystals

Abstract. The Hexagonality Law (#66) says that in a stacking of close-packed layers, where space is shared out by weight among atoms and holes, each kind of piece totals exactly what its layers hold in their own pure stackings. DICTO proposed to look for the conditions where this matters in real matter. Two follow at once, and both are exact. **Theorem 1 (any stacking):** however the layers are stacked, repeating, faulted or random, with any correlations, each kind of piece averages exactly V(cc) + h·(V(hh) – V(cc)) per layer, where h is the share of hexagonal layers; so the single number that diffraction measures in randomly stacked colloids and in stacking-disordered ice fixes every hole volume exactly. **Theorem 2 (faults):** a stacking fault changes the total of every kind of piece by exactly the number of layers it turns hexagonal (or cubic) times V(hh) – V(cc); the classical faults of cubic crystals (twin, intrinsic, extrinsic) turn 1, 2 and 2 layers hexagonal, those of hexagonal crystals (I1, I2, E) turn 1, 2 and 3 layers cubic. In rock salt the balls gain at each fault exactly what the holes lose. The one piece the law's proof does not cover, the Laves foot atom that reads three layers, keeps the law exactly on every stacking up to eight layers (390 of 390 exact tests).

1. Where the law matters

Paper #66 proved the Hexagonality Law for every kind of piece fixed by its two neighbouring layers, and checked it exactly in 1064 cases across pyrochlore, rock salt, cadmium iodide, zinc blende and the Laves phases. DICTO proposed the next step: to find the conditions in real matter where the law, the Swap Rule and the rule of halves play a significant part. Two such conditions are everywhere in crystals: stackings that are not periodic at all, and faults in otherwise perfect stackings. For both the law gives exact statements with no further computation, and this chapter sets them out.

2. Setting

As in #57, #64 and #66: close-packed layers in positions A, B, C; a layer is cubic (c) when its two neighbours differ and hexagonal (h) when they are the same (Jagodzinski); the sites of each gap (a ball site, two tetrahedral sites, an octahedral site) share space by weight in a power diagram, atoms with weight λ, empty holes 0. A *role* is a kind of site, and the law applies to every role whose piece in the gap between layers of types x and y has a volume F(x, y) fixed by those two types; #64 found this for every role computed except the Laves foot atom above λ = 17/96. For such a role, #66 proved that the Swap Rule F(c, h) + F(h, c) = F(c, c) + F(h, h) holds exactly when F(x, y) = f(x) + g(y), so that over any closed run of layers the total is $n_c F(c, c) + n_h F(h, h)$.

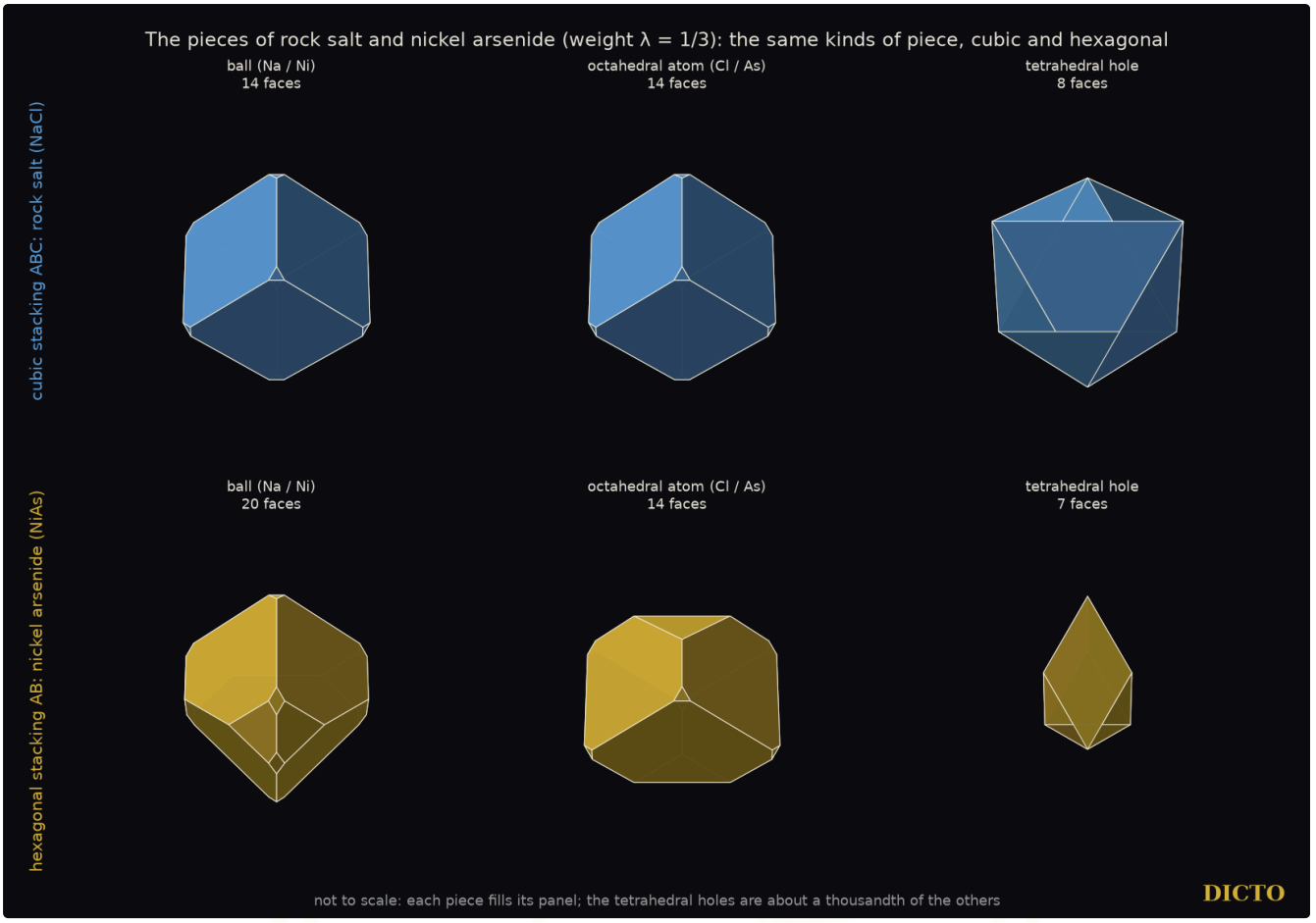


Figure 6.1. The pieces of rock salt (cubic stacking ABC, NaCl) and nickel arsenide (hexagonal stacking AB, NiAs) at weight λ = 1/3: the ball, the octahedral atom and a tetrahedral hole, each the power cell of its site. Their volumes agree with the exact volumes of the run to nine decimals. Not to scale: each piece fills its panel; the tetrahedral holes are in fact about a thousandth of the others. A mixed stacking is built from halves of these pieces, and by the Hexagonality Law each layer holds exactly the share of its own row.

3. Any stacking

> **Theorem 1.** Let a role satisfy the Swap Rule. For any sequence of N consecutive layers of any stacking, periodic or not, the total volume T of that role's pieces in the N – 1 gaps between them satisfies $|T - (n_c F(c, c) + n_h F(h, h))| \le 2 \cdot \max|f| + 2 \cdot \max|g|$, where n_c, n_h count the cubic and hexagonal layers. Hence, in any stacking whose share of hexagonal layers tends to h, the average volume per layer tends to $F(c, c) + h \cdot (F(h, h) - F(c, c))$, whatever the order of the layers and whatever the correlations between them.

Proof. With $F(x, y) = f(x) + g(y)$, the gaps between layers 1, …, N give $T = \Sigma_{k<N} f(t_k) + \Sigma_{k>1} g(t_k) = \Sigma_k (f + g)(t_k) - g(t_1) - f(t_N)$, and $(f + g)(c) = F(c, c)$, $(f + g)(h) = F(h, h)$. The two end terms are bounded; dividing by N gives the limit. ∎

So the total needs only h. The *split* of the pieces among the four kinds cc, ch, hc and hh does depend on how neighbouring layers are correlated, and the Markov descriptions of disordered stackings (Hart, Hansen and Kuhs, 2021) give it; the totals are independent of all of that.

Where h is measured. In crystals of nearly hard colloidal spheres the layers are often randomly stacked (random hexagonal close packing): each new layer takes either of the two positions not used by the layer below, with probability α for the cubic choice; α = ½ is completely random stacking, and then h = ½ (Pusey et al., 1989; later work discusses whether faulted-twinned cubic crystals are the stable form, Kegel and Dhont, 2000, and how such crystals age, Martelozzo et al., 2002). In ice frozen from supercooled water the layers are cubic and hexagonal in random order, stacking-disordered ice I_{sd} (Malkin et al., 2012), whose cubicity 1 – h is measured by diffraction:

73.3% in the most cubic bulk sample (Malkin et al., 2015), 78% in 20 nm particles (Amaya et al., 2017). The framework of ice and of diamond is the stacking of #60's pyrochlore run (zinc blende with both atoms alike), where the law held in all 96 exact tests (#66).

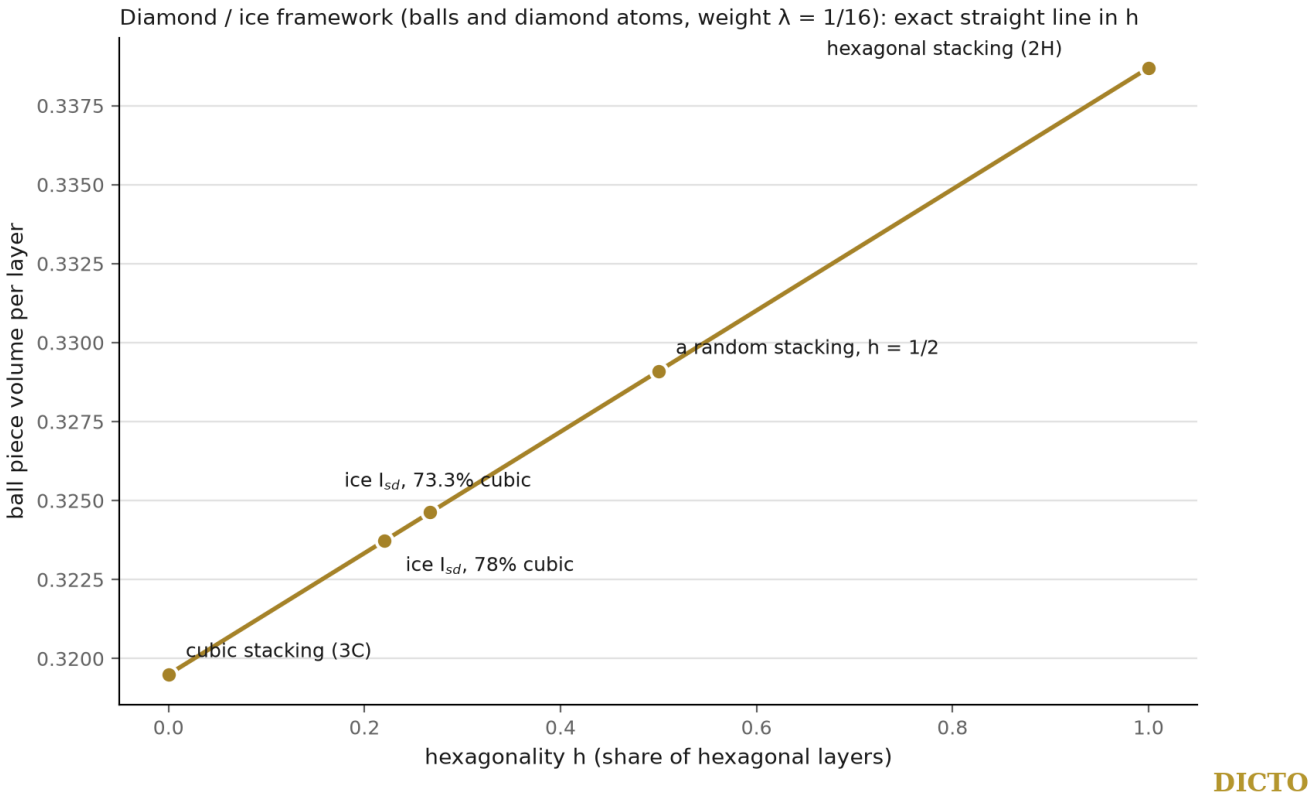


Figure 6.2. The diamond / ice framework (#60's pyrochlore run: balls and diamond atoms of weight λ = 1/16, holes 0): the volume of the ball pieces per layer is an exact straight line in the hexagonality h (Theorem 1). Marked: the two pure stackings, a completely random stacking (h = 1/2), and the measured cubicities of stacking-disordered ice (73.3%, Malkin et al. 2015; 78%, Amaya et al. 2017).

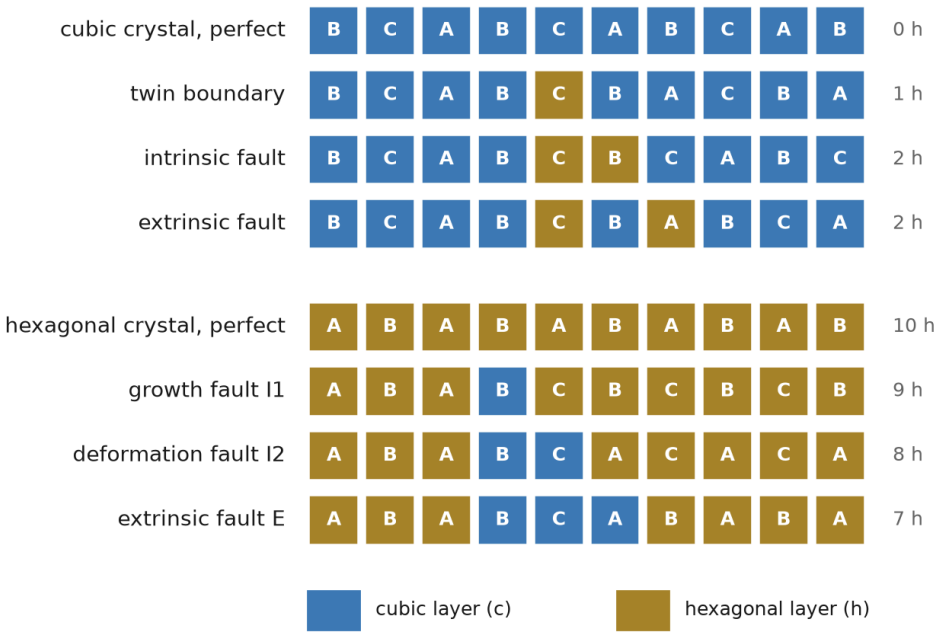
4. Stacking faults

> **Theorem 2.** Let a role satisfy the Swap Rule. Compared with a perfect crystal of the same number of layers, a stacking fault that changes Δ layers from cubic to hexagonal (Δ < 0: from hexagonal to cubic) changes the total volume of that role's pieces by exactly Δ·(F(h, h) – F(c, c)).

Proof. Theorem 1 with the same N and the same end layers: T changes by $(\Delta n_h)F(h, h) + (\Delta n_c)F(c, c)$ with $\Delta n_c = -\Delta n_h$. ∎

Counting the layers with Jagodzinski's rule on the classical sequences of each fault (perfect and faulted runs of 18 layers, read away from the ends):

Fault	Sequence	Layers changed
cubic crystal: twin boundary	…ABCABC**B**ACBAC…	1 becomes hexagonal
cubic crystal: intrinsic fault (a layer removed)	…ABCABC**BC**ABC…	2 become hexagonal
cubic crystal: extrinsic fault (a layer inserted)	…ABCABC**B**ABC…	2 become hexagonal
hexagonal crystal: growth fault I1	…ABAB**C**BCB…	1 becomes cubic
hexagonal crystal: deformation fault I2	…ABAB**CA**CA…	2 become cubic
hexagonal crystal: extrinsic fault E	…ABAB**C**ABAB…	3 become cubic



DICTO

Figure 6.3. The six classical faults as strips of layers, each layer coloured by its type (blue cubic, gold hexagonal; Jagodzinski's rule), ten layers round the fault. A twin turns one layer of a cubic crystal hexagonal, an intrinsic or an extrinsic fault two; the faults of a hexagonal crystal turn one, two and three layers cubic.

So the faults of cubic crystals stand 1 : 2 : 2 and those of hexagonal crystals 1 : 2 : 3, in their effect on every kind of piece. In the physics of stacking faults the energies are ordered in the same way, in terms of the number of layers of the other kind that a fault creates, to lowest order (ANNNI descriptions of fault energies); here the ratios are exact, for the space each atom and hole occupies.

Rock salt. In rock salt and nickel arsenide (balls and octahedral atoms of weight λ, #64), at λ = 1/3 (in the window 7/24 < λ < 3/8), each layer that a fault turns hexagonal changes the pieces by exactly:

Piece	V(cc)	V(hh)	Change per hexagonal layer
ball	0.49961	0.52016	+71/3456
octahedral atom	0.49961	0.47888	–215/10368
tetrahedral hole (lower)	0.00039	0.00048	+1/10368
tetrahedral hole (upper)	0.00039	0.00048	+1/10368

The four changes add up to exactly zero (213/10368 – 215/10368 + 2/10368): a fault moves space from the octahedral atoms to the balls and the small tetrahedral holes, and creates or loses none.

5. The Laves foot atom on longer stackings

The proof of the law needs each piece to be fixed by its two neighbouring layers. In the Laves phases (#64) the foot atom, the big atom M0, reads three layer types above λ = 17/96, so the proof does not apply to it; #66 found the law exact for it on every stacking up to six layers and left it as an open problem. Faults and random stackings bring longer runs of layers, so we extended the test to the nine further close-packed stackings of seven and eight layers (ABACACB, ABABACB, ABCBACB, ABACBACB, ABCBABAB, ABCBCBAB, ABCBCACB, ABCACBAB, ABCABACB), each walked exactly in rational arithmetic with #57's census engine (all census checks passed), and tested every role in every window of λ against $n_c \cdot T(ABC)/3 + n_h \cdot T(AB)/2$.

> **Result.** On all 13 non-pure close-packed stackings up to eight layers, in all five windows of λ, every role of the Laves phases, the foot atom M0 included, totals exactly $n_c F(c, c) + n_h F(h, h)$: 390 of 390 exact tests.

So the law reaches beyond its proof: a piece that depends on three layers still adds up layer by layer on every stacking tested. Whether it does so on every stacking remains an open problem.

6. Conclusion

Wherever the Swap Rule holds, the space of a stacking is counted layer by layer: each layer brings exactly its own pure share. A random stacking is therefore fixed, in its totals, by one number, the share of hexagonal layers, which is what diffraction measures in randomly stacked colloids and in stacking-disordered ice; and every stacking fault moves space by a whole number of layer shares, 1, 2 and 2 for the faults of cubic crystals and 1, 2 and 3 for those of hexagonal crystals. Where a count breaks the Swap Rule, exactly one more number enters at two layers, the share p_{hh} of neighbouring hexagonal pairs (Chapter 15).

7. Prior art and credit

Random hexagonal close packing of colloids and its stacking probability α are due to Pusey and coworkers (1989); stacking-disordered ice and its cubicity to Malkin, Murray, Salzmann and coworkers (2012, 2015); Markov descriptions of stacking disorder are reviewed by Hart, Hansen and Kuhs (2021); correlations in randomly stacked solids are studied by Ganesh and Nasr (2023). The classical fault sequences are textbook material, and the ANNNI description of fault energies in terms of polytype energies is standard. We found no statement that the volumes of the weighted pieces of a stacking are fixed exactly by its hexagonality, or that faults change them by whole multiples of one layer share. The Hexagonality Law is #66; the rule of halves is #57 (with #61 for squashed stackings and #64 for crystals), and the Swap Rule is #61 (with #64 for crystals).

Chapter 7. The Hexagonality Law and the Fibonacci Stacking

From paper #69, DICTO DOI 10.5281/zenodo.23721038

The golden ratio in the volumes of a never-repeating crystal

Abstract. DICTO's long-standing aim is a bridge between repeating and never-repeating order. Close-packed layers can be stacked in a never-repeating Fibonacci order: layer types cubic (c) and hexagonal (h) following the substitution c → ch, h → c. We prove that in this stacking the share of hexagonal layers is exactly $1/\varphi^2$ and that hexagonal layers never touch, so of the four kinds of piece between neighbouring layers only three occur, in the golden proportions $1/\varphi^3 : 1/\varphi^2 : 1/\varphi^2$ (cubic–cubic, cubic–hexagonal, hexagonal–cubic). By the Hexagonality Law (#66), every kind of piece then averages exactly $V(cc) + (V(hh) - V(cc))/\varphi^2$ per layer: the golden ratio enters the volumes of a crystal exactly. In rock salt at weight $\lambda = 1/3$ the ball piece averages $10999/20736 - (71/6912)\sqrt{5}$ per layer. The Fibonacci order closes into periodic stackings of 15, 16, 26 and 63 layers with hexagonalities 2/5, 3/8, 5/13 and 8/21, ratios of Fibonacci numbers. On the first three, the Laves phases keep the law exactly in all 90 tests, including the foot atom for which the law is not proved. The stackings in which hexagonal layers never touch are exactly the configurations of the supersymmetric lattice fermions of Fendley, Schoutens and de Boer, whose ground states have the hexagonality of 6H.

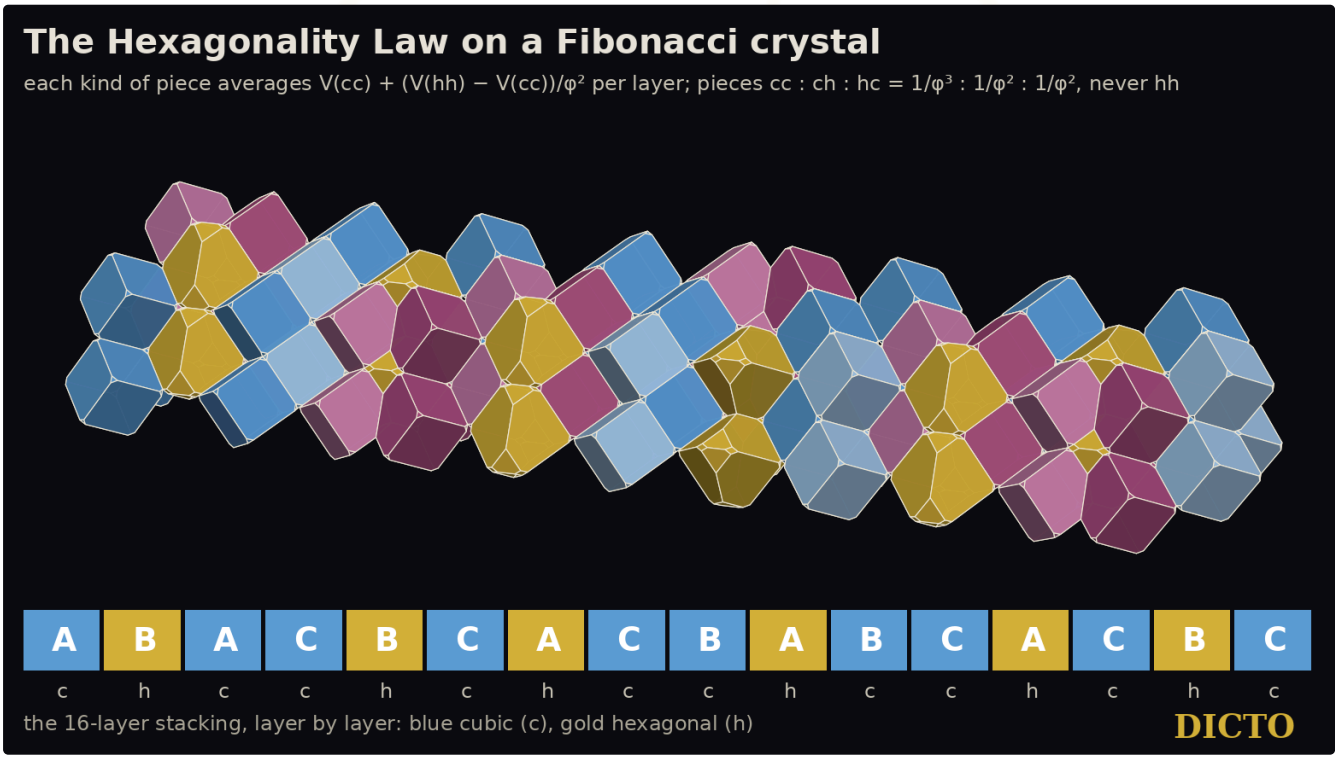


Figure 7.1. The Hexagonality Law on a Fibonacci crystal: rock salt on the 16-layer Fibonacci stacking ABACBCACBABCACBC at weight $\lambda = 1/3$, three neighbouring columns, the stacking running left to right. Each piece is the exact power cell of its site. Balls: cubic layers blue, hexagonal layers gold. Octahedral atoms between layers: cubic–cubic pale blue, cubic–hexagonal and hexagonal–cubic rose; hexagonal–hexagonal never occurs. Volumes agree with the exact run to nine decimals.

1. From repeating to never-repeating

A stacking of close-packed layers is a sequence of positions A, B, C, each layer in one of the two positions not taken by the layer below. The pure stackings repeat: ABC (cubic) and AB (hexagonal). The rule of halves (#57), the Swap Rule (#61, #64) and the Hexagonality Law (#66) hold on every stacking, and #67 drew from them that the totals of a random stacking are fixed by its share of hexagonal layers alone. DICTO's aim, to bridge the periodic and the aperiodic, asks for the most ordered of the never-repeating stackings: the Fibonacci one. In *A Fibonacci Spine*

(#30) the Fibonacci chain appeared in a five-fold arrangement of the DICTO Ogdoecontahedron; here it is the order of the layers themselves.

2. The Fibonacci stacking

A layer is cubic (c) when its two neighbours sit in different positions and hexagonal (h) when they sit in the same (Jagodzinski). Any sequence of types is a stacking: with steps $s_k = \pm 1$ between layers, layer k is c when $s_{k-1} = s_k$ and h when they differ, so the types fix the steps up to one sign. The **Fibonacci stacking** takes the types in the order of the Fibonacci word, the limit of the substitution c → ch, h → c started from c: chcchchcchcch… (Figure 7.2).

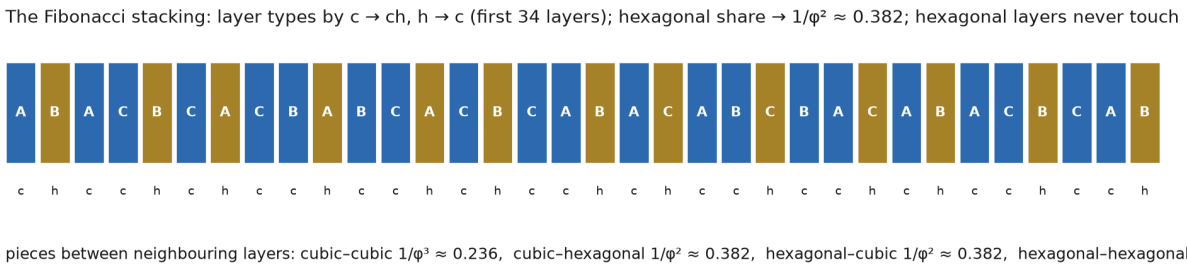


Figure 7.2. The first 34 layers of the Fibonacci stacking, coloured by type (blue cubic, gold hexagonal), with the position A, B, C of each layer. The hexagonal layers never touch.

> **Theorem.** In the Fibonacci stacking (i) the share of hexagonal layers is $h = 1/\varphi^2 = (3 - \sqrt{5})/2$; (ii) no two hexagonal layers are neighbours; (iii) among the pairs of neighbouring layers, cubic–cubic has frequency $1/\varphi^3$, cubic–hexagonal and hexagonal–cubic $1/\varphi^2$ each, hexagonal–hexagonal 0. (iv) For every kind of piece to which the Hexagonality Law applies, the average volume per layer is exactly $V(cc) + (V(hh) - V(cc))(3 - \sqrt{5})/2$.

Proof. (i) The substitution has matrix [[1, 1], [1, 0]] on the counts (c, h); its eigenvalue φ gives letter frequencies $1/\varphi$ for c and $1/\varphi^2$ for h, the classical result for the Fibonacci word. (ii) Every h in a substituted word is the image of c → ch, the second letter of a block, and every block begins with c; so each h is followed by c, and hh never occurs. (iii) Each h is therefore preceded and followed by c, so ch and hc each have the frequency of h, $1/\varphi^2$, and cc has $1 - 2/\varphi^2 = 1/\varphi^3$ (using $\varphi^2 = \varphi + 1$). (iv) By #66, over any long run of layers the total of such a piece is $n_c V(cc) + n_h V(hh)$ up to two end terms (#67, Theorem 1); divide by the number of layers and use (i). ∎

So a Fibonacci crystal contains the pure cubic piece and the two mixed pieces only, in golden proportions, and never the pure hexagonal piece; and the pieces of each kind average to a number in $\mathbb{Q}(\sqrt{5})$.

3. Rock salt in the Fibonacci order

In rock salt and nickel arsenide (balls and octahedral atoms of weight λ, tetrahedral holes empty, #64), at $\lambda = 1/3$ the exact averages per layer are:

Piece	Average per layer, exact	Decimal
ball	10999/20736 – (71/6912)√5	0.507461
octahedral atom	9715/20736 + (215/20736)√5	0.491693
each tetrahedral hole	11/20736 – (1/20736)√5	0.000423

They add up to the volume of one layer, as they must; the √5 parts cancel in the sum.

4. Repeating approximants, computed exactly

A periodic stacking must close: an even number of sign changes, and steps adding up to a multiple of 3. The Fibonacci words of Fibonacci lengths close after repeating them once, twice or three times: 5 types (chcch) three times gives the 15-layer stacking ABACBCACBABCBAC; 8 types twice the 16-layer ABACBCACBABCACBC; 13 types twice the 26-layer ABACBCACBABCACBCABACABCBAC; 21 types three times a 63-layer stacking. Their hexagonalities 2/5, 3/8, 5/13, 8/21 are ratios of Fibonacci numbers and tend to $1/\varphi^2$.

For the pieces fixed by their two neighbouring layers the law is a theorem (#66), so these stackings follow it. The Laves foot atom M0 reads three layers above $\lambda = 17/96$, where the law is not proved, and #67 found it exact on every stacking up to eight layers. We walked the Laves phases exactly on the 15-, 16- and 26-layer Fibonacci approximants with #57's census engine (all census checks passed) and tested every role in every window of λ against $n_c \cdot T(ABC)/3 + n_h \cdot T(AB)/2$.

> **Result.** On the 15-, 16- and 26-layer Fibonacci approximants, every role of the Laves phases, the foot atom M0 included, keeps the Hexagonality Law exactly: 90 of 90 tests. The 26-layer approximant (hexagonality 5/13 ≈ 0.385) is the closest to the golden $1/\varphi^2$ computed so far.

5. Hard-core fermions

Read a hexagonal layer as a particle on a ring of sites, one site per layer. A stacking in which no two hexagonal layers touch, the Fibonacci stacking among them, is then a configuration of particles with hard cores: never two on neighbouring sites. These are exactly the configurations of the supersymmetric lattice fermions of Fendley, Schoutens and de Boer (2003), whose Hamiltonian is $H = \{Q^+, Q^-\}$ with fermions that may not sit next to each other. On the periodic chain of N sites their ground states have int((N + 1)/3) fermions, a filling of 1/3, two of them when N is a multiple of 3. We counted, for every N from 2 to 15, the hard-core configurations by particle number: they agree with theirs (for N = 6: 1, 6, 9 and 2 with 0 to 3 particles, total 18), and so does their Witten index, $2(-1)^p$ for $N = 3p$ and $(-1)^p$ otherwise.

A crystal adds two conditions, because a periodic stacking must close: the step signs close only for an even number of hexagonal layers, so stackings lie in the sector of even particle number, and the layer positions close only when the steps add up to a multiple of 3. For N = 6 the closing configurations are the cubic stacking (no particle) and the three positions of 6H (two particles, h c c h c c); and 6H, the commonest polytype of silicon carbide, has hexagonality 1/3, the ground-state filling of the supersymmetric chain, in the same particle-number sector. The two pictures differ in their physics: the supersymmetric fermions hop and repel, while by the Hexagonality Law the volume each hexagonal layer carries does not depend on the others. We record the correspondence of configurations and counts only; the Fibonacci stacking's density $1/\varphi^2 \approx 0.382$ lies between the ground-state filling 1/3 and the hard-core limit 1/2.

6. Conclusion

The Fibonacci stacking is the never-repeating crystal of close-packed layers in its most ordered form. Its share of hexagonal layers is $1/\varphi^2$, its hexagonal layers never touch, and its pieces come in the golden proportions $1/\varphi^3 : 1/\varphi^2 : 1/\varphi^2$. By the Hexagonality Law the golden ratio then enters every average piece volume exactly. One kit of pieces, borrowed from the two repeating stackings, builds the never-repeating one: the bridge between the two orders that DICTO set out to find, here at the level of the pieces themselves. Its hexagonal layers never touch, so its share p_{hh} of neighbouring hexagonal pairs is zero, and every count of two neighbouring layers, Swap Rule or not, is read on a straight line in h; Chapter 15 carries this to the repeating approximants of quasicrystals, window after window.

7. Prior art and credit

The Fibonacci chain is the most studied never-repeating one-dimensional order: Baake (1999) obtains it by cutting and projecting the square lattice along a line of slope 1/τ, and calls it the most common and best studied non-periodic 1D point set; its theory is set out in Baake and Grimm, *Aperiodic Order* (2013), and its Fibonacci numbers and words in the combinatorics of Lothaire (as cited by Kramer, 2011). Our sequence of layer types is this chain. Periodic structures converging to a quasicrystal are its approximants (Goldman and Kelton, 1993), which our 15-, 16-, 26- and 63-layer stackings are; a fault in a Fibonacci stacking is the local flip by which an approximant and its quasicrystal differ (Kléman, 2013). Hard-core lattice fermions with N = 2 supersymmetry are Fendley, Schoutens and de Boer (2003). Disordered stackings of layers are described by Markov chains (Hart, Hansen and Kuhs, 2021; Ganesh and Nasr, 2023). The Fibonacci word and its letter and pair frequencies are classical. Fibonacci chains have been used to describe faulted close-packed crystals: Farkas-Jahnke proposed approximating heavily faulted structures such as ZnS by Fibonacci chains of five-layer structure elements (IUCr congress abstract). Fibonacci multilayers (for example quasiperiodic Fe/Cr magnetic multilayers) are studied in physics. We found no statement that a Fibonacci order of the layer types gives the golden proportions of the stacking's pieces, or exact golden-ratio volumes. The Fibonacci chain in DICTO's own work is #30; the laws used are #57, #61, #64, #66 and #67.

Part II. Real materials

Chapter 8. Real Polytypes and the Hexagonality Law

From paper #68, DICTO DOI 10.5281/zenodo.23719275

Silicon carbide, silicon, germanium and cerium against the straight line

Abstract. The Hexagonality Law (#66) says that in ideal stackings every kind of piece totals a straight line in the share h of hexagonal layers. DICTO proposed to test it against real crystals. In the language physicists use for stackings, the axial Ising (ANNNI) model, we prove that a property is a straight line in h exactly when its couplings between layers two and three apart vanish, J2 = J3 = 0, and that its stacking faults then stand intrinsic : extrinsic : twin = 2 : 2 : 1, the ratios of #67. We then read published data. The computed volumes of the silicon carbide polytypes lie on the line within the precision of the published numbers; the measured ones scatter between laboratories by more than the whole effect. Silicon and germanium bend slightly but clearly, by about 0.04% of the volume: their volume couplings J2 are of the size of J1, a relaxation that ideal geometry does not have. Germanium's stacking-fault energies come close to the ideal 2 : 2 : 1 (2.25 : 2.06 : 1), silicon's do not (4.1 : 2.4 : 1). Silicon carbide's band gap is far from straight. In cerium, the double-hexagonal form at h = 1/2 has 1.2% more volume than the cubic form. The straight line is the ideal part of every stacking; the departures measure what real crystals add.

1. From the ideal to the real

The Hexagonality Law was proved for the pieces of ideal stackings, where space is shared by weight among atoms and holes at fixed positions (#66), and #67 drew its consequences for random stackings and faults. DICTO proposed the next step: to look at real crystals. Real crystals relax: their layers move apart or together a little, and their atoms shift. What matters is how close real stackings come to the straight line, and what the departures mean. We use only published numbers, and say for each how precise it is.

2. The straight line in the physicists' language

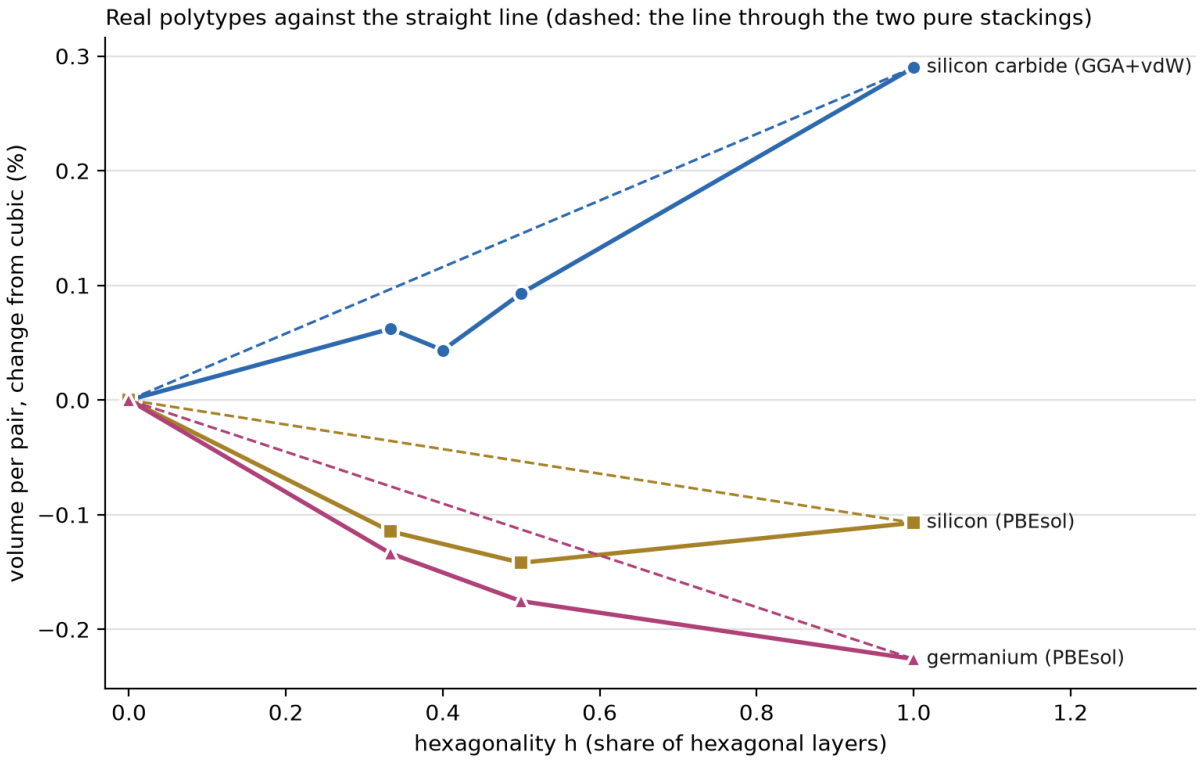
A stacking is a sequence of steps $\sigma_i = \pm 1$ from layer to layer (Hägg); a layer is cubic when the steps on either side agree and hexagonal when they differ, so the hexagonality is $h = (1 - \langle \sigma_i \sigma_{i+1} \rangle)/2$. In the axial next-nearest-neighbour Ising (ANNNI) model of Heine and coworkers a property X of a stacking, per atom pair and relative to the cubic stacking 3C, is written with couplings J_1, J_2, J_3 between layers one, two and three apart. With the convention of Keller et al. (2023), checked against their energies: $X(2H) = 2(J_1 + J_3)$, $X(4H) = J_1 + 2J_2 + J_3$, $X(6H) = (2/3)(J_1 + 2J_2 + 3J_3)$.

Proposition. (i) A property X of the polytypes 3C, 6H, 4H and 2H is a straight line in h, $X = s \cdot h$, exactly when $J_2 = J_3 = 0$ (and then $J_1 = s/2$). (ii) For such a property the stacking faults of a cubic crystal stand X(ISF) : X(ESF) : X(TSF) = 2 : 2 : 1, where, in the same model, $X(ISF) = 4(J_1 + J_2 + J_3)$, $X(ESF) = 4(J_1 + 2J_2 + 2J_3)$ and $X(TSF) = 2(J_1 + 2J_2 + 3J_3)$.

Proof. (i) The three equations have the matrix with rows (2, 0, 2), (1, 2, 1), (2/3, 4/3, 2), of determinant $16/3 \neq 0$, so the couplings are fixed by the three values. $X = s \cdot h$ gives $X(2H) = s$, $X(4H) = s/2$, $X(6H) = s/3$, which $J_1 = s/2$, $J_2 = J_3 = 0$ satisfies, and by uniqueness only it. Conversely $J_2 = J_3 = 0$ gives $2J_1$, J_1, $(2/3)J_1$, which is $s \cdot h$ with $s = 2J_1$. (ii) With $J_2 = J_3 = 0$ the three fault values are $4J_1$, $4J_1$ and $2J_1$. ∎

So the ideal pieces of #66, which obey the law, have only a nearest coupling, and #67's fault ratios (a twin turns one layer hexagonal, an intrinsic and an extrinsic fault two each) are this 2 : 2 : 1. In a real crystal, J_2 and J_3 measure how far a property bends away from the line.

3. Volumes



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Figure 8.1. Volume per atom pair of real polytypes, as the change from the cubic stacking, against the hexagonality; dashed, the straight line through the two pure stackings. Silicon carbide (density-functional calculation with van der Waals correction, Kawanishi and Mizoguchi): within the precision of the published numbers (about 0.05%). Silicon and germanium (PBEsol, Keller et al.): a clear bend of up to about 0.04%.

Silicon carbide (2H, 4H, 15R, 6H, 3C). Kawanishi and Mizoguchi (2015) list measured lattice constants from six sources and three calculated sets. The volume per Si–C pair is $V = (\sqrt{3}/2)a^2(c/p)$. In every calculated set it lies on a straight line in h to within the rounding of the published numbers (largest departure 0.008–0.011 Å³ against a rounding of about 0.011 Å³, on about 20.6 Å³). The measured values differ between laboratories by up to 0.13 Å³, more than the whole change from cubic to hexagonal (under 0.3%), so they neither confirm nor refute the line.

Silicon and germanium (2H, 4H, 6H, 3C), the diamond framework of #60's pyrochlore run and of ice. Keller et al. (2023) give volumes per pair to three decimals. They do not lie on a straight line: silicon departs by up to 0.019 Å³ (on 40.1 Å³), about forty times the rounding, with its smallest volume at 4H, as the authors note; germanium by up to 0.016 Å³. Solving for the volume couplings:

Data set	J_1 (Å³/pair)	J_2	J_3	J_2/J_1, volume	J_2/J_1, energy
Si, PBEsol	−0.0155	−0.0178	−0.0060	+1.14	−0.23
Si, their ref. [46]	−0.0297	−0.0132	+0.0012	+0.45	−0.25
Ge, PBEsol	−0.0457	−0.0143	−0.0058	+0.31	−0.06

The volume couplings beyond the nearest are of the size of the nearest one, relatively larger than the energy couplings and of the opposite sign; the two silicon sets disagree on their size, so the robust statement is their order of magnitude: real relaxation adds layer couplings of about 0.01–0.02 Å³ per pair, which the ideal geometry of the law does not have.

4. Stacking-fault energies

Keller et al. give the fault energies (mJ/m²) from their energy couplings. Against the ideal 2 : 2 : 1 of the Proposition:

Data set	intrinsic : extrinsic : twin
Ge, PBEsol (69.4, 63.3, 30.8)	2.25 : 2.06 : 1
Ge, their ref. [46]	2.32 : 2.07 : 1
Si, PBEsol (36.5, 21.2, 8.9)	4.10 : 2.38 : 1
Si, their ref. [46]	4.68 : 2.45 : 1

Germanium, whose energy coupling J_2 is small, is close to the ideal; silicon, with a larger J_2, is far from it. Measured fault energies of silicon (intrinsic about 55–69 mJ/m², extrinsic about 60, cited there) are higher than the calculated ones; the authors note that they are hard to measure.

5. Band gaps

The band gap of the silicon carbide polytypes rises with the hexagonality (Choyke, Hamilton and Patrick, 1964): about 2.36 eV (3C), 3.02 (6H), 3.23 (4H) and 3.33 (2H). The straight line through these misses them by up to 0.2 eV, and the gap couplings are $J_1 = 0.43$, $J_2 = 0.19$, $J_3 = 0.06$ eV ($J_2/J_1 = 0.45$): the classical "linear" law is a rough trend, as later work found (Matsushita and Oshiyama, 2014). The band gap is an electronic property, not a share of space, and we compare it here only for its form.

6. A metal: cerium

At room temperature cerium exists both as cubic γ-cerium (fcc, a = 5.1612 Å, h = 0) and as double-hexagonal β-cerium (dhcp, ABAC, a = 3.6811 Å, c = 11.857 Å, h = 1/2). Their volumes per atom are 34.37 and 34.79 Å³: the double-hexagonal form is 1.2% larger. With two stackings a line cannot be tested, but the size of the change from cubic to half hexagonal is thirty times that of silicon carbide or silicon.

7. Conclusion

In the language of stacking physics the Hexagonality Law is the statement that a property has only a nearest coupling between layers, and its fault ratios are then exactly 2 : 2 : 1. Real crystals follow the straight line to within their data in silicon carbide and to about 0.04% in silicon and germanium, whose small bend is a relaxation coupling of the size of the nearest one; germanium's fault energies are close to 2 : 2 : 1, silicon's are not; band gaps bend much more. The law's straight line is the ideal part of every stacking, and the departures of real crystals from it are what they add. In the terms of Chapter 15, with $J_3 = 0$ a property is the plane $X = (2J_1 + 4J_2)h - 4J_2p_{hh}$ in h and the share p_{hh} of neighbouring hexagonal pairs: J_2 is the Swap defect ($\Delta = -4J_2$), and J_3 needs a window of three layers.

8. Prior art and credit

The ANNNI description of polytypes is Heine's (Cheng, Needs and Heine, 1988); the fault-energy formulas and all numbers used are those of the sources below. That a property is linear in the hexagonality exactly when its couplings beyond the nearest vanish is a direct consequence of the model, which we state with its proof for completeness; we did not find it stated in this form, nor its link to the volumes of the weighted pieces of #66. The Hexagonality Law is #66; random stackings and faults are #67.

Chapter 9. The Hexagonality Law in Real Materials II

From paper #70, DICTO DOI 10.5281/zenodo.23723565

Entropy and pressure

Abstract. The Hexagonality Law (#66) says that in ideal stackings every kind of piece is a straight line in the share h of hexagonal layers; in the physicists' layer model this is a property with only a nearest term (#68). DICTO proposed to test the law's form on two quantities that decide which stacking a crystal takes: its entropy and its enthalpy under pressure. **Entropy:** Mau and Huse wrote the stacking entropy of hard spheres in the same layer model; at close packing their nearest term is 55 and the next 6 (in 10^{-5} k_B per sphere), so the entropy is about 90% of the law's form, and the farther terms grow toward melting (37, 18 and 9). **Pressure:** we prove that if a crystal's volume follows the law's form, pressure shifts only the balance between cubic and hexagonal layers and never reorders the longer stackings among themselves, and that it keeps the fault ratios 2 : 2 : 1. From published volumes and energies, at first order, pressure strengthens cubic 3C silicon carbide against 6H, 4H and 2H, as experiments find (6H turns into 3C above about 4.5 GPa); in silicon and germanium, whose volumes bend away from the straight line, pressure tilts them toward hexagonal-type stackings, mainly through the farther term.

1. Two quantities that choose the stacking

Which stacking a crystal takes is decided by its free energy: energy, entropy, and, under pressure, the term pressure × volume. Paper #68 compared real volumes with the law's straight line. DICTO proposed to go further and test the law's form on the two other quantities: the entropy of hard spheres, where energy plays no part at all, and the enthalpy under pressure, where the volume enters directly.

2. The layer model

As in #68: in the axial Ising (ANNNI) model a property X of a stacking, per atom pair and relative to 3C, is $X(2H) = 2(J_1 + J_3)$, $X(4H) = J_1 + 2J_2 + J_3$, $X(6H) = (2/3)(J_1 + 2J_2 + 3J_3)$ (convention of Keller et al., 2023), and X is a straight line in h exactly when $J_2 = J_3 = 0$ (#68, Proposition). Mau and Huse (1999) use the same idea with the layer types as spins: a field h for each layer, a coupling J between neighbouring layers, and h′, J′ for longer ranges. A straight line in the hexagonality is the case of the field alone.

3. Entropy of hard spheres

Hard spheres have no energy; their stackings differ only in entropy, by about 10^{-3} k_B per sphere, with cubic stacking the highest (fcc – hcp = 0.00115 k_B per sphere at close packing, Mau and Huse; 0.001164(8), Elser 2014). Mau and Huse measured the layer couplings of this entropy:

Density	field h	coupling J	h′	J/h
close packing	55	6	—	0.11
near melting (about 74% of close packing)	37	18	9	0.49

(in 10^{-5} k_B per sphere; all terms favour cubic stacking). So at close packing the stacking entropy is about 90% of the law's form, a straight line in the hexagonality, and the farther terms grow as the spheres rattle more freely toward melting. Elser points out why the entropy differs at all: if every sphere moved in a fixed cage of its neighbours, the difference between cubic and hexagonal stacking would be exactly zero; it comes from collective motion. The law describes the fixed geometry; the entropy departs from its form only as far as the spheres move together. Entropy is not volume, and we compare the form only, not the numbers.

4. Pressure

At pressure P a stacking's enthalpy per pair is E + PV, so in the layer model the volume couplings J_k^V add $P \cdot J_k^V$ to the energy couplings.

> **Proposition.** If a crystal's volume per pair follows the law's form (a straight line in the hexagonality, $J_2^V = J_3^V = 0$), then at first order pressure changes only J_1: it shifts the balance between cubic and hexagonal layers, adds the same amount per hexagonal layer to every stacking, keeps the fault ratios intrinsic : extrinsic : twin of its own contribution at 2 : 2 : 1, and leaves J_2 and J_3, which order the longer stackings among themselves, unchanged.

Proof. The volume term is P·V, and V has couplings (J_1^V, 0, 0) by assumption; so P·V adds $P \cdot J_1^V$ to J_1 and nothing to J_2, J_3. A property with nearest term only is s·h with $s = 2J_1$ (#68), the same amount per hexagonal layer, and its fault values are $4J_1$, $4J_1$, $2J_1$ (#68). ∎

Real crystals are not exactly of this form, so we computed, at first order (couplings and volumes at zero pressure), the margin by which cubic 3C beats 6H, 4H and 2H, from the same published data as #68 (silicon carbide with van der Waals correction: energies 15, 2.7 and 1.3 meV per pair above 3C for 2H, 4H and 6H, Kawanishi and Mizoguchi; silicon and germanium, Keller et al.).

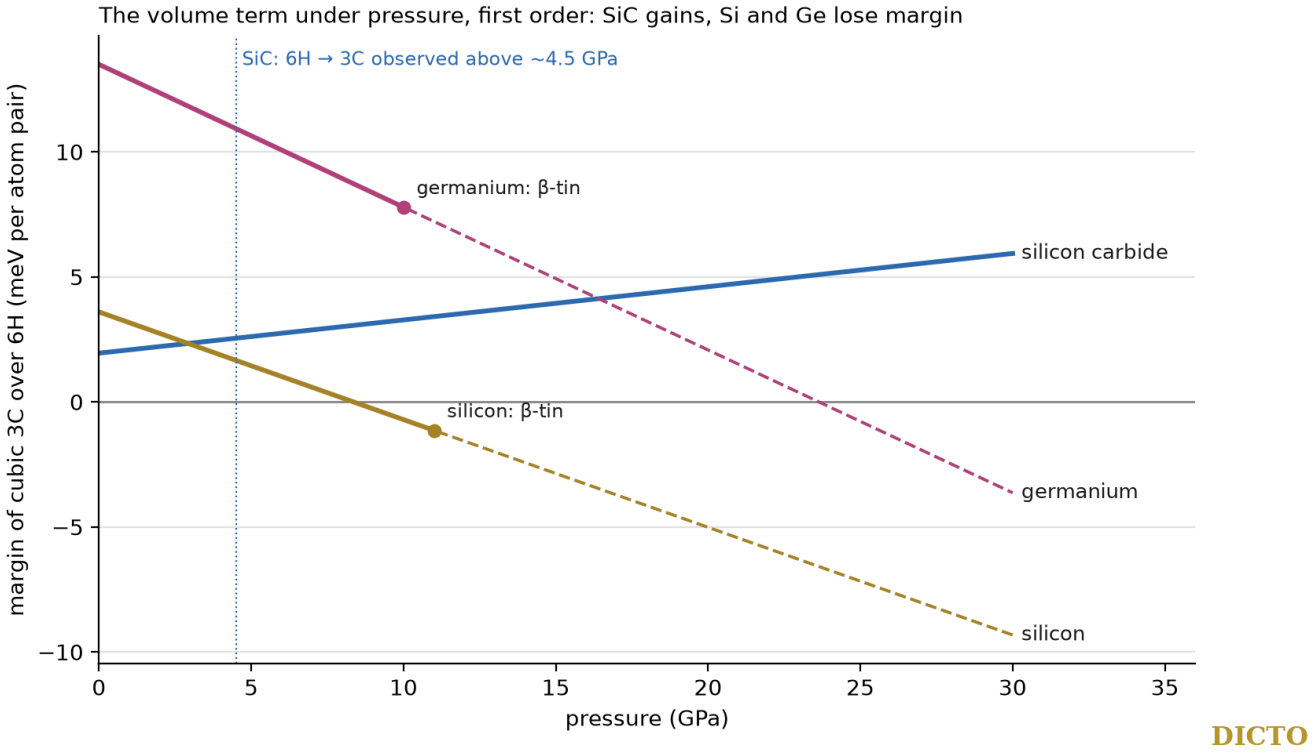


Figure 9.1. The margin of cubic 3C over 6H, energy plus pressure × volume per atom pair, against pressure, at first order. Silicon carbide gains margin; silicon and germanium lose it; their lines are dashed beyond their own transitions to β-tin (near 11 and 10 GPa). The dotted line marks the pressure above which 6H silicon carbide is observed to turn into 3C.

Material	nearest share of the volume couplings	3C over 6H	3C over 4H	3C over 2H
silicon carbide	0.78	+0.13	+0.11	+0.17
silicon	0.39	−0.43	−0.36	−0.13
germanium	0.70	−0.57	−0.50	−0.32

(change of the margin in meV per atom pair per GPa; nearest share = $|J_1^V| / (|J_1^V| + |J_2^V| + |J_3^V|)$.) Silicon carbide's volume is close to the law's form and rises slightly with the hexagonality, so pressure strengthens the

cubic stacking. That agrees with experiment: 3C becomes more stable with pressure, and 6H turns into 3C above about 4.5 GPa at high temperature (Sugiyama and Togaya, 2001; review, 2018). In silicon and germanium the volume falls with the hexagonality and bends away from the straight line, and pressure tilts them toward hexagonal-type stackings; at first order silicon's margin over 6H would close near 8 GPa, but silicon turns into β-tin near 11 GPa and the first-order estimate ignores how compression changes the couplings, so this is a tilt, not a predicted phase. For germanium the margins close only at 24–50 GPa, far beyond its own transition.

5. Conclusion

The law's straight line describes the ideal part of what chooses a stacking. Near close packing the entropy of hard spheres is about 90% of that form. Under pressure, a volume of that form could only tilt the balance between cubic and hexagonal; silicon carbide, close to it, is pushed toward cubic as experiments see; silicon and germanium, further from it, are tilted the other way, through the farther couplings their relaxation adds.

6. Prior art and credit

The layer (ANNNI) description of polytypes is Heine's; the entropy couplings of hard spheres are Mau and Huse's (1999), with Bolhuis, Frenkel and others before them; the net entropy difference and its phonon origin are Elser's (2014); the energies and volumes are those of Kawanishi and Mizoguchi (2015) and Keller et al. (2023); the high-pressure behaviour of silicon carbide is from Sugiyama and Togaya (2001) and a 2018 review. The statement that a volume of the law's form shifts only J_1 under pressure is an immediate consequence of the model, which we state with its proof. The Hexagonality Law is #66; real polytypes are #68.

Chapter 10. The Hexagonality Law in Real Materials III

From paper #71, DICTO DOI 10.5281/zenodo.23726064

Alloys and hydrogen

Abstract. Hydrogen is the enemy of titanium implants and of the zirconium tubes that hold nuclear fuel: it gathers in the metal and turns it brittle. We show that the places hydrogen can sit follow the Hexagonality Law (#66) exactly. In a close-packed metal of any stacking, per metal atom, the pairs of tetrahedral holes that share a face, and the pairs of octahedral holes that share a face, both number h, the share of hexagonal layers. Two hydrogen atoms cannot sit in one face-sharing tetrahedral pair, so at most 2 – h tetrahedral sites per atom can be filled, and a hydride MH_x with hydrogen in tetrahedral sites needs the metal locally to have $h \le 2 - x$: the brittle dihydrides need fully cubic stacking. DICTO proposed to test the law on alloys, starting from Ti-6Al-4V and Ti-Nb-Zr. Across six alloy families the law sorts them on one scale: exact for the hole and hydrogen counts; the composition of the magnesium LPSO alloys is a straight line in h in the authors' own model; the energies of cobalt and iron stackings come close (second-neighbour share 0.15–0.17); the faults of the high-entropy alloy CrCoNi depart clearly. In the standard fault model of these steels and alloys, Olson and Cohen's interface energy σ is exactly twice the second-neighbour layer coupling, so it measures the distance from the law.

1. Why alloys

Papers #67–#70 tested the Hexagonality Law on stackings of single elements and simple compounds. Engineering metals are alloys, and their most important stacking events are faults: the turning of a few layers from hexagonal to cubic or back. DICTO proposed to test the law there, starting from the titanium alloys Ti-6Al-4V (aircraft, implants, printed parts) and Ti-Nb-Zr (low-stiffness implants), and to include their spherical powders. We added four families in which cubic against hexagonal stacking is the main story: the long-period stacking (LPSO) magnesium alloys, Fe–Mn and high-entropy alloys, cobalt alloys, and the zirconium alloys of nuclear fuel cladding. A powder bed or a printed part is made of grains; inside each close-packed grain the results below apply unchanged.

2. Hydrogen sites

Take balls of diameter a in close-packed layers A, B, C, any stacking, and let h be the share of hexagonal layers (a layer is hexagonal when the layers above and below it sit in the same position). Every stacking has 2 tetrahedral and 1 octahedral hole per ball (#66).

> **Theorem.** Per ball, in any close-packed stacking: (i) the pairs of tetrahedral holes sharing a face number h; (ii) the pairs of octahedral holes sharing a face number h. Consequently (iii) if two hydrogen atoms cannot occupy one face-sharing tetrahedral pair, at most 2 – h tetrahedral sites per ball can be filled, and (iv) a hydride MH_x with its hydrogen in tetrahedral sites needs local hexagonality $h \le 2 - x$.

Proof. (i) A tetrahedral hole has its base on a triangle of one layer and its apex on a ball of the next. Two tetrahedra can share a face only across a layer i, on the same triangle, one with its apex in layer i – 1 and one in layer i + 1. For each ball of layer i there is exactly one triangle of i that lies under a ball of i + 1, and its partner below exists exactly when layer i – 1 sits where i + 1 sits, that is when layer i is hexagonal. So each hexagonal layer gives one pair per ball and each cubic layer none: h pairs per ball. (ii) The octahedral holes between two layers sit over the third position, the one neither layer uses. The holes between i – 1 and i and between i and i + 1 share a face of layer i exactly when they sit over the same position, that is when i – 1 and i + 1 coincide: again one pair per ball of each hexagonal layer. (iii) Each pair loses one of its two sites, and the pairs are disjoint (each tetrahedron has one base layer), so 2 – h remain. (iv) follows. ∎

We checked (i) and (ii) on explicit coordinates for six stackings (3C, 2H, 4H, 6H and the 10H and 14H of §3), periodic in all three directions; all counts are exact. The same run finds the tetrahedral pairs that share only an edge (centres 0.707a apart) at 6 − 3h per ball, also a straight line (computed, not proved here).

A face-sharing tetrahedral pair has its centres 0.408a apart: 1.20 Å in α-titanium (a = 2.95 Å), 1.32 Å in α-zirconium (a = 3.23 Å), far inside the 2.1 Å that hydrogen atoms in metals keep apart (Switendick's criterion), so (iii) applies to both. The edge-sharing pairs are 2.28 Å apart in zirconium (allowed) and 2.09 Å in titanium (at the limit; no claim made). Real titanium and zirconium are slightly flattened (c/a ≈ 1.59 against the ideal 1.633): the counts are exact, the distances shift by about 3%.

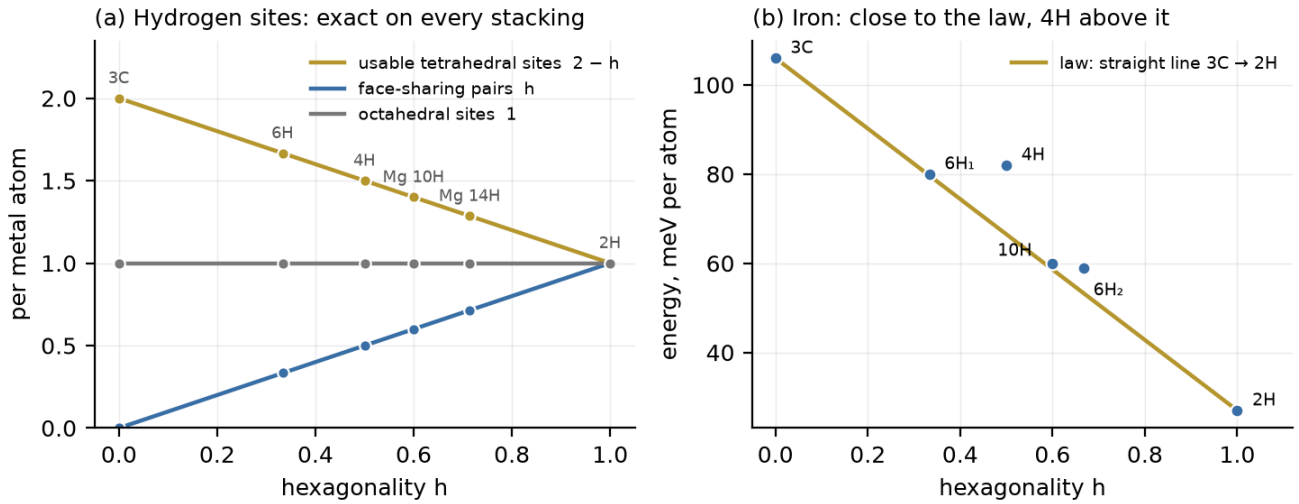


Figure 10.1. (a) Hydrogen sites per metal atom against the hexagonality, from explicit coordinates: usable tetrahedral sites 2 − h, face-sharing pairs h, octahedral sites 1, exact on every stacking tested. (b) Energies of six iron stackings (non-magnetic, Tsumuraya et al.) against the law's straight line through 3C and 2H: close, with 4H above the line.

3. Magnesium LPSO alloys

The Mg–Zn–Y alloys owe their strength to long-period stacking ordered phases: hexagonal magnesium with a fault every k layers, each fault making two cubic layers that collect the zinc and yttrium (Abe, Ono, Itoi, Yamasaki and Kawamura, from atom-resolved microscopy). Built from their description and checked with Jagodzinski's rule:

Phase	fault every	stacking (one period)	h
10H	5 layers	ABCBCBACAC	3/5
18R	6 layers	ABCBCBCABABABCACAC	2/3
14H	7 layers	ABCBCBCBACACAC	5/7
24R	8 layers	ABCBCBCBCABABABABCACACAC	3/4

Abe and co-workers estimate the compositions by giving each cubic layer the same composition $Mg_{85}Zn_5Y_{10}$ and each hexagonal layer pure magnesium, and their microscopy shows the same cubic-layer intensity in every phase. That model is the law's form: the zinc-plus-yttrium content is 15% × (1 − h), giving their 6.0, 5.0, 4.3 and 3.7%. By the Theorem, the hydrogen-site counts of these phases are straight lines in the same h.

4. Faults and the second neighbour

In the layer (ANNNI) model used in #68, a property relative to cubic stacking is X(2H) = $2(J_1 + J_3)$ per layer, and the fcc faults cost ISF = $4(J_1 + J_2 + J_3)$, ESF = $4(J_1 + 2J_2 + 2J_3)$, twin $2(J_1 + 2J_2 + 3J_3)$ (Denteneer and van Haeringen). The law holds exactly when $J_2 = J_3 = 0$ (#68). Steels and alloys that deform by turning cubic layers hexagonal (TRIP) or by twinning (TWIP) are described by Olson and Cohen's formula, ISF = 2ρΔG + 2σ, where ρΔG is the cost per area of one hexagonal layer and σ an interface energy.

> **Proposition.** In the layer model, $\rho\Delta G = 2(J_1 + J_3)$ and $2\sigma = 4J_2$. So $\sigma = 0$ exactly when the second-neighbour coupling vanishes, and σ measures how far a material's faults are from the law.

Proof. $\rho\Delta G$ is X(2H) per layer, $2(J_1 + J_3)$; subtract twice that from ISF $= 4(J_1 + J_2 + J_3)$. ∎ (An immediate consequence of the standard expressions; not claimed as new.)

Material (data)	what was tested	result	$\lvert J_2/J_1 \rvert$
hcp cobalt (Aas et al., four computed faults)	energy per cubic layer made: I1 16, T2 39, I2 48, E 62 meV for 1, 2, 2, 3 layers	16.0, 19.5, 24.0, 20.7 meV: within ±20% of one constant	0.15
iron, six stackings (Tsumuraya et al., non-magnetic)	energy and volume against h	volume linear within 0.1 Bohr³ of 69 (rounding); energy close, 4H 12 meV/atom above the line	0.17
CrCoNi high-entropy alloy (Ding et al.)	ISF against ESF (the law says equal)	random –42.9 / –27.8, ordered 30 / 66 mJ/m²; $2\sigma \approx 18$ mJ/m²	≈ J_1-sized

Cobalt and iron are close to the law: about four fifths of a fault's cost is the per-cubic-layer term. The 10H of iron in that set is the same stacking as the magnesium 10H of §3. In CrCoNi the local chemical order reaches across layers and the law's prediction ESF = ISF misses by 15 to 36 mJ/m².

5. Hydrides of zirconium and titanium

By the Theorem (iv):

Hydride	x	needs h ≤
γ-ZrH	1	1 (the count allows hexagonal)
δ-$ZrH_{1.5}$	1.5	0.50
δ-$ZrH_{1.66}$	1.66	0.34
ε-ZrH_2, δ-TiH_2	2	0 (fully cubic)

The hydrogen-rich hydrides that embrittle zirconium cladding and titanium therefore need the metal locally at least two-thirds cubic. This agrees with the accepted mechanism, in which the metal turns from hexagonal to cubic by partial dislocations gliding on alternate basal planes (Carpenter), each leaving a fault that makes cubic layers (#67). The bound is a necessary condition only: γ-ZrH is itself cubic-stacked though the count would allow hexagonal.

6. Where the law stops

Body-centred cubic metals are not close-packed stackings, and the law does not apply to them: the β phase of Ti-Nb-Zr, ferritic steels such as chromium–molybdenum steel, and the ω phase of titanium alloys. The α″ martensite of Ti-Nb-Zr is a distorted hexagonal stacking: the counts of §2 apply to it, the distances shift. The cobalt–chromium–molybdenum alloys of implants are close-packed and belong with cobalt in §4.

7. Conclusion

The law sorts alloys on one scale. Exact: the hole and hydrogen counts of every stacking, hence the bound $h \le 2 - x$ for hydrides. Exact in the authors' own model: the composition of the magnesium LPSO phases. Close: the stacking energies of cobalt and iron. Broken: the faults of the high-entropy alloy CrCoNi, where Olson and Cohen's σ, twice the second-neighbour coupling, is large. In the terms of Chapter 15, that second-neighbour coupling is a Swap defect: the part of a stacking energy that needs, besides h, the share p_{hh} of neighbouring hexagonal pairs.

8. Prior art and credit

The LPSO stackings and the composition model are Abe, Ono, Itoi, Yamasaki and Kawamura's; the iron polytype energies and volumes are Tsumuraya, Watanabe and Sawaguchi's; the cobalt fault energies are Aas, Szunyogh, Evans and Chantrell's; the CrCoNi fault energies are Ding, Yu, Asta and Ritchie's; the fault model is Olson and Cohen's; the layer-model fault expressions are Denteneer and van Haeringen's; the hydride mechanism is Carpenter's; the hydrogen distance criterion is Switendick's. That face-sharing tetrahedral pairs occur around hexagonal layers is standard crystallography; the count h per ball for any stacking, the resulting bound for hydrides, and the reading of σ as $2J_2$ are stated here with proofs, and the proofs are short. DICTO proposed the test on alloys and the titanium cases; the further families and the computations are joint work.

Chapter 11. The Hexagonality Law in Real Materials IV

From paper #72, DICTO DOI 10.5281/zenodo.23728287

Oxides: hexagonal perovskites and battery cathodes

Abstract. Chemists judge oxide structures with two rules of thumb: Pauling's third rule (octahedra that share faces are unstable) and Goldschmidt's tolerance factor (a large A cation pushes a perovskite towards hexagonal stacking). DICTO proposed that three-dimensional stacking rules can replace such rules of thumb. We show that both rules are measured by one number, the hexagonality h of the oxygen stacking (#66). In a hexagonal perovskite ABO_3 of any stacking, each B octahedron shares faces with 2h neighbours and corners with 6(1 – h). In a layered battery cathode $LiMO_2$, the lithium–metal octahedra that share a face number 2h per lithium site on average. Measured data follow the law: the volumes of four $BaRuO_3$ polytypes lie on one straight line in h (within 0.2% for the three precisely measured ones), and pressure walks the structure down h, 9R → 4H → 6H → 3C. In Li_xCoO_2 on charge, the H1-3 phase puts all its face-sharing into the empty gaps and keeps lithium only where it shares no face, giving the bound $x \le 1 - h$ for the stable phases; its layer spacing, 4.52 Å, lies on the law line between O3 and O1 (4.525 Å). The law has limits: in $BaMnO_3$ the volume follows it but the energy does not.

1. Why oxides

Papers #67–#71 tested the Hexagonality Law on elements, simple compounds and alloys: there the atoms themselves are the close-packed balls. In most oxides the oxygen ions are the close-packed layers and the metal ions sit in the holes between them, mostly in octahedra. Whether two neighbouring octahedra share a corner, an edge or a face then depends only on the stacking of the oxygen layers, and Pauling's third rule says the face-sharing ones cost energy, because their metal ions come close. DICTO proposed that three-dimensional stacking rules (the Hexagonality Law, the Rule of Halves, the Swap Rule) can stand in for rules of thumb used in chemical modelling. Here we test the first of them on two families where the stacking is the main story: the hexagonal perovskites, and the layered oxides that hold the lithium in rechargeable batteries.

2. The count

Let h be the share of hexagonal layers in the stacking of the close-packed oxygen (or AO_3) layers: a layer is hexagonal when the layers above and below it sit in the same position, A, B or C. An octahedral hole between two layers sits on the third position, the one neither layer uses.

Theorem. (i) In a perovskite ABO_3 built of close-packed AO_3 layers with B in the octahedra formed by oxygen alone, each B octahedron shares a face with 2h neighbours and a corner with 6(1 – h) neighbours, for any stacking. (ii) In a layered oxide $LiMO_2$ (oxygen layers in pairs around the M layers, lithium in the gaps), the face-sharing lithium–metal pairs number 2h per lithium site; a gap bounded by two cubic layers (O3 type) has none, a gap bounded by two hexagonal layers (O1 type) has two, a mixed gap (O2 type) has one.

Proof. Take one oxygen layer and the octahedra just above and just below it. Each sits on the position missing from its own two layers. If the layer is hexagonal, the layers above and below it are on the same position, so the two octahedra are on the same position too and share the three oxygen atoms of the layer between them: a face. If the layer is cubic, the two octahedra are on different positions and meet in an edge (layered oxides) or, in the perovskite, where only one hole in four is filled, in single corners with three neighbours. Each B octahedron of a perovskite has one layer above and one below; each lithium gap has one bounding layer on each side. Averaging over the stacking gives 2h and 6(1 – h). ∎ The counts were checked from explicit coordinates for 3C, 2H, 4H, 6H and 9R perovskites and for the O3, O1, O2 and H1-3 layered stackings (study files perovskite_counts.py, cathodes.py).

3. Hexagonal perovskites

$BaRuO_3$ is made in four polytypes; the stacking can be read from the interfaces between neighbouring octahedra (C corner, F face) listed by Jin et al.: 9R (CFF repeated, h = 2/3), 4H (CF, h = 1/2), 6H (FCC, h = 1/3) and 3C (all C, h = 0). The face-sharing share equals h in every case, as Theorem (i) requires. Their volumes per formula unit lie on one straight line:

polytype	h	volume measured (Å^3)	law line 64.25 + 6.82h	off
3C	0	64.28	64.25	+0.06%
6H	1/3	65.88 (c known to two figures)	66.52	−0.97%
4H	1/2	67.51	67.66	−0.22%
9R	2/3	68.91	68.79	+0.16%

Since the volume rises with h, pressure favours lower h at first order (#70, Proposition). Jin et al. made the polytypes at increasing pressure: 9R at ambient, 4H at 3 GPa, 6H at 5 GPa, 3C at 18 GPa. The sequence walks down h step by step: 2/3, 1/2, 1/3, 0. The same direction is seen with chemistry: in $Ba_{1-x}Sr_xRuO_3$, the smaller strontium takes the structure 9R → 4H → 3C. The tolerance factor says which way the structure leans; the law says by how much the volume and the face-sharing change at each step.

$BaMnO_3$, computed by Chen and Millis (DFT, antiferromagnetic, per Mn), gives a second case. The volumes of 3C (61.9), 4H (66.1) and 2H (71.0 Å^3) are close to the law: 4H lies 0.5% below the straight line through 3C and 2H. The energies are not: 2H is lowest, 4H is 109 meV higher and 3C 775 meV higher, so 4H lies 278 meV below the line. The barium ion is too large for cubic $BaMnO_3$ (tolerance factor above 1), which makes 3C unusually costly; in the language of #68, the second-neighbour coupling is large here.

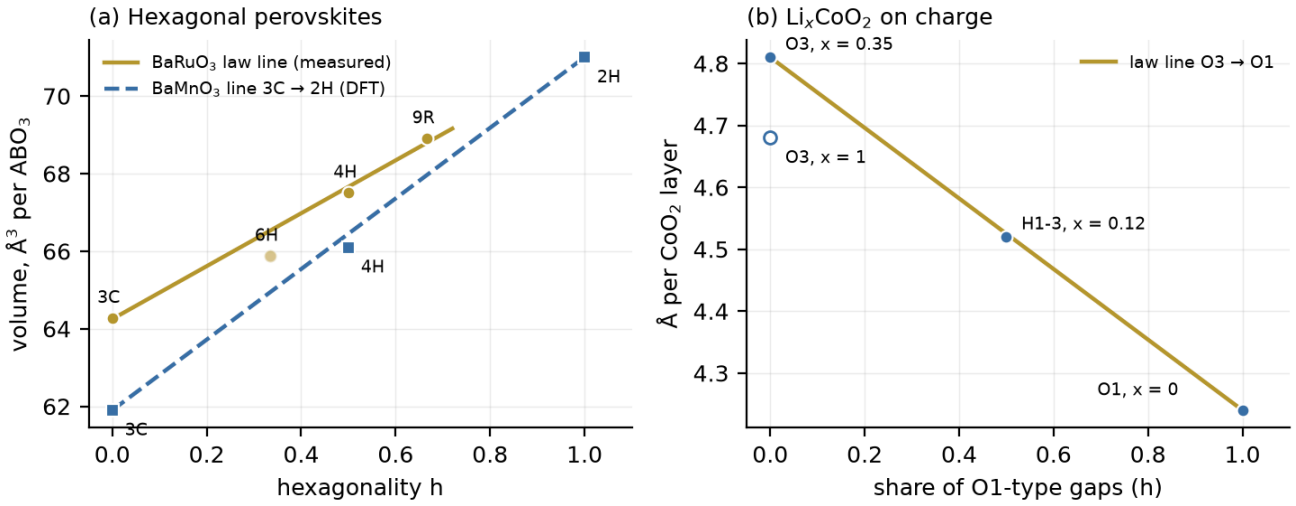


Figure 11.1. (a) Volume per ABO_3 against the hexagonality h: $BaRuO_3$ measured (Jin et al.; 6H pale, its c known to two figures) with the law line, and $BaMnO_3$ computed (Chen and Millis) with the straight line from 3C to 2H. (b) Layer spacing of Li_xCoO_2 (measured, Motohashi et al.) against the share of O1-type gaps: the H1-3 phase sits on the line from O3 to O1; open circle: O3 with full lithium.

4. Lithium cobalt oxide on charge

$LiCoO_2$, the cathode of most phone batteries, has cubic O3 stacking (h = 0): every lithium octahedron shares only edges with the cobalt octahedra. On charge the lithium leaves. Motohashi et al. measured the full series: O3 down to x = 0.35, then the H1-3 phase at x = 0.12, then CoO_2 with O1 stacking (h = 1) at x = 0. H1-3 has alternating O3-type and O1-type gaps (h = 1/2), and the lithium sits only in the O3-type gaps (Van der Ven et al. predicted this staging; Motohashi et al. confirm the structure). By Theorem (ii) the face-sharing count per gap runs 0, 2, 0, 2, …: H1-3 puts all its face-sharing into the gaps it leaves empty.

> **Proposition (lithium bound).** If lithium occupies only gaps in which it shares no face with a metal octahedron, then $x \le 1 - h_{gap}$, where h_{gap} is the share of O1-type gaps. All stable phases in Motohashi et al.'s series obey it: O3 ($h_{gap} = 0$, $x \le 1$), H1-3 (1/2, $x = 0.12$), O1 (1, $x = 0$).

This is the oxide counterpart of the hydride bound of #71 ($h \le 2 - x$): in both, the number of guest atoms is limited by the faces the stacking creates. One known structure breaks it: O2-$LiCoO_2$, in which every gap is mixed and every lithium shares one face with cobalt. It is made only by ion exchange from a sodium cobalt oxide (Delmas, Braconnier and Hagenmuller) and is metastable (Carlier et al.). A face-sharing pair therefore costs energy rather than being forbidden, as Pauling's rule says; the law counts the pairs exactly.

The layer spacing tests the law as a number. Motohashi et al. give 4.81 Å per CoO_2 layer for O3 at $x = 0.35$–0.50 and 4.24 Å for O1. The law line at $h = 1/2$ gives 4.525 Å; H1-3 measures 4.52 Å ($c/6 = 4.522$ Å), within 0.11%. Spacings in staged intercalation compounds are known to add layer by layer (as in graphite intercalation), so this agreement is expected once H1-3 is read as half O3 and half O1; what the law adds is the face count that tells which half holds the lithium.

5. Where the law stops

(1) The $BaMnO_3$ energies are far from the straight line (Section 3), although the volumes are close. (2) The sizes of face-sharing clusters (dimers in 4H, trimers in 9R, endless chains in 2H) are not linear in h; they depend on runs of hexagonal layers and drive the magnetism Jin et al. report. (3) The layer spacing of Li_xCoO_2 also depends on x at fixed stacking (4.68 Å at $x = 1$ against 4.81 Å at $x = 0.35$), so the law describes the stacking part only. (4) In nickel-rich cathodes the H2 → H3 change keeps O3 stacking ($h = 0$ throughout); its large shrinkage of the layer spacing is outside the law.

6. Conclusion

Pauling's third rule and the tolerance factor become one count: the faces shared by oxygen octahedra are 2h per octahedron in perovskites and 2h per lithium site in layered cathodes, for any stacking. Measured volumes of $BaRuO_3$ lie on the law line, pressure walks them down h, and charged $LiCoO_2$ keeps its lithium where the count gives zero, $x \le 1 - h$. Chapter 15 extends the count: per octahedron, face-sharing clusters number $1 - h$ and A neighbours $8 - 2h$ in every stacking, while the octahedra with exactly 0, 1 or 2 shared faces break the Swap Rule and are planes in h and p_{hh}.

7. Prior art and credit

The stacking notation for hexagonal perovskites (h and c layers, nH and nR) goes back to Katz, Ward and co-workers (Donohue, Katz and Ward); the O/P notation for layered oxides is Delmas, Fouassier and Hagenmuller's; H1-3 as an O1/O3 hybrid with lithium in every other gap is Van der Ven et al.'s prediction and Motohashi et al.'s measurement; that face-sharing costs energy is Pauling's third rule. The counts 2h and $6(1 - h)$ for any stacking, the split of the face count between gaps, the lithium bound and the comparison with the law line are stated here with proofs; the proofs are short. DICTO proposed replacing chemical rules of thumb with three-dimensional stacking rules; the choice of perovskites and cathodes and the computations are joint work.

Chapter 12. The Hexagonality Law in Real Materials V

From paper #73, DICTO DOI 10.5281/zenodo.23728921

Ice, diamond and nanowires

Abstract. Ice, diamond and the semiconductor nanowires of new lasers and solar cells are all built from the same piece: a tetrahedron of bonds, in close-packed layers that stack cubic or hexagonal. We test the Hexagonality Law (#66) on them. The geometry is exact: in any stacking, the eclipsed bonds (whose neighbours face each other) number h/2 per atom, where h is the share of hexagonal layers. For six III-V semiconductors in four stackings each, computed by Belabbes et al., the conduction-band offset lies on the law's straight line within 4% of its full change and the crystal-field splitting within 7%. The polytype energies of the tetrahedral elements rank them by their distance from the law: tin 0.02, germanium 0.08, diamond 0.15, silicon 0.32, and silicon carbide 1.03; silicon carbide, the best-known material with many stacking forms, is the one furthest from the law. For ice, Tonauer et al. measured –37.7 J/mol for pure cubic ice turning hexagonal; the law predicts that the stacking-disordered ice of clouds releases 37.7 J/mol times its cubicity, and that its excess vapour pressure scales the same way. This closes the series on real materials.

1. One piece, three materials

In diamond each carbon atom bonds to four others at the corners of a tetrahedron. Silicon, germanium and grey tin do the same; in gallium arsenide, indium phosphide and their relatives the two kinds of atom alternate; in ice the oxygen atoms of water hold the same network through hydrogen bonds. In all of them the atoms form close-packed double layers, and the double layers stack cubic (zinc blende, diamond, ice Ic) or hexagonal (wurtzite, lonsdaleite, ice Ih), or any mixture. Grown as thin wires, GaAs and InP switch between the two, and the stacking can be set layer by layer during growth (Algra et al.; Caroff et al.); a few cubic layers in a hexagonal wire act as a quantum well (Spirkoska et al.). DICTO chose these three families as the last test of the law in real materials.

2. The count

Each double layer has its lower atoms on one of the positions A, B, C and its upper atoms straight above them, joined by a bond along the stacking axis. The upper atom bonds to three atoms of the next double layer, and the lower atom to three of the previous one. A layer is hexagonal when the layers on either side of it sit on the same position.

> **Theorem.** In a tetrahedral network on any close-packed stacking, per atom: (i) the eclipsed bonds number h/2 (all other bonds are staggered); (ii) the eclipsed third-neighbour pairs number 3h/2, at 5/3 of the bond length (staggered third neighbours lie at 1.915 and 2.517 bond lengths).

Proof. Only the bond along the axis can be eclipsed; the three slanted bonds of every atom are staggered in all stackings. The axial bond of double layer k joins a lower atom whose other bonds point down to position L_{k-1} and an upper atom whose other bonds point up to position L_{k+1}. The two sets of three bonds face each other (eclipsed) exactly when $L_{k-1} = L_{k+1}$, that is when layer k is hexagonal, and then each of the three pairs of end atoms is a short third-neighbour pair. There is one axial bond per two atoms, so the eclipsed bonds are h/2 per atom and the short pairs 3h/2. ∎ Both counts were checked from explicit coordinates on 3C, 2H, 4H, 6H, 9R and 15R (tetra_counts.py).

The eclipsed bond is the only geometric difference between cubic and hexagonal stacking up to third neighbours, so any property that is a sum of local bond terms is a straight line in h. That is the Hexagonality Law for these materials. How far real properties stay on it is a question of data.

3. Nanowire semiconductors

Belabbes, Panse, Furthmüller and Bechstedt computed the band structures of GaP, GaAs, GaSb, InP, InAs and InSb in the stackings 3C, 6H, 4H and 2H (h = 0, 1/3, 1/2, 1) with one method throughout. We compare each quantity at 6H and 4H with the straight line from 3C to 2H, and give the distance as a share of the full change from 3C to 2H:

quantity	mean distance from the law line	worst case
conduction-band offset ΔE_c	4%	InP 4H, –8%
crystal-field splitting Δ_{cf}	7%	InP and GaP, +13%
valence-band offset ΔE_v	10%	InP 4H, –27%; GaSb 6H, +27%
band gap E_g (GaP left out: indirect in 3C)	11%	GaAs 6H, +23%

The conduction-band offset decides how electrons collect in a cubic segment of a hexagonal wire. On the law line, a wire with hexagonality h has its conduction band raised by h times the full wurtzite–zinc-blende offset (for GaAs 149 meV, for InP 183 meV), whatever the order of the layers, to within a few per cent. The crystal-field splitting, which comes from the hexagonal symmetry itself, follows the line almost as well.

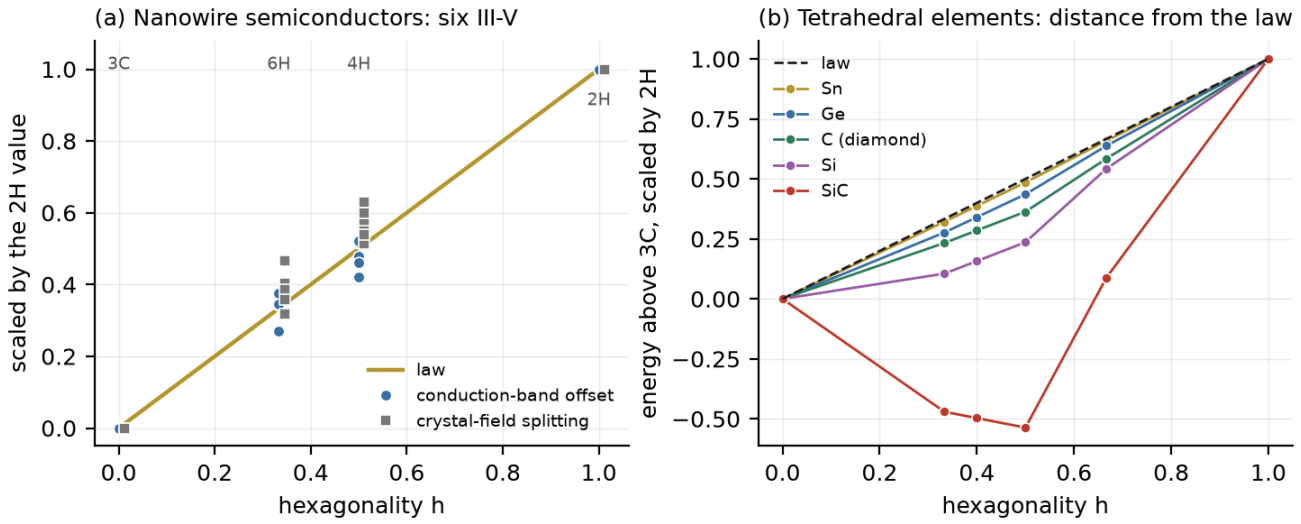


Figure 12.1. (a) Conduction-band offsets (circles) and crystal-field splittings (squares) of six III-V semiconductors in four stackings (Belabbes et al.), each scaled by its 2H value, against the hexagonality h: the law is the diagonal. (b) Energies above 3C, scaled by the 2H value, for six stackings of tin, germanium, diamond, silicon and silicon carbide, from the ANNNI parameters of Ziembicki et al.: tin and germanium lie on the law, silicon carbide far from it.

4. Diamond and the tetrahedral elements

In the layer model used in #68, the energy of a stacking depends on couplings J_1, J_2, J_3 between double layers one, two and three apart; the law holds exactly when $J_2 = J_3 = 0$. Ziembicki et al. fitted the couplings for the group-IV elements and their compounds with one method. The ratio $(|J_2| + |J_3|)/J_1$ measures the distance from the law:

material	J_1, J_2, J_3 (meV per double-layer pair)	distance from the law
tin (grey)	15.18, –0.22, –0.05	0.02
germanium	16.82, –1.05, –0.34	0.08
diamond	25.45, –3.44, –0.34	0.15
silicon	10.59, –2.59, –0.77	0.32
silicon carbide	2.74, –2.42, –0.40	1.03

Keller et al. computed the energies directly for silicon and germanium. Germanium's 6H and 4H lie within 6% of the law line (4.5 and 7.1 meV per atom against 5.3 and 8.0); silicon's lie 21–24% below it (1.2 and 2.5 against 3.2 and 4.9). Diamond sits at the same distance as cobalt and iron in #71. Silicon carbide, the best-known polytypic material, is the one furthest from the law, where J_2 nearly cancels J_1 and 4H and 6H fall below 3C. Near the law one stacking wins clearly; far from it, many compete.

5. Ice

Ice formed in clouds and from deeply supercooled water (Murray, Knopf and Bertram) is neither cubic nor hexagonal but stacking-disordered (ice I_{sd}, Malkin et al.; Hansen, Koza and Kuhs), and stacking disorder takes part in how ice nucleates (Lupi et al.); its make-up is given by the cubicity, the share of cubic layers, which is 1 − h. Tonauer, Yamashita, del Rosso, Celli and Loerting measured the enthalpy of pure cubic ice turning hexagonal, using the fault-free cubic ice made only recently (del Rosso et al.; Komatsu et al.): –37.7 ± 2.3 J/mol, at 226 K. Earlier values for stacking-disordered samples scatter from less than –10 to about –160 J/mol, partly because the samples differed in cubicity. Tonauer et al. note that ice with many hexagonal layers must release less heat and have a smaller excess vapour pressure.

> **Prediction.** To the extent that the law holds for ice, stacking-disordered ice of cubicity Φ releases 37.7 Φ J/mol on turning hexagonal, and its excess vapour pressure over hexagonal ice is Φ times that of cubic ice. The eclipsed hydrogen bonds number (1 − Φ)/2 per molecule.

The prediction is testable with samples whose cubicity is measured by diffraction; we found no published set that pairs cubicity with enthalpy. Its weight is in the clouds: the vapour pressure decides whether cubic-rich ice crystals grow at the expense of hexagonal ones, and measured vapour pressures of cubic-rich ice (Murray and Bertram) differ more than the calorimetry allows, as Tonauer et al. point out.

6. Where the law stops

(1) Silicon carbide, and to a lesser degree silicon, are far from the law in energy (Section 4). (2) The volumes of silicon and germanium polytypes change by only 0.1–0.2% across the stackings and do not follow the line. (3) Valence-band offsets of InP and GaSb depart by up to 27% of the full change. (4) The ice prediction is untested. (5) Nanowire surfaces, strain and the electric fields of hexagonal segments are outside the bulk numbers used here.

7. Conclusion

The tetrahedral networks carry the exact count of the law: eclipsed bonds h/2 per atom in every stacking. Their electronic properties come close to the line, the conduction-band offset of nanowire semiconductors closest. Their energies sort the elements from tin, on the law, to silicon carbide, far from it, and the distance tells how many stackings compete. For ice the law turns a qualitative remark into a number to measure. In the terms of Chapter 15, the J_2 part of that distance is a plane in h and the share p_{hh} of neighbouring hexagonal pairs, and only J_3 needs a window of three layers. The series on real materials (Chapters 8 to 12) was followed by a companion on magnets, Chapter 13.

8. Prior art and credit

The hexagonality h as a stacking measure goes back to Jagodzinski; the layer model of polytype energies (ANNNI) to Cheng, Needs and Heine; the band data are Belabbes et al.'s, who also plot their structural results against the hexagonality; the polytype energies are Keller et al.'s and Ziembicki et al.'s; the ice measurements are Tonauer et al.'s and the description of stacking-disordered ice Malkin et al.'s. The eclipsed-bond count for any stacking with its proof, the straight-line tests with numbers, the ranking by distance from the law, and the ice prediction are stated here. DICTO chose the three families and set the end of the series; the computations are joint work.

Chapter 13. Spin Ice and Kagome Magnets on Every Stacking

From paper #74, DICTO DOI 10.5281/zenodo.23730421

Loops, kagome planes and the Hexagonality Law

Abstract. Two families of magnets are built from corner-sharing triangles and tetrahedra: spin ice, whose magnetic atoms sit on the pyrochlore network of DICTO Janus c/h Pyrochlore (#60), and the kagome antiferromagnets Mn_3X, whose manganese atoms form kagome nets. Both are close-packed layers, so both can stack cubic, hexagonal or any mixture. We count exactly what changes with the share h of hexagonal layers. On the pyrochlore network of any stacking, three quarters of the sites lie in kagome layers and one quarter in triangular layers; per tetrahedron the boat-shaped six-loops number 3h/2 and the chair-shaped 2 – 3h/2. Yet the closed loops of the network do not see the stacking at all: on five stackings the numbers of closed non-backtracking loops of every length from 6 to 34 bonds are identical, while the shells of neighbours differ by up to 5%. We call this the Loop Equality (computed, conjecture for all lengths). It gives a counting reason why the frozen-in entropy of spin ice and of water ice is the same for cubic and hexagonal stacking, as recent tensor-network work finds, and predicts the same value for every stacking, the disordered ice of clouds included. In the kagome magnets each manganese atom lies on exactly 1 + 3(1 – h) kagome planes: four in cubic Mn_3Ir, kagome in all directions, one in hexagonal Mn_3Sn, kagome in layers only.

1. Triangles, tetrahedra and stacking

In spin ice ($Dy_2Ti_2O_7$, $Ho_2Ti_2O_7$) the magnetic atoms sit at the corners of tetrahedra that touch only at corners: the pyrochlore network. Each tetrahedron must have two spins pointing in and two out, the same rule as the hydrogen atoms of water ice, and like ice the material keeps a frozen-in entropy at the lowest temperatures (Bramwell and Gingras); its defects behave as magnetic monopoles (Castelnovo, Moessner and Sondhi). In a field along the stacking axis the spins of the triangular layers are held and the kagome layers carry the disorder (Isakov et al.); ice on a single kagome layer has its own exact entropy (Lin and Tang). The pyrochlore network is the network of bond midpoints of a diamond-like crystal (#60), so it follows the stacking of the diamond layers: cubic, hexagonal or any mixture, as in #73. In the kagome antiferromagnets Mn_3X the manganese atoms make kagome nets of corner-sharing triangles inside close-packed layers, the same AX_3 layers as the hexagonal perovskites of #72; Mn_3Ir and Mn_3Pt stack them cubic, Mn_3Sn and Mn_3Ge hexagonal. Their spins on the kagome triangles give a large anomalous Hall effect without net magnetisation, found at room temperature in Mn_3Sn (Nakatsuji, Kiyohara and Higo; reviewed by Šmejkal et al.). DICTO chose these two families as the last materials study.

2. The pyrochlore network on any stacking

Theorem 1. On the pyrochlore network of any close-packed stacking with hexagonality h: (i) there are two sites per atom of the parent network, one quarter in triangular layers and three quarters in kagome layers; (ii) per tetrahedron, h/2 of the corner-sharing pairs are eclipsed (mirror images across the shared corner) and the rest staggered (inversion images).

Proof. Each site is the midpoint of one bond; there are two bonds per atom. The bond along the stacking axis, one per two atoms, has its midpoint in a triangular layer; the three slanted bonds have theirs in kagome layers. Two tetrahedra sharing a corner are the tetrahedra of the two atoms of one bond; they are eclipsed exactly when the bond is eclipsed, which by #73 happens for h/2 bonds per atom. ∎

The loops of the network change shape with the stacking. Counted from explicit coordinates on 3C, 2H, 4H, 6H and 9R, each tetrahedron has 3h/2 boat-shaped six-loops and 2 – 3h/2 chair-shaped ones, two in all, and three eight-loops, the same in every stacking.

3. The Loop Equality

A closed non-backtracking loop is a walk along the network that returns to its start and never turns straight back. These loops carry the corrections to Pauling's estimate of the ice-rule entropy. We counted them on 3C, 2H, 4H, 6H and 9R, from one column of sites in a cell 36 by 36 sites wide and at least 36 layers high, so that no loop up to length 34 can wrap round the cell, with every bond checked against its length.

> **Loop Equality (computed; conjecture for all lengths).** On every close-packed stacking, the number of closed non-backtracking loops of each length per site is the same. Computed for all even lengths from 6 to 34 on the five stackings: 24, 48, 480, 3432, 25704, 194256, 1509072, 11820000, …, 27416593190640, identical to the last digit.

The networks themselves differ. The numbers of atoms 3, 4, 5 and 6 bonds away are 24, 42, 64, 92 for the cubic stacking and 25, 44, 67, 96 for the hexagonal, the known values for diamond and lonsdaleite, with 4H in between (Figure 13.1a). The shells see the stacking from three bonds on; the loops do not see it up to 34.

Entropy. Pauling's estimate gives the same residual entropy, (1/2) ln(3/2) per spin in spin ice, for every stacking. Onsager proved that the hexagonal value is at least the cubic one; Nagle's series found them equal within its precision, and Xu, Lin and Zhang's tensor-network computation finds w = 1.5074584 for both, as numerical evidence that they are equal. The Loop Equality gives a counting reason: the loops that correct Pauling's estimate are the same on both, through loops of 34 bonds. It also extends the statement from the two pure stackings to all of them: to the extent that the entropy is set by these loops, spin ice and water ice have the same frozen-in entropy in every stacking, stacking-disordered ice included. In the language of the Hexagonality Law, this is the law with zero slope.

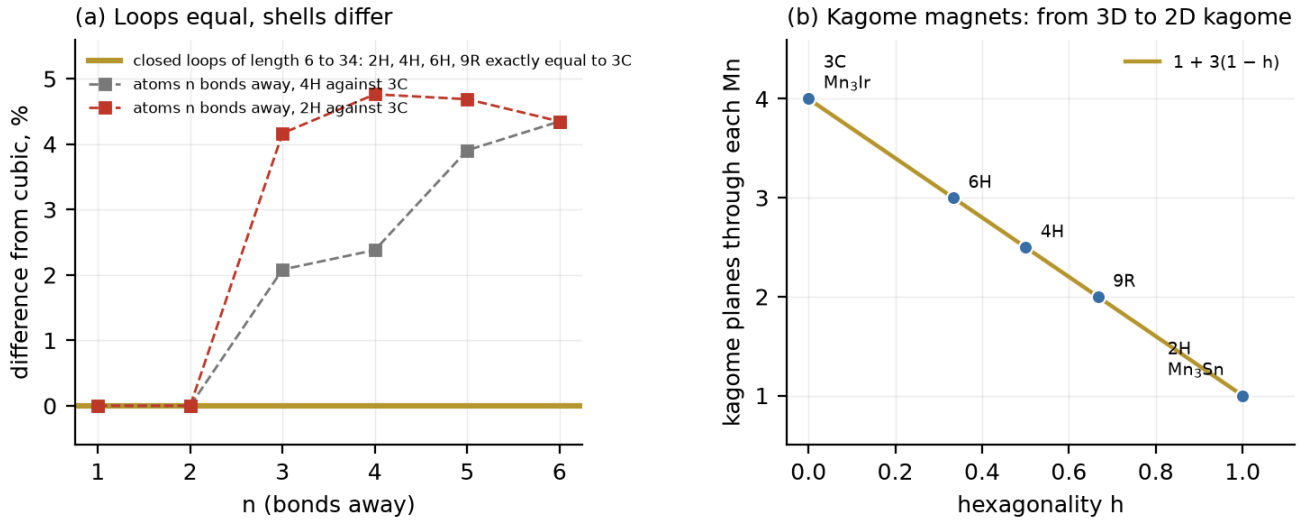


Figure 13.1. (a) The closed non-backtracking loops of lengths 6 to 34 are exactly equal on 3C, 2H, 4H, 6H and 9R (gold line at zero difference), while the numbers of atoms n bonds away differ by up to 5% from the cubic values for 4H and 2H. (b) Kagome planes through each manganese atom of the A_3B magnets against the hexagonality: 1 + 3(1 – h), from four in cubic Mn_3Ir to one in hexagonal Mn_3Sn, exact on all five stackings.

4. Kagome magnets

In Mn_3X each close-packed layer holds manganese on a kagome net and X on the remaining quarter of the sites, and the layers stack so that no two X atoms touch. We counted from explicit coordinates on 3C, 2H, 4H, 6H and 9R. Many things do not change: each manganese has 8 manganese and 4 X neighbours, 2 triangles of manganese in its layer and 6 tilted ones, and one third of an octahedral hole surrounded by manganese only. What changes is how the tilted triangles join.

> **Theorem 2.** In an A_3B close-packed stacking with hexagonality h, each A atom lies on 1 + 3(1 − h) kagome planes: always its own layer, and three tilted planes if its layer is cubic, none if it is hexagonal.

Proof. A kagome plane through the atom needs two triangles of the plane meeting at the atom, one on each side. For a tilted plane, one triangle reaches into the layer below and the other into the layer above, and the two lie in one plane exactly when the atoms above are the images of the atoms below under inversion through the atom. That is so when the layers below and above sit on different positions, a cubic layer, for each of the three tilted directions; when they sit on the same position, a hexagonal layer, the images are mirror images and no tilted pair is coplanar. Averaging over the layers gives 1 + 3(1 − h). ∎ The count was checked on all five stackings (1 + 3(1 − h) = 4, 3, 2.5, 2, 1 for 3C, 6H, 4H, 9R, 2H).

So the law counts the dimension of the kagome network. In cubic Mn_3Ir every manganese lies on four kagome planes, one in each of the four (111) directions, as Chen, Niu and MacDonald describe; in hexagonal Mn_3Sn on one, the basal plane. The intermediate stackings, such as the 4H ($D0_{24}$) type, would have kagome planes in between, 2.5 per atom on average: half the atoms with four, half with one. The Swap Rule of #61 and #64 applies to these layers unchanged, since they are the close-packed AX_3 layers treated there and in #72; we do not test it again.

5. Where the results stop

(1) The Loop Equality is computed through length 34 on five stackings; we have no proof for all lengths and all stackings. (2) The step from equal loops to equal entropy is not exact: the corrections to Pauling's estimate come from clusters of loops, not only from loop counts; the agreement with the tensor-network values supports it but does not prove it. (3) Theorem 2 counts geometry only; which kagome planes carry the spin order and the Hall effect depends on the magnetic structure, and for Mn_3Sn the basal-plane Hall response is forbidden by symmetry (Liu et al.).

6. Conclusion

The Hexagonality Law reaches magnets in two ways. The kagome planes through each manganese atom of the Mn_3X magnets fall on its straight line, 1 + 3(1 − h), from three-dimensional kagome in Mn_3Ir to two-dimensional in Mn_3Sn. The loops of the pyrochlore network do not depend on the stacking at all through length 34, so the frozen-in entropy of spin ice and ice is the same for every stacking: the law with zero slope. This chapter closes the series on single materials; Part IV returns to the perovskites and adds the Frank–Kasper phases and the approximants of quasicrystals, under one theorem (Chapter 15).

7. Prior art and credit

The ice rule and its entropy estimate are Pauling's; the inequality between the hexagonal and cubic entropies Onsager's; the series Nagle's; the numerical values Herrero and Ramírez's and Xu, Lin and Zhang's, who give evidence that the two are equal; the kagome description of Mn_3Ir is Chen, Niu and MacDonald's. The pyrochlore network as bond midpoints of a stacked diamond network is DICTO Janus c/h Pyrochlore (#60). Stated here: the counts of Theorems 1 and 2 with proofs, the loop shapes, and the Loop Equality as computed. We found no earlier statement that diamond-like networks of different stacking have equal loop counts. DICTO chose the two families and set the end of the materials line; the computations are joint work.

Part III. HexOct



Chapter 14. HexOct: Two Tiles for Every Stacking

From paper #75, DICTO DOI 10.5281/zenodo.23737639

Every close-packed stacking is a string of two layer tiles, the c-tile and the h-tile

The Law gave the pieces,
The pieces prove the law.
DICTO

Abstract. In the Rule of Halves (#57) every hole of every close-packed stacking gets a piece, and each piece is made from the pieces of the two pure stackings, cubic and hexagonal. Here we cut space into layer tiles: the tile of a layer is the share of space of that layer, made of four pieces, the top half of the octahedral piece below it, the two tetrahedral pieces next to it and the bottom half of the octahedral piece above it. We prove, exactly and along the whole weight walk of #57, that there are only two tiles. Every layer tile is carried onto the c-tile of the cubic stacking or the h-tile of the hexagonal stacking by one motion, all four pieces at once, and which of the two is decided by the layer's own type. Any tile sits on any tile, since the joint between two layers has the same shape for both types. So every close-packed stacking, periodic or not, is a string of two tiles, and a string of two tiles in any order is a stacking. The tiles meet on flat planes for weights up to $\lambda = -7/24$ and interlock above it, their tetrahedral pieces reaching into the neighbouring layer. At each weight all stackings together use five whole shapes. Version 2 adds HexOct-K, keyed HexOct tiles: with keys on the axis faces of one layer the tiles form a Wang tiling of the square grid, so HexOct-K layers can compute; a layer of four keyed tiles counts in binary. Where the tiles interlock, every tile touches the layer below and the layer above at exactly three positions along the layers' common diagonal, in every stacking (192 of 192 cases), so HexOct-K tiles run any elementary cellular automaton with one step per layer: the stacking itself is the clock, and with Rule 110 the stack is a universal computer.

> *Stack layers of balls and the gaps between them can be shared out among the holes. Paper #57 found that every piece is built from two kinds of half. Here the halves are packed into layers, and only two kinds of layer come out, one for each way a layer can sit between its neighbours. Pick any sequence of the two and you have built a crystal stacking; the pieces fit at every joint, endlessly.*

1. Where it started

The Euclid–Kepler–Pacioli cell (#8 [9]) holds all five Platonic solids in one unit, and copies of the unit join across their faces so that the cycle of solids runs on from cell to cell without end. DICTO proposed that the pieces of the Rule of Halves (#57) behave in the same way: that they fill one unit and join across units, cycling without end. The natural unit is one layer of balls with its share of the holes on either side. This chapter proves that the idea holds exactly, and finds the shape of the two units.

2. The set-up

As in #57: unit balls sit in close-packed layers (triangular lattices of spacing 1) at heights kd, $d = \sqrt{(2/3)}$; layer k sits on one of the three positions A, B, C, with no two consecutive layers on the same position. Layer k is *h* if layers $k-1$ and $k+1$ sit on the same position and *c* otherwise. Between layers k and $k+1$ lie an octahedral hole at height $(k + ½)d$ and two tetrahedral holes, at $(k + ¼)d$ and $(k + ¾)d$. One weight λ, given to the tetrahedral holes (the octahedral holes have weight 0), decides the power diagram of the holes, whose cells are the pieces; the walk is $-11/24 \le \lambda \le 3/8$. By #57, a tetrahedral piece is the cubic or hexagonal tetrahedral piece named by its nearest layer, and an octahedral piece cut by the horizontal plane through its hole has its bottom half named by the layer below and its top half by the layer above.

Definition. The *tile* of layer L, for one ball of the layer, is made of four pieces: the top half of the octahedral piece of the gap below (hole at $(L - ½)d$), the tetrahedral pieces at $(L - ¼)d$ and $(L + ¼)d$, and the bottom half of the

octahedral piece of the gap above (hole at (L + ½)d). The *c-tile* and *h-tile* are the tiles of the pure cubic stacking ABC… and the pure hexagonal stacking AB…. Every piece and half of the diagram belongs to exactly one tile, so the tiles of all balls of all layers fill space without overlap.

3. Main theorem

> **Theorem (two tiles).** For every close-packed stacking and every λ with $-11/24 \le \lambda \le 3/8$, the tile of each layer L is carried onto the c-tile, if L is c, or onto the h-tile, if L is h, by one motion that maps the layer to itself: a symmetry of the triangular lattice about the vertical axis through the ball, followed by a translation. The same motion carries all four pieces.

Corollaries. (i) *Any tile sits on any tile.* The face where two tiles meet is the section of an octahedral piece by the horizontal plane through its hole; for every λ this section has one shape, shared by the c- and h-tiles. Every sequence of c- and h-tiles is therefore a stacking, and every stacking is such a sequence. (ii) *Flat or interlocking.* For $\lambda \le -7/24$ every tile lies between the planes through the octahedral holes above and below, and the tiles meet on flat planes; for $\lambda > -7/24$ the tetrahedral pieces cross these planes and the tiles interlock. The value $-7/24$ is the same for both tiles and is not a stop of the walk. (iii) *Five shapes.* At each λ all close-packed stackings together use five whole pieces: the c- and h-tetrahedral pieces and three octahedral pieces, with two c-halves, two h-halves, or one of each.

The Rule of Halves matched each piece separately, each with its own motion. The theorem here is stronger: one motion places the whole tile, so the tile is a rigid unit that can be picked up and stacked.

4. Proof

Step 1: seven layers decide. The two octahedral halves and two tetrahedral pieces of tile L belong to the gaps L – 1 and L. By Step 1 of #57 the pieces of a gap depend only on the six layers around it, so the tile depends only on the layers L – 3, …, L + 3. Up to relabelling A, B, C these are fixed by the types of the five inner layers L – 2, …, L + 2: 32 contexts. For each we took a periodic stacking containing the seven layers; the 32 stackings are listed in prove.out [S].

Step 2: the stretches. For each context we found exactly, with the census engine of #53 and #57, every λ at which one of the four cells of the tile, or of the pure tile of the same type, changes its faces. Between two such breakpoints every corner of every cell is an affine function of λ, and the corners of a half, where an edge crosses the cutting plane, are rational functions of λ. On each sub-interval we found the motion at one sample λ, applied it to the tile, and compared the result with the pure tile as rational functions of λ (computer algebra, exact): they are identical, so the tiles are equal on the whole sub-interval. All 32 contexts, every sub-interval: equal.

Step 3: the breakpoints. At each breakpoint the tile was computed directly, in rational arithmetic, and the same comparison holds. □

Corollary (i). In a stacking with a c-layer below an h-layer, the octahedral piece between them has a c-half below and an h-half above. Its section by the plane through its hole is the top face of the c-half and the bottom face of the h-half at once. By the theorem every c-tile's top face is a copy of this c-face and every h-tile's bottom face a copy of this h-face, and they are the same shape. The same holds the other way up, and for c on c and h on h trivially. **Corollary (ii).** On each sub-interval the heights of the tetrahedral corners are affine in λ; solving where they meet ±½ (in layer units) gives $\lambda = -7/24$ in all 32 contexts, and no other crossing. **Corollary (iii).** In a c-layer the ball is a centre of symmetry of the cubic stacking, which swaps the two tetrahedral pieces of the tile and its two octahedral halves; in an h-layer the layer plane is a mirror of the hexagonal stacking, which does the same. So each tile holds one tetrahedral shape and one half shape, and the octahedral pieces are made of two halves of these four kinds: c + c, h + h or c + h (the mixed pieces of #57, Chucky and Fang among them). □

5. The two tiles

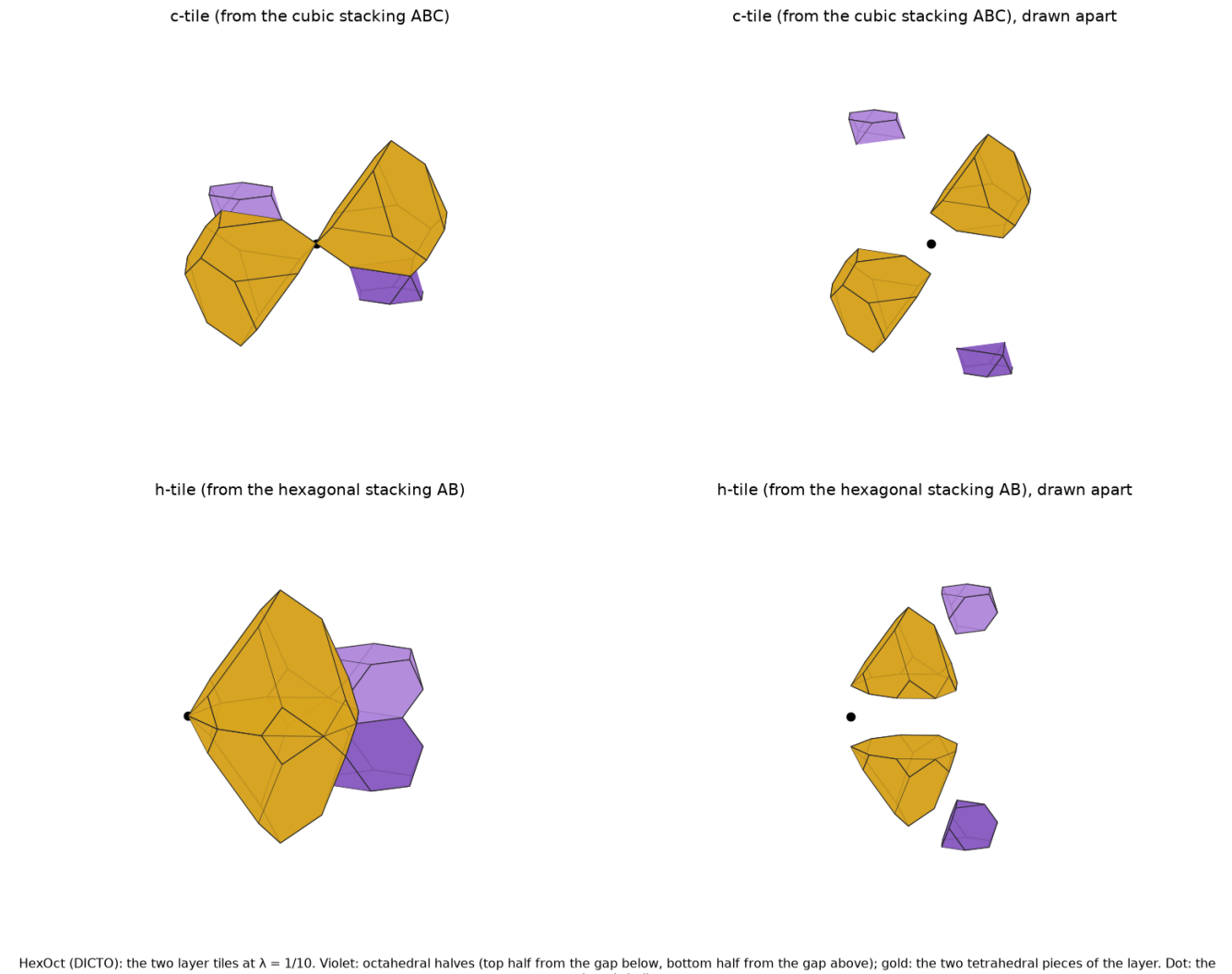


Figure 14.1. The c-tile (top) and h-tile (bottom) at λ = 1/10, whole (left) and drawn apart (right). Gold: the two tetrahedral pieces, meeting at the layer's ball (black dot); light violet: the bottom half of the octahedral piece above; dark violet: the top half of the octahedral piece below. In the h-tile the two halves lie on one vertical line, because the layers above and below sit on the same position; in the c-tile they are set apart, because those layers sit on different positions. That is the whole difference between the two tiles.

A tile is one ball's share of its layer; side by side, copies of a tile fill the layer as a slab whose upper and lower faces are flat for λ ≤ –7/24 and toothed above. At λ = 1/10 the tetrahedral pieces reach a full layer up and down, to the height of the next layer's balls, so each layer keys into the next (Figure 14.2).

So there is a cubic tile and a hexagonal tile, and either fits on the top or bottom face of either. Sideways a layer is made of one tile only, since all its balls have the same type. The type of a tile does not limit what goes above it; it decides where the next layer sits: above a c-tile the next layer takes the third position, above an h-tile it returns to the position of the layer below.

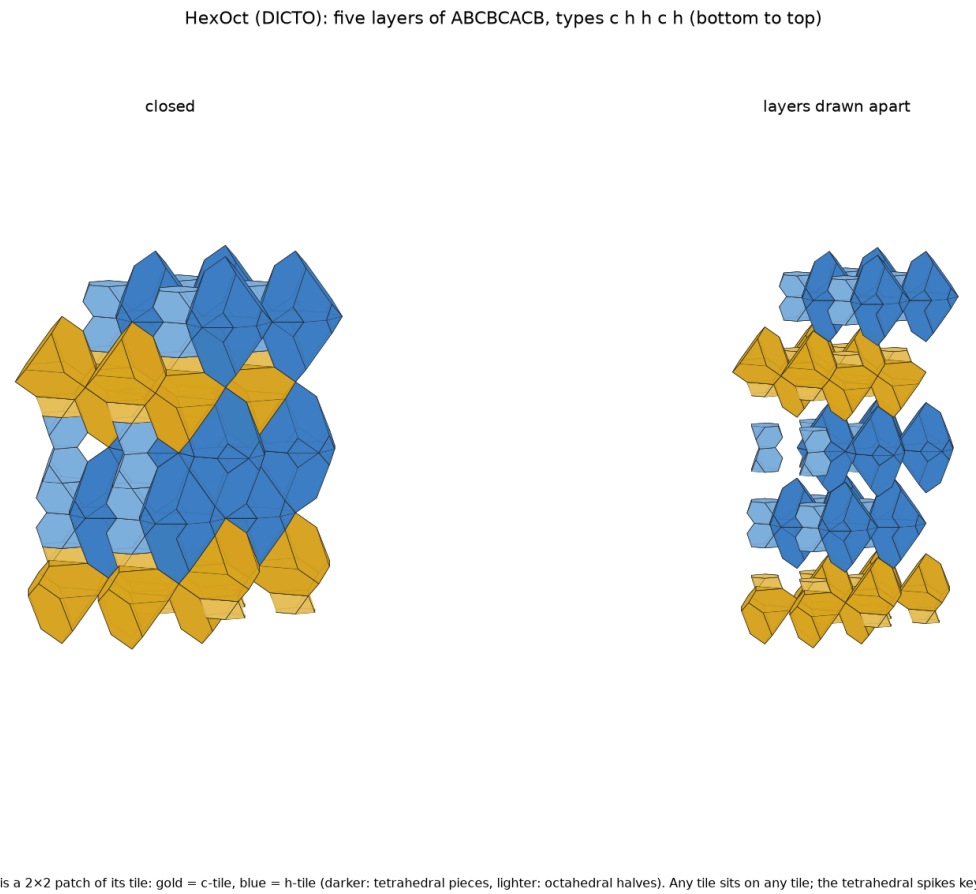


Figure 14.2. Five layers of the stacking ABCBCACB, types c h h c h from the bottom, at λ = 1/10; each layer a 2 × 2 patch of its tile, gold for c-tiles and blue for h-tiles (darker: tetrahedral pieces; lighter: octahedral halves). Left closed, right with the layers drawn apart.

6. What is known, and what was not found

Known and credited. Close-packed stackings, polytypes and the h/c notation [4]; power diagrams [5]; the pieces and the Rule of Halves [1] on the walk of #53 [2], with the twinned octet pieces of #56 [3]. Order–disorder (OD) theory, from Dornberger-Schiff, describes polytypic crystals as stacks of layers of one or more kinds, any two neighbouring layer pairs being geometrically equivalent (the vicinity condition) [6]: the general frame in which a stacking is a string of layers. Tilings of space by regular tetrahedra and octahedra, including non-periodic ones obtained by sliding along planes, are enumerated by Gabbrielli, Jiao and Torquato [7]; the unit cell of the octet as a rhombohedron of two tetrahedra and one octahedron, and its layered sections, are in Lalvani [8]. The Euclid–Kepler–Pacioli cell network is DICTO's [9].

Stated here. The layer tiles built from the power-diagram pieces of the holes; the proof that there are only two, placed each by one motion, for all stackings and the whole walk; the shared joint; the switch from flat to interlocking at λ = –7/24; the five shapes. We found no earlier statement of tiles of this kind.

7. Where the results stop

(1) The theorem covers the walk of #57, $-11/24 \le \lambda \le 3/8$, with one weight for all tetrahedral holes; at the ends some pieces shrink to points and the tiles are read as limits. (2) With different weights for the two kinds of tetrahedral hole the tiles may split into more kinds; not tested. (3) The tiles are geometric; which of them a real crystal prefers is the subject of the Hexagonality Law papers, not of this one.

8. HexOct-K

DICTO proposed that data could be built into HexOct configurations, the ways the tiles may connect setting the limits, as the basis of a new kind of programming language. On their own the two tiles carry one free choice per layer: any tile sits on any tile, so a stack is a string of c and h, a perfect store of bits that can never refuse a tile. To compute, a set of tiles must be able to refuse. That is what keys do: marks on the faces where tiles meet, with a tile allowed only where every key matches its neighbour's. DICTO names keyed HexOct tiles HexOct-K.

Where keys can go. We measured, for every tile, which neighbouring tiles share a face with it and how much face area each, in six stackings covering both tile types and mixed neighbourhoods (ABC, AB, ABAC both layers, ABABC, ABACBC) and five weights across the walk ($\lambda = -2/5, -1/3, 0, 1/10, 1/4$): points sampled on every face of every piece of the tile, pushed a hair outward, and looked up in the pieces of every tile within two layers; no face area was left unmatched. Three facts hold in every case. (i) Within its own layer every tile shares faces with its four neighbours along the lattice axes, $(\pm 1, 0)$ and $(0, \pm 1)$, each face of area at least 0.28 (ball spacing 1), and with up to four more. (ii) Where the tiles meet on flat planes ($\lambda \le -7/24$), each tile touches exactly one tile above and one below, through the whole joint of area $\sqrt{3}/2$: the stack falls into columns. (iii) Where they interlock, each tile touches four to six tiles in the layer above and in the layer below.

Keys on the joints alone count, but cannot compute. With keys only on the joints between layers, a column of tiles is a string read once from bottom to top, each tile checking only the one below. Such a set of tiles accepts exactly the strings a finite machine with a fixed memory can recognise (a regular language): it can enforce patterns such as "no two h-tiles in a row", but it cannot count without bound or compute.

Keys on the side faces compute anything. Put keys on the four axis faces of the tiles of one layer, the other side faces left blank (one key that always matches). The layer is then a Wang tiling of the square grid: square tiles with marked edges, placed without turning, edges matching [10]. Wang tilings can carry out any computation, the rows of a tiling being the successive steps of a Turing machine; and no general method can decide whether a given set of Wang tiles fills the plane [11, 12]. Both statements therefore hold for HexOct-K layers, at every weight of the walk, for c-tiles and h-tiles alike.

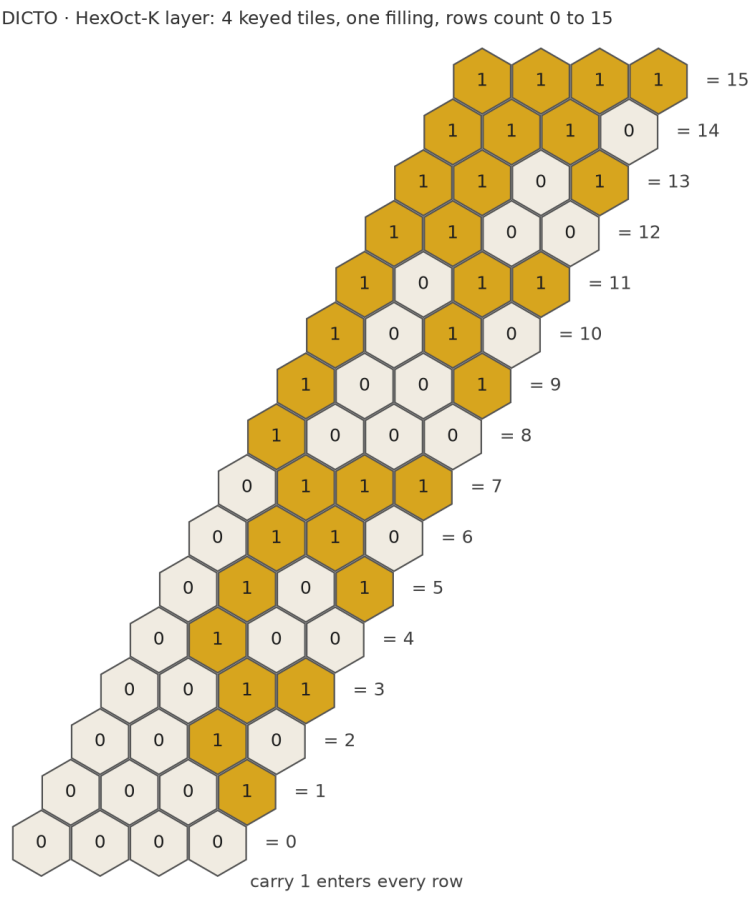


Figure 14.3. A HexOct-K layer that counts. Four keyed tiles: each holds a bit; its lower axis face reads the bit of the tile below in the layer, its right face takes a carry, its left face passes the carry on (new bit = old bit xor carry, carry out = old bit and carry). With row 0 all zeros and a carry of 1 entering every row at the right, an exhaustive search finds exactly one filling, and its rows are 0, 1, 2, … 15 in binary. Tiles drawn schematically as hexagons at their balls.

A layer that counts. The smallest working example uses four keyed tiles (Figure 14.3). An exhaustive search over windows of 3 × 8, 4 × 16 and 5 × 20 tiles found exactly one filling in each, and in each the rows count up in binary. Where the tiles meet on flat planes, the same four tiles can count up the stack instead, one number per layer, with the carry passing along the layer and each number handed to the layer above through the single joint (fact ii): the stacking itself becomes the clock.

The stacking as the clock. Where the tiles interlock ($\lambda > -7/24$), measure the lattice offsets of the touching tiles along the layers' common diagonal, $s = du - dv$; every layer shift, $B = (1/3, 1/3)$ or $C = (2/3, 2/3)$, lies along $(1, 1)$, so s does not change from layer to layer. In all 32 five-layer contexts at six weights across the interlocking range ($\lambda = -1/4, -1/6, 0, 1/10, 1/4, 1/3$; 192 cases, face areas sampled as above, none unmatched), every tile touches the tiles of the layer below, and of the layer above, at exactly three positions, $s - 1$, s and $s + 1$, and at no others [S3]. Give each tile one state, written on the faces it shares with tiles above and read on the faces it shares with tiles below, the faces within a layer blank, and take every layer constant along the diagonal: a set of keyed tiles with one tile for each triple of states below then runs an elementary cellular automaton, one step per layer, forced layer by layer, whatever the order of c- and h-tiles. Rule 110 is universal [19], so a HexOct-K stack can compute anything, with the stacking itself as the clock. Each tile is decided by its own layer, so the tiles are the Window Law's window of one layer (Chapter 15): every count of a stacking read tile by tile is a straight line in h. The side faces that carry the keys grow with the weight: their smallest area is 0.013 at $\lambda = -1/4$, 0.33 at $\lambda = 0$, 0.40 at $\lambda = 1/10$, 0.48 at $\lambda = 1/4$ and 0.50 at $\lambda = 1/3$, where the three contacts are nearly equal (Table 2); physical keyed pieces belong in that upper range.

λ	−1/4	−1/6	0	1/10	1/4	1/3
smallest side contact (s ± 1)	0.013	0.11	0.33	0.40	0.48	0.50
smallest centre contact (s)	0.86	0.86	0.70	0.60	0.52	0.50

Table 2. Contact areas of a HexOct tile with the layer below and above, smallest over all 32 contexts (ball spacing 1).

What is known, and what was not found. Wang tiles and their computing power are classical [10–12]; tiles that compute by assembling themselves were made from DNA [13], and three-dimensional tile assembly can simulate Turing machines [14]; Wang cubes have been used to build aperiodic material samples [15], and triangular and hexagonal tiles have been studied for self-assembly [16]. Space-time diagrams of cellular automata are tilings of this kind, and Wang cubes have been made aperiodic (K. Culik II and J. Kari, Journal of Universal Computer Science 1 (1995) 675–686). Sources checked: these works; web searches for cellular automata as layered three-dimensional tilings and for close-packed stacking layers as computation (one study models the change of a stacking itself by cellular automaton rule 232: D. P. Varn and J. P. Crutchfield, arXiv cond-mat/0307296); web searches for three-dimensional polyhedral tiles with matching rules and computation, and for close-packed stacking layers as tiles with keys or computation; and DICTO's reference library (7,730 passages) for Wang tiles and self-assembly, which returned the aperiodic monotile [17] and interlocking tetrahedron–octahedron blocks [18]. We found no use of close-packed stacking layers, or of tiles built from their holes, as a medium for computation. The computing power here is Wang's; what HexOct adds is the geometry that makes it available: two tiles that fit every stacking, with axis faces of positive area in every layer, and one-to-one joints that make the stacking a clock.

Status. HexOct-K is a first study (2026-10-29, [S2]): the contact measurements are sampled, not exact; the counting layer is checked exhaustively on the windows above. The stacking-as-clock contacts are measured in the same way, in all 192 cases [S3]; the cellular automaton follows from them by construction.

9. Conclusion

Every close-packed stacking, at every weight, is a string of two tiles, and any string of the two tiles is a stacking. Each tile is one rigid unit of four pieces, decided by its own layer alone; the tiles meet on flat planes up to λ = −7/24 and interlock beyond it. As DICTO proposed, the pieces fill one unit and join across units without end. Keyed, as HexOct-K, the tiles compute: within a layer as a Wang tiling, and, where they interlock, up the stack, one cellular-automaton step per layer, with the stacking itself as the clock.

Plates

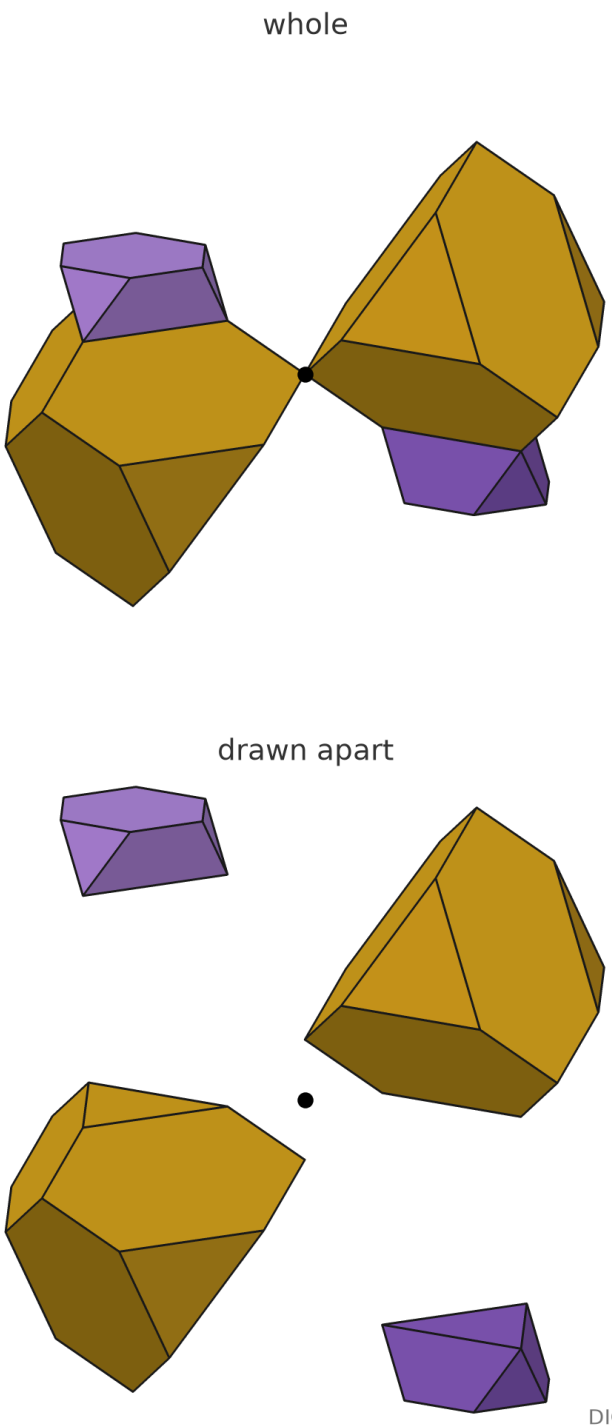
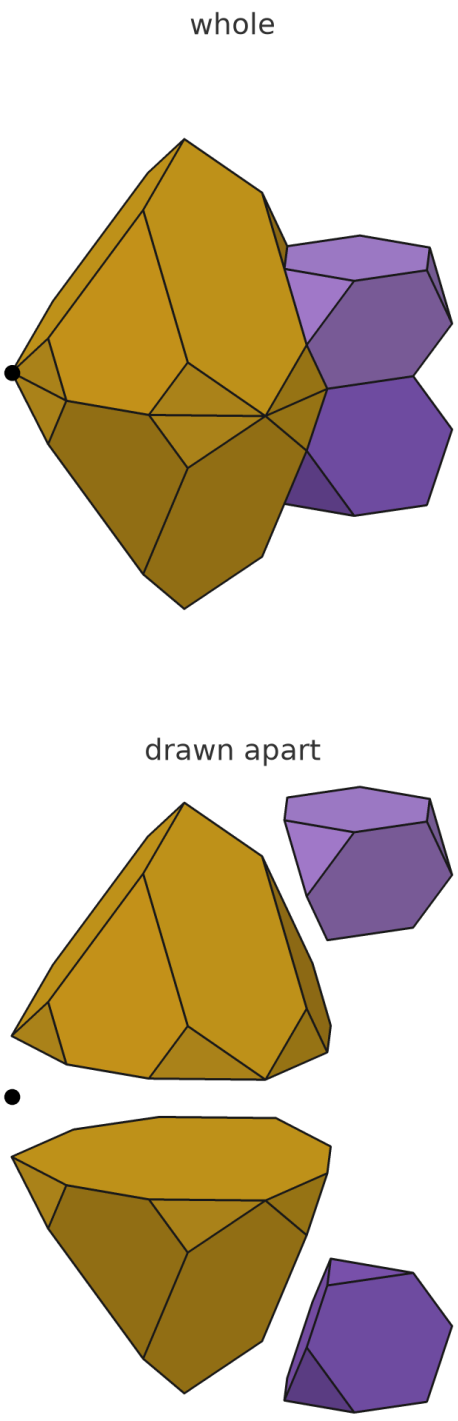


Plate 1. The c-tile

The tile of a cubic layer, from the cubic stacking ABC…, at λ = 1/10 (as Figure 14.1) · Chapter 14

- One ball's share of its layer, made of four pieces: the top half of the octahedral piece below (dark violet), the two tetrahedral pieces (gold, meeting at the ball, black dot) and the bottom half of the octahedral piece above (light violet).
- In every close-packed stacking, at every weight of the walk $-11/24 \le \lambda \le 3/8$, the tile of every c-layer is carried onto this tile by one motion, all four pieces at once (the Two-Tile Theorem).
- The two octahedral halves are set apart, because the layers above and below sit on different positions. The ball is a centre of symmetry of the cubic stacking and swaps the two tetrahedral pieces and the two halves.
- Above a c-tile the next layer takes the third position.

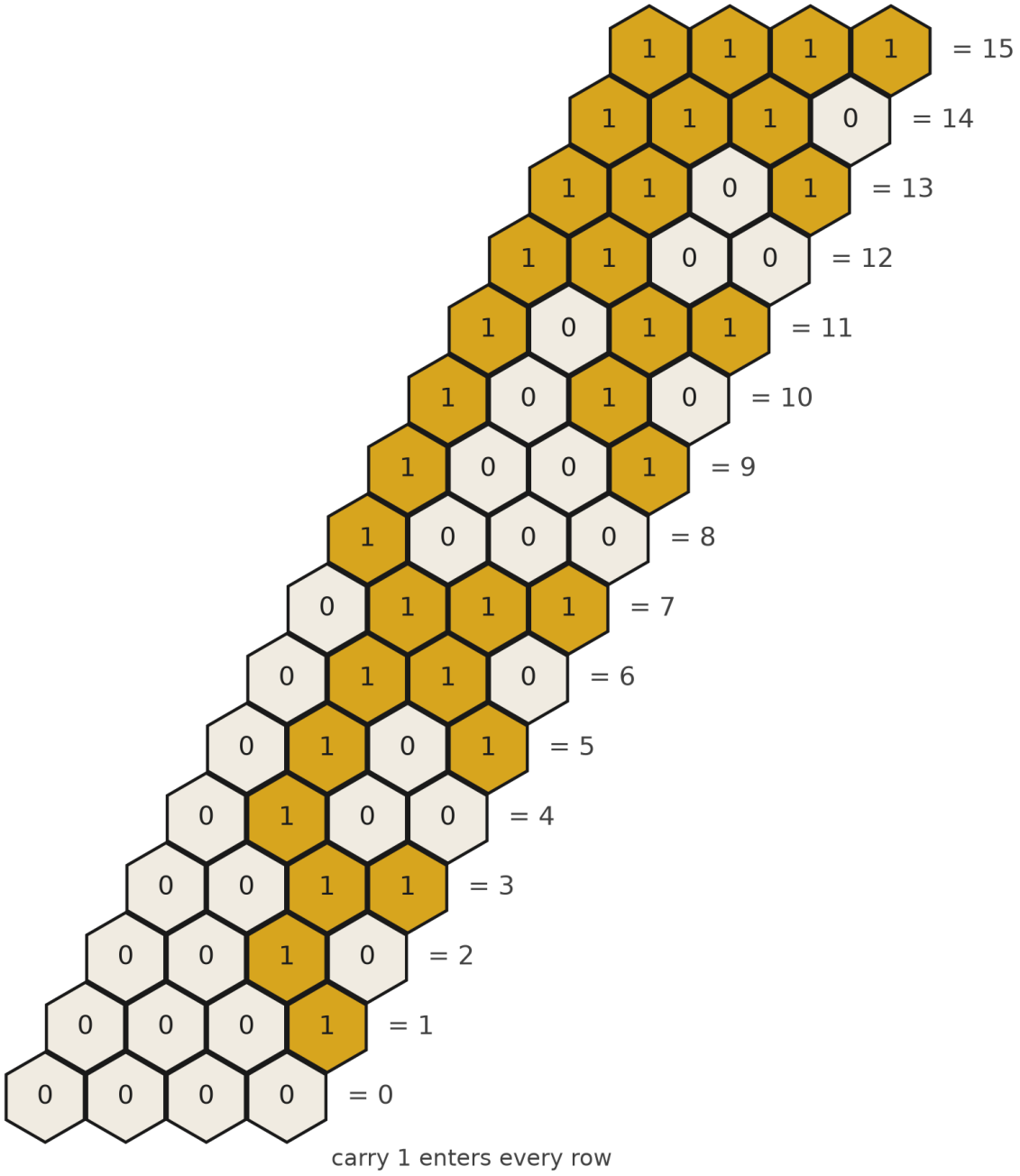


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Plate 2. The h-tile

The tile of a hexagonal layer, from the hexagonal stacking AB…, at λ = 1/10 (as Figure 14.1) · Chapter 14

- The same four kinds of piece as the c-tile: two octahedral halves (violet) and two tetrahedral pieces (gold); black dot, the layer's ball.
- The two halves lie on one vertical line, because the layers above and below sit on the same position; that is the whole difference between the two tiles. The layer plane is a mirror of the hexagonal stacking.
- Any tile sits on any tile: the joint between two layers, the section of an octahedral piece through its hole, has one shape for both tiles. Every string of c- and h-tiles is a stacking, and every stacking is such a string.
- The tiles meet on flat planes for λ ≤ –7/24 and interlock above; at λ = 1/10 the tetrahedral pieces reach a full layer up and down. Above an h-tile the next layer returns to the position of the layer below.



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Plate 3. HexOct-K: a layer that counts

Keyed HexOct tiles (DICTO's name), as Figure 14.3; tiles drawn as hexagons at their balls · Chapter 14, Section 8

- Four keyed tiles, each holding a bit (gold 1, pale 0): the lower axis face reads the bit of the tile below in the layer, the right face takes a carry, the left face passes it on (new bit = old bit xor carry, carry out = old bit and carry).
- With row 0 all zeros and a carry of 1 entering every row at the right, an exhaustive search finds exactly one filling (3 × 8, 4 × 16 and 5 × 20 windows); its rows are 0, 1, 2, … 15 in binary.
- With keys on the four axis faces a HexOct-K layer is a Wang tiling of the square grid, so it can carry out any computation. Where the tiles interlock, each touches the layer below and the layer above at exactly three positions along the common diagonal (192 of 192 cases): the stack runs a cellular automaton, one step per layer, with the stacking itself as the clock.

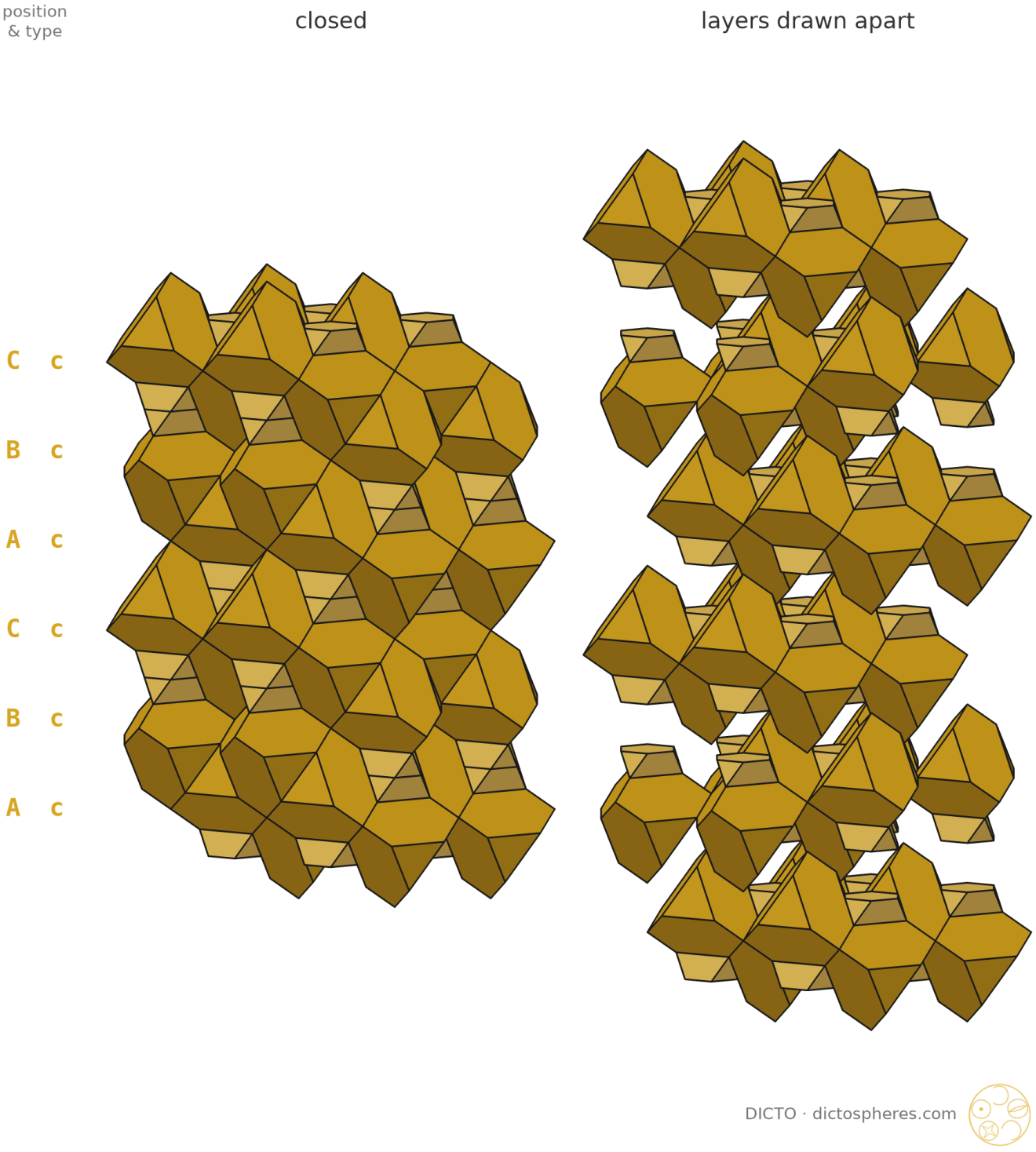


Plate 4. 3C, the cubic stacking

ABC… · all layers c · h = 0 · six layers, each a 2 × 2 patch of its tile, λ = 1/10 · Chapters 1, 11, 14

- A string of c-tiles only (gold; darker the tetrahedral pieces, lighter the octahedral halves). Its pieces are those of the octet of #51.
- One of the two pure stackings from which, by the Rule of Halves, every piece of every stacking is borrowed, whole or in half.
- $BaRuO_3$ takes 3C at 18 GPa, the last step of the walk down h under pressure, 9R → 4H → 6H → 3C (Chapter 11).

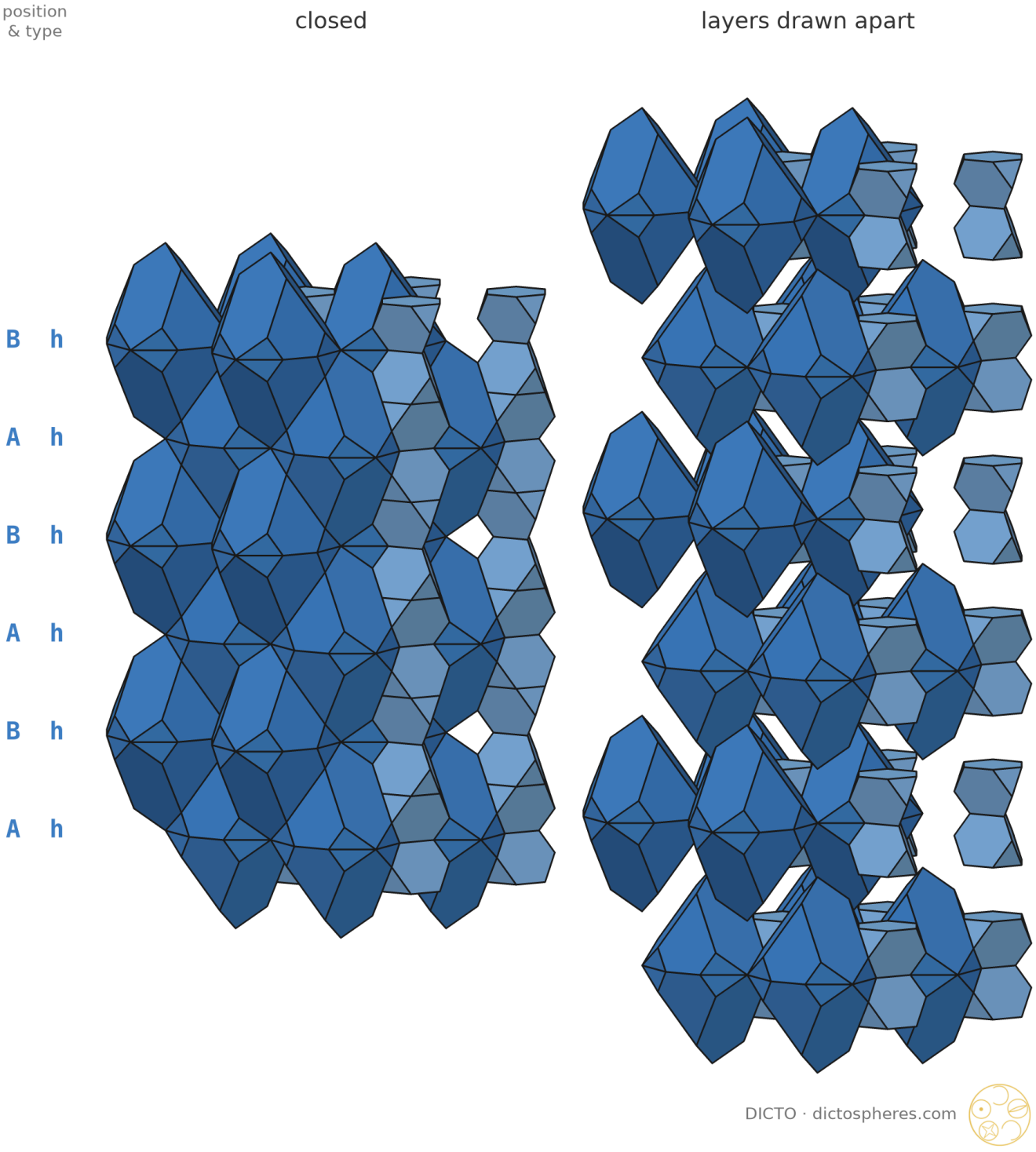


Plate 5. 2H, the hexagonal stacking

AB… · all layers h · h = 1 · six layers, each a 2 × 2 patch of its tile, λ = 1/10 · Chapters 1, 11, 14

- A string of h-tiles only (blue; darker the tetrahedral pieces, lighter the octahedral halves). Its pieces are those of the twinned octet of #56.
- The other pure stacking of the Rule of Halves; with 3C it fixes the two ends of every straight line of the Hexagonality Law.
- In hexagonal perovskites the face-sharing octahedra of 2H form endless chains (Chapter 11).

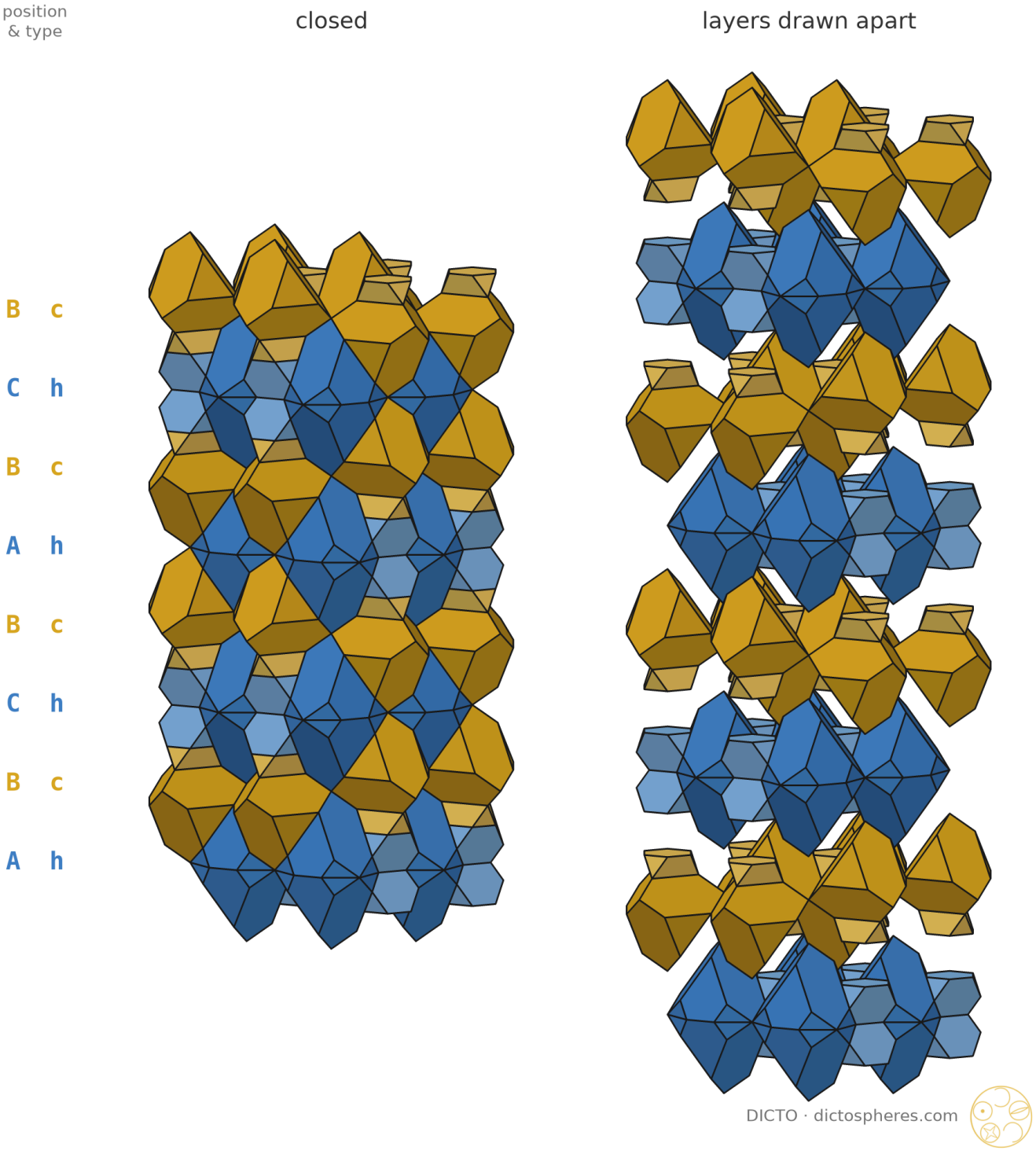


Plate 6. 4H

ABCB… (the double-hexagonal stacking, written ABAC in Chapter 1) · types h c h c · h = 1/2 · eight layers, λ = 1/10 · Chapters 1, 5, 8, 11

- h- and c-tiles in turn (blue and gold). By the Rule of Halves every octahedral piece here is a Janus c/h piece, a cubic half and a hexagonal half.
- Chucky, Fang and the Triskelion Tetrahedron (Plates 11 to 13) were found on this stacking (Chapter 1).
- In the proof that the Swap Rule gives the Hexagonality Law, 4H is the test case: its four gaps give T = 2F(h, c) + 2F(c, h), which the Swap Rule makes 2F(c, c) + 2F(h, h) (Chapter 5).
- β-cerium is double-hexagonal (h = 1/2), 1.2% larger per atom than cubic γ-cerium (Chapter 8); $BaRuO_3$ takes 4H at 3 GPa (Chapter 11).

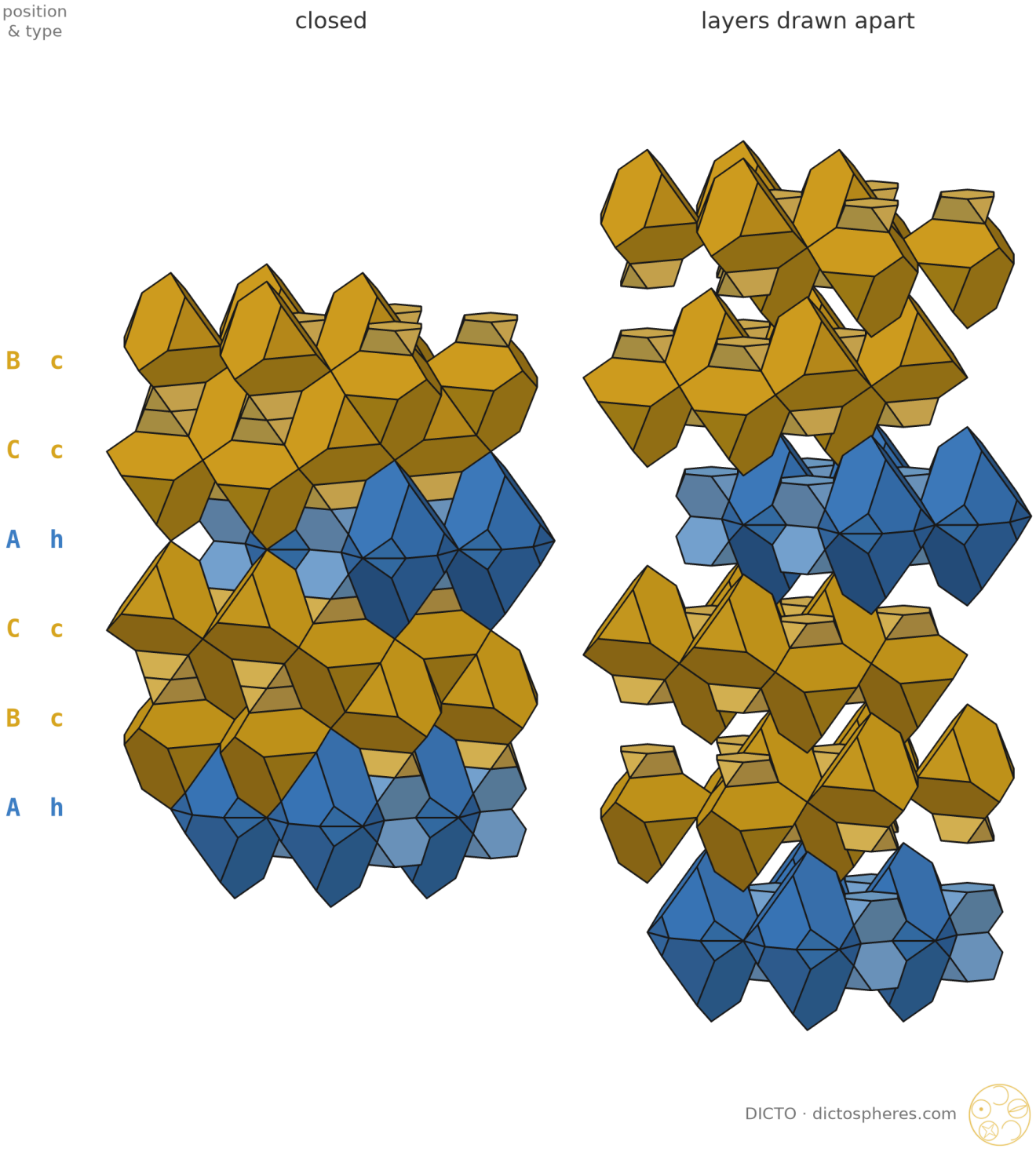


Plate 7. 6H

ABCACB… · types h c c h c c · h = 1/3 · six layers, λ = 1/10 · Chapters 1, 7, 8, 11

- One h-tile after every two c-tiles. The six-layer stacking on which the walk first mixed the cubic and hexagonal pieces (Chapter 1).
- The commonest polytype of silicon carbide; its hexagonality 1/3 is the ground-state filling of the supersymmetric lattice fermions of Fendley, Schoutens and de Boer (Chapter 7).
- $BaRuO_3$ takes 6H at 5 GPa (Chapter 11).

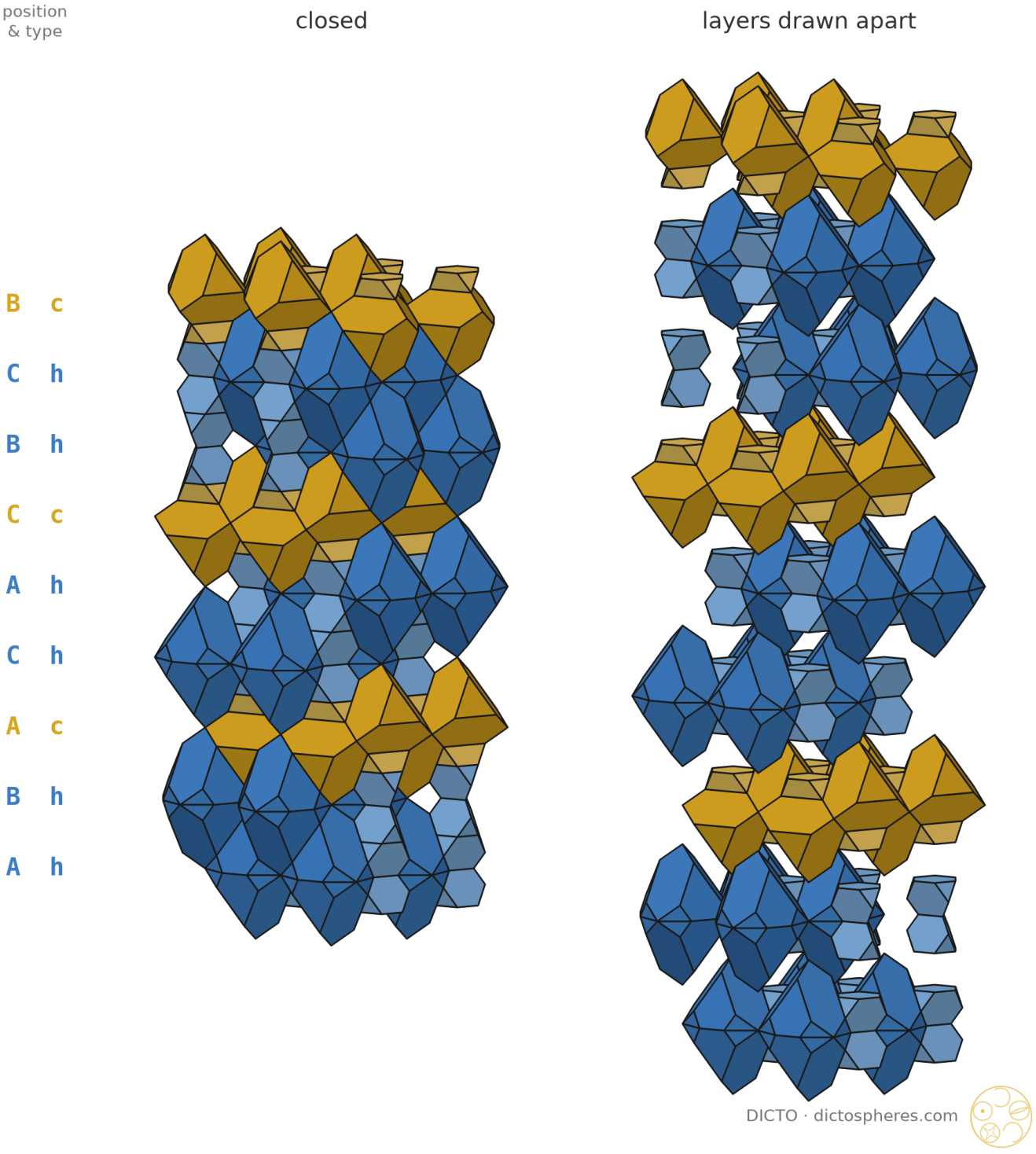


Plate 8. 9R, the samarium stacking

ABACACBCB… · types h h c, three times · h = 2/3 · nine layers, λ = 1/10 · Chapters 1, 11

- Two h-tiles to every c-tile. The nine-layer samarium stacking added no piece beyond those of the mixed stackings before it, which pointed to the local Rule of Halves (Chapter 1).
- $BaRuO_3$ is 9R at ambient pressure (octahedra joined corner, face, face, repeated); its face-sharing octahedra form trimers (Chapter 11).

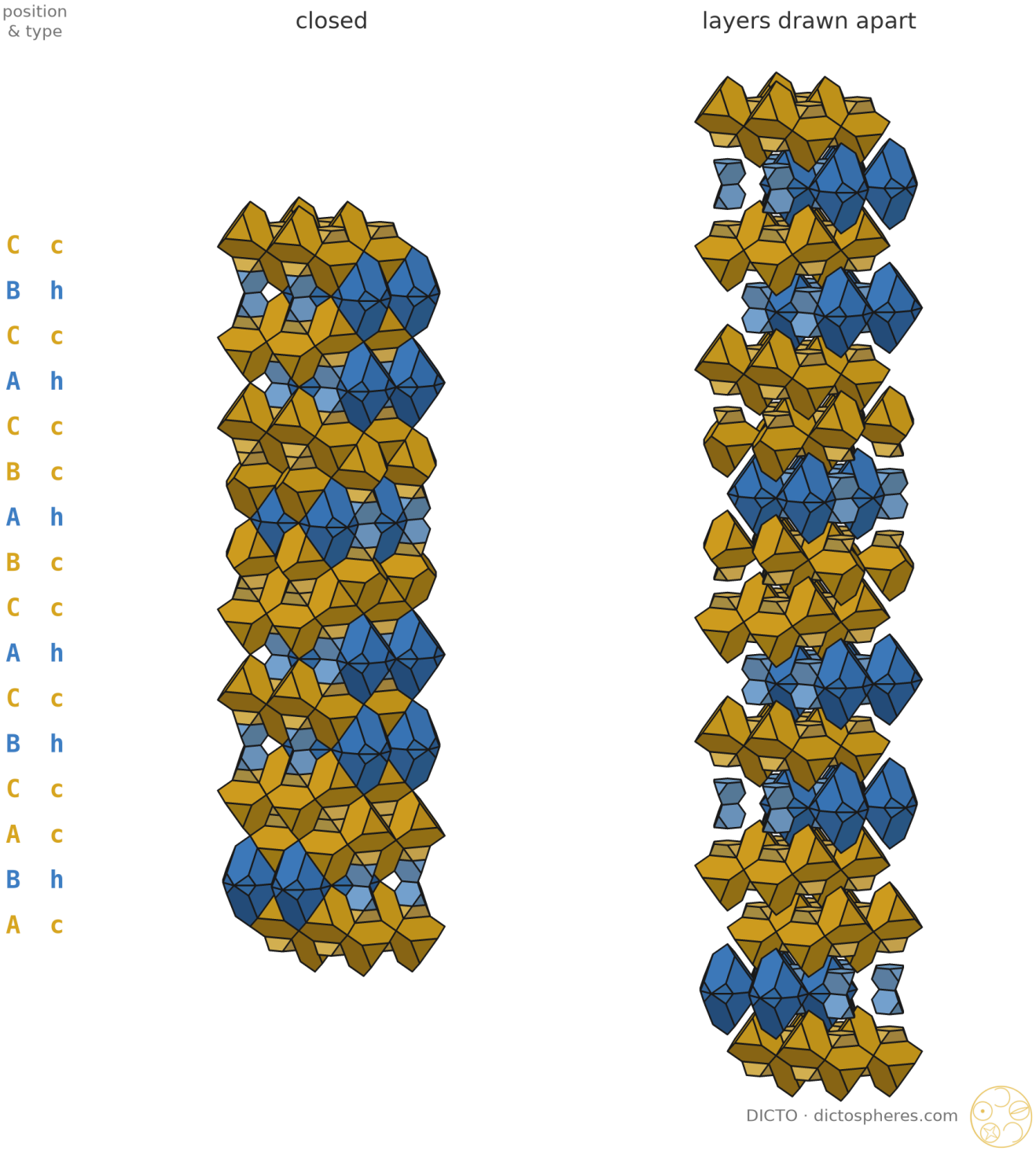


Plate 9. The Fibonacci stacking

16 layers ABACBCACBABCACBC (as Figure 7.1) · types in the order of the Fibonacci word, chcchchcchcch… · λ = 1/10 · Chapter 7

- The types follow the substitution c → ch, h → c. In the never-repeating stacking the share of hexagonal layers is h = $1/\varphi^2 = (3 - \sqrt{5})/2$, and no two h-tiles are neighbours.
- Of the four kinds of piece between neighbouring layers only three occur, cubic–cubic, cubic–hexagonal and hexagonal–cubic, in the proportions $1/\varphi^3 : 1/\varphi^2 : 1/\varphi^2$; every kind of piece averages exactly $V(cc) + (V(hh) - V(cc))/\varphi^2$ per layer.

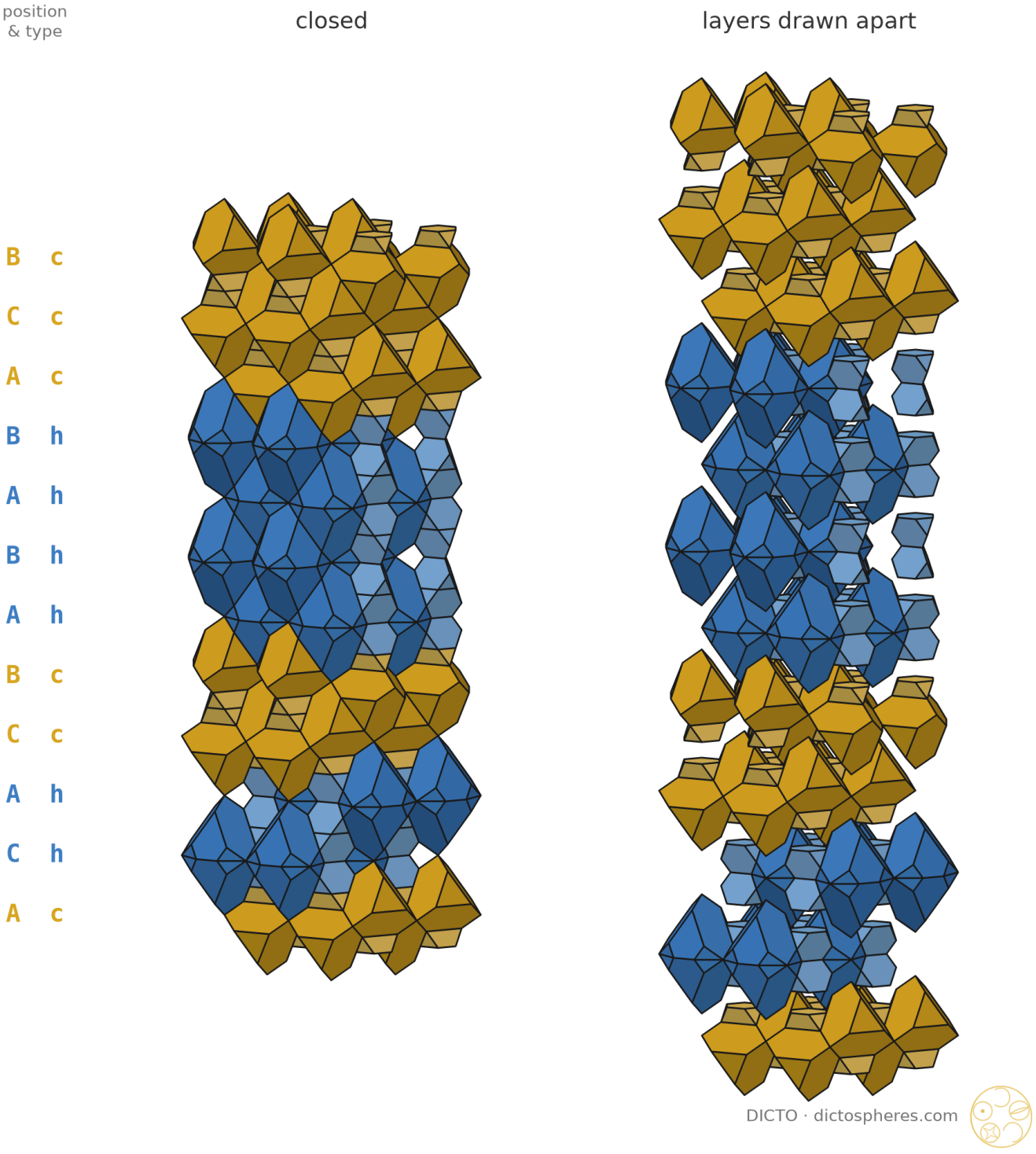
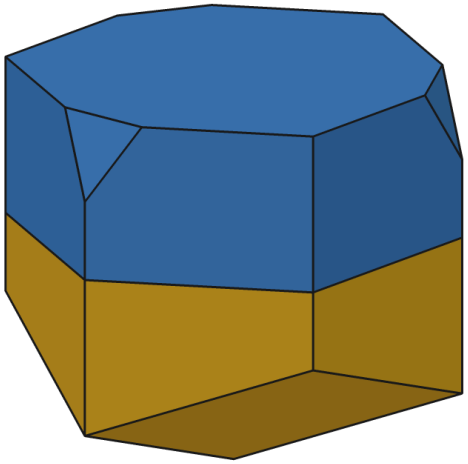


Plate 10. A random stacking

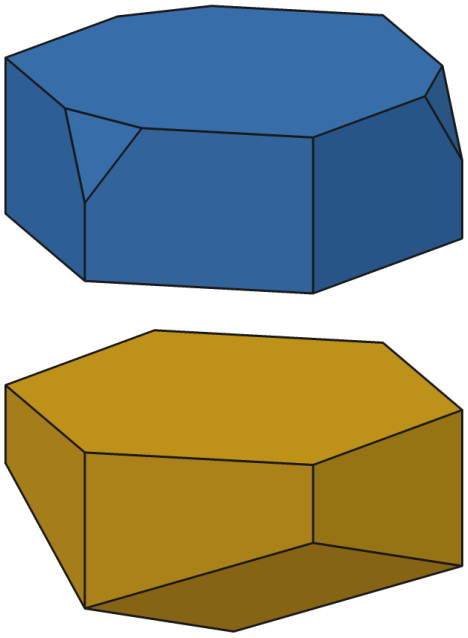
12 layers ACACBABABACB, drawn by coin toss · types c h h c c h h h h c c c · λ = 1/10 · Chapter 6

- Each new layer takes either of the two positions not used by the layer below, with equal chance; the draw was repeated until the twelve layers close into a repeat, so that the stack can be computed as a crystal. This draw has 6 hexagonal layers of 12.
- With equal chances (α = ½, completely random stacking) the expected hexagonality is h = 1/2. Wherever the Swap Rule holds, the totals of a random stacking are fixed by its share of hexagonal layers alone.
- Whatever the order, the string is made of the same two tiles.

whole



drawn apart at the mid-plane



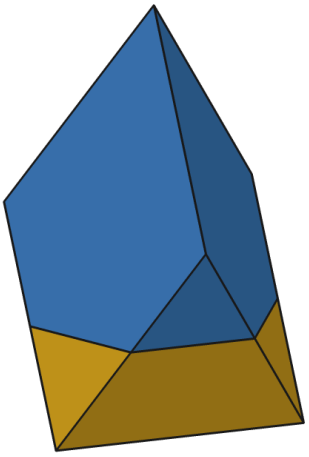
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Plate 11. Chucky

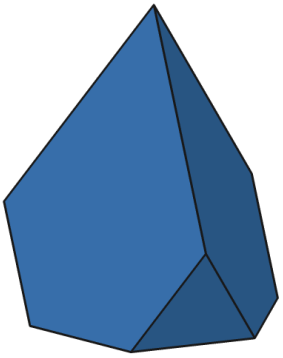
DICTO's name · the octahedral piece at λ = –3/8 with a hexagonal layer above and a cubic layer below (ABAC) · Chapter 1

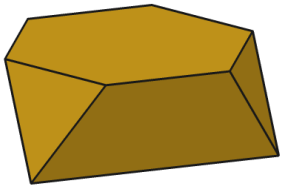
- 19 corners; 3 triangles, 3 squares, 6 pentagons and one nonagon.
- A Janus c/h piece: hexagonal top half (blue), cubic bottom half (gold). Volume $383\sqrt{2}/768$, exactly the mean of the cubic rhombic dodecahedron $\sqrt{2}/2$ and the hexagonal piece $191\sqrt{2}/384$.
- With the cubic layer above and the hexagonal one below, the stacking gives its mirror image.

whole



drawn apart at the mid-plane





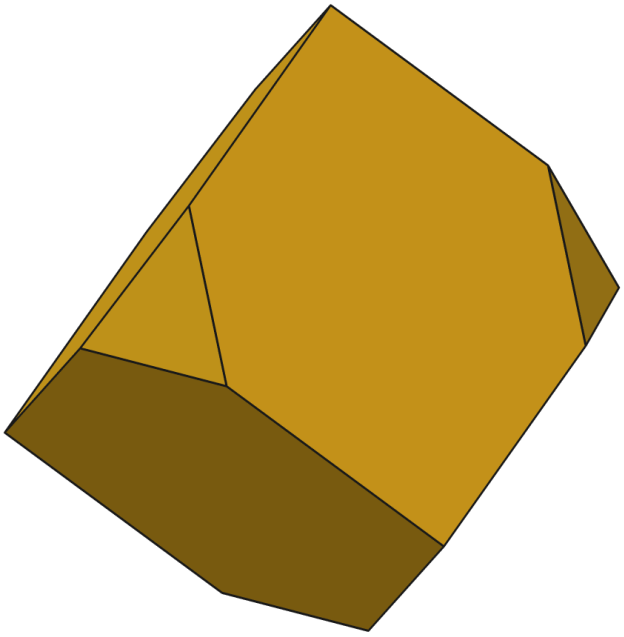
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Plate 12. Fang

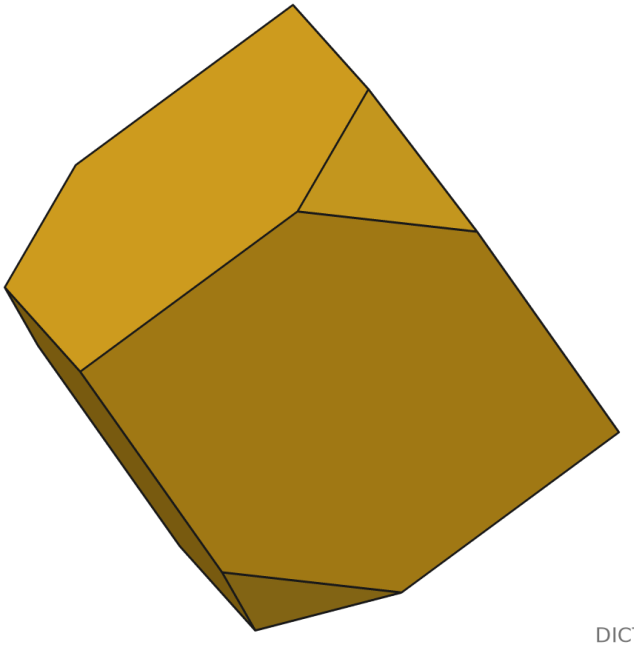
DICTO's name · the octahedral piece at λ = 5/24 with a hexagonal layer above and a cubic layer below (ABAC) · Chapter 1

- 7 corners; 4 triangles and 3 squares.
- A Janus c/h piece: hexagonal top half (blue), cubic bottom half (gold). Volume 5√2/324, exactly the mean of the cubic octahedron √2/81 and the hexagonal rhombohedron √2/54.
- Chucky and Fang are the named members of the Janus c/h pieces; two further families, of 9 and of 27 corners, complete it.

from above



from below, turned



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Plate 13. Triskelion Tetrahedron (TT)

DICTO's name · the tetrahedral piece of ABAC at λ = 5/24 · Chapter 1

- 16 corners; 4 triangles and 6 hexagons. Volume 79√2/324.
- By the Rule of Halves it is the cubic tetrahedral piece at that weight, a member of the sliding family of Samson in #51; the name marks that member.
- In every tile each tetrahedral piece is the cubic or the hexagonal tetrahedral piece named by its nearest layer (gold in Plates 1 and 2, there at λ = 1/10); TT is the cubic one at λ = 5/24.

Part IV. The Window Law

Chapter 15. The Window Law

From paper #84, DICTO DOI 10.5281/zenodo.23848832. Numbers in brackets refer to the chapter's list of outside works in the Appendix.

The Hexagonality Law beyond one layer: one elementary theorem for every count of a two-letter stacking, tested on perovskites, Frank–Kasper phases and quasicrystal approximants

Abstract. DICTO's Hexagonality Law says that in every close-packed crystal structure, however its layers are stacked, every count of its pieces, holes, bonds and swaps is a straight line in its hexagonality h, the share of hexagonal layers. We prove the law that contains it and marks its exact reach: every count read from one layer, and every pair count that obeys the Swap Rule. For a layer sequence over two letters, c and h, any quantity fixed by g consecutive layers has a mean that is a linear function of the shares of the c/h words of length g, which have exactly 2^{g-1} free values; the mean drops to a window of g − 1 layers exactly when certain Swap defects vanish. At g = 1 this is the Hexagonality Law. At g = 2 exactly one new number enters, the share p_{hh} of neighbouring hexagonal pairs, and its coefficient is Δ = F(cc) + F(hh) − F(ch) − F(hc), the Swap defect: DICTO's Swap Rule is exactly the condition for a pair count to stay a straight line in h. The proof is elementary, and the frame is classical in substance. We test it on three kinds of crystal, with exact counts throughout. In the hexagonal perovskites (all 505 polytypes up to twelve layers and 50 random stackings) 16 of 36 counts are lines in h, 9 are planes in (h, p_{hh}) with coefficient Δ, and 11 need longer windows; new exact counts include face-sharing clusters per B = 1 − h, A neighbours per B = 8 − 2h, the same holes in every gap, and the B–B distance shells. In the Frank–Kasper phases (18 real phases and 181 layered stackings) the classical identities and the 1974 Yarmolyuk–Kripyakevich line hold, and three hypothetical structures leave that line through a single defect, the same shape; the layered families carry Swap defects in the pattern +1, –2, +1 and cannot see the c/h letter of their Laves slabs. In the periodic approximants of quasicrystals one number, the share of one tile, orders everything: on the chamber that holds the quasicrystal the defect is zero and the quasicrystal lies on every line; beyond its walls the defect appears, in one dimension as a kink of twice the Swap defect, in the Ammann–Beenker tiling as a forbidden vertex star growing as a square. Approximants carry zero defect window after window, up to a length that grows with their period; zero defect at window 2 alone does not single them out. This is the bridge to DICTO's aim.

1. Introduction

> *Look at a stack of layers through a window one layer tall, and every count you can read there is a straight line in one number, the share of hexagonal layers. That is DICTO's Hexagonality Law. Open the window to two layers and one more number comes into view, the share of hexagonal pairs; how strongly a count sees it is the Swap Rule's defect. Open it further and more numbers come in, a window at a time. In the approximants of a quasicrystal the second number is held at zero, and the straight lines run all the way to the never-repeating limit.*

DICTO's Hexagonality Law is a law of structural crystallography: in every close-packed crystal structure, however its layers are stacked, every count of the structure (its pieces, holes, bonds and swaps) is a straight line in the structure's hexagonality, the crystallographers' share of hexagonal layers. It grew out of DICTO's Rule of Halves (#57), which builds every hole piece of every stacking from the pure cubic and hexagonal ones, and his Swap Rule (#61, #64): the two mixed pieces add up to the two pure ones. The Focus (#66) proved the law from the Swap Rule wherever a piece is fixed by its two neighbouring layers; The Hexagonality Law and the Fibonacci Stacking (#69) carried it to a never-repeating stacking, and The Hexagonality Law in Real Materials IV (#72) to the perovskite oxides. This book gathers the whole line of work, and this chapter closes it.

DICTO's Splash Rules (27 October) look for one simple rule behind a whole family of exactly computed values. Three outside runs on 31 October, on perovskite polytypes, Frank–Kasper phases and quasicrystal approximants, found the same rule behind every local rule in all three. This chapter states it as a theorem and proves it (Section 2), draws its corollaries (Section 3), and reports the three test grounds with their exact results and their status, known or new (Sections 4–6). Section 7 sets out what is known and what is new, Section 8 concludes.

2. The Window Law

A *stacking* is a sequence of letters c and h, one per layer; a layer is h when its two neighbours sit on the same position (A, B or C) and c otherwise, which is Jagodzinski's notation [1]. Every such sequence is the layer sequence of a stacking: each new position is either the one two layers down (h) or the third one (c). We work with periodic stackings $w = w_1 \dots w_n$, indices taken mod n; everything holds also for any infinite sequence in which the shares below exist, such as the Fibonacci stacking. The *share* f_u of a word u of length g is the number of positions i at which $w_i \dots w_{i+g-1} = u$, divided by n. Thus $h = f_h$, and $p_{hh} = f_{hh}$ is the share of neighbouring hh pairs.

A quantity is *fixed by g consecutive layers* if it attaches a number F_i to every position i (a layer, a gap, a formula unit) and there is a table F on the words of length g and an offset k such that $F_i = F(w_{i+k} \dots w_{i+k+g-1})$. Its mean is $\bar{F} = (1/n)\,\Sigma\, F_i$. For a word v of length g − 2 the *Swap defect* of F at v is

$$\Delta_v(F) = F(cvc) + F(hvh) - F(cvh) - F(hvc);$$

at g = 2 the word v is empty and Δ = F(cc) + F(hh) − F(ch) − F(hc), the quantity whose vanishing is DICTO's Swap Rule.

> **Main Theorem (the Window Law).** Let F be a quantity fixed by g consecutive layers of a two-letter stacking.
>
> (a) $\bar{F} = \Sigma_{|u|\,=\,g}\, F(u)\cdot f_u$: the mean is a linear function of the shares of the c/h words of length g.
>
> (b) These shares obey $\Sigma\, f_u = 1$ and, for every word v of length g − 1, $f_{cv} + f_{hv} = f_{vc} + f_{vh}$. They leave exactly 2^{g-1} free values, and no other linear relation holds for all stackings; so the linear function in (a) is unique once these free values are chosen.
>
> (c) For g ≥ 2, $\bar{F}$ is a linear function of the shares of the words of length g − 1 for every stacking if and only if every Swap defect $\Delta_v(F)$, |v| = g − 2, is zero.
>
> (d) For g = 2, with h and p_{hh} as free values,
>
> $$\bar{F} = F(cc) + [F(ch) + F(hc) - 2F(cc)]\cdot h + \Delta\cdot p_{hh}, \quad \Delta = F(cc) + F(hh) - F(ch) - F(hc),$$
>
> so $\bar{F}$ is a straight line in h for every stacking if and only if Δ = 0, the Swap Rule.

Proof. (a) Group the positions i by the word they read: the word u occurs $n\cdot f_u$ times and contributes F(u) each time.

(b) Each occurrence of a word v of length g − 1 is followed by exactly one letter and preceded by exactly one letter, and in a periodic word every occurrence of a word of length g contains one occurrence of its first g − 1 letters and one of its last g − 1. So $f_{vc} + f_{vh} = f_v = f_{cv} + f_{hv}$ (Kirchhoff's law for factor frequencies). There are 2^g shares and 2^{g-1} balance relations; the balance relations are the node equations of the de Bruijn graph B(2, g − 1) (nodes: the 2^{g-1} words of length g − 1; arrows: the words of length g), which is strongly connected, so its incidence matrix has rank $2^{g-1} - 1$: the relations add up to 0 = 0 and are otherwise independent. $\Sigma\, f_u = 1$ is independent of them, since every combination of balance relations vanishes on the indicator of a cycle while the sum of shares does not. Together they have rank 2^{g-1}, leaving $2^g - 2^{g-1} = 2^{g-1}$ free values. To see that no further relation holds, read the words of length g as the arrows of the de Bruijn graph (from their first g − 1 letters to their last g − 1). The non-negative solutions of the relations are the circulations of total 1 on this graph; each is a mixture of simple cycles (the classical decomposition of a flow), and a simple cycle is the share vector of the periodic word that reads it. The de Bruijn word of order g has every share equal to 2^{-g}, a solution with no zero entry, so the solutions form a polytope of full dimension 2^{g-1} spanned by share vectors of stackings.

(c) If every Δ_v is zero, set A(xv) = F(xvc) and B(vy) = F(cvy) − F(cvc) for letters x, y. Then A(xv) + B(vy) − F(xvy) = F(xvc) + F(cvy) − F(cvc) − F(xvy), which is zero when x or y is c and equals $-\Delta_v = 0$ when x = y = h. So F(u) = A(first g − 1 letters of u) + B(last g − 1 letters), and by the balance relations $\bar{F} = \Sigma_{|w| = g-1}\,[A(w) + B(w)]\cdot f_w$. Conversely, suppose $\bar{F} = \Sigma_w\, L(w)\cdot f_w$ for every stacking (a constant term is L summed against $\Sigma\, f_w = 1$). Then G(u) = F(u) − L(first g − 1 letters of u) has $\Sigma\, G(u)\, f_u = 0$ for every stacking, hence by (b) along every direction inside the polytope. For a fixed v the vector δ_v with +1 on cvc and hvh and −1 on cvh and hvc keeps every balance relation (each of the four end words cv, hv, vc, vh meets one +1 and one −1) and has total 0, so it is such a direction. Therefore $0 = \Sigma\, G\cdot\delta_v = \Delta_v(F) - [L(cv) + L(hv) - L(cv) - L(hv)] = \Delta_v(F)$. The 2^{g-2} defects are a complete set of invariants, so the condition is minimal: the functions of g letters form a space of dimension 2^g, the sums of a function of the first g − 1 and a function of the last g − 1 letters a subspace of dimension $2^{g-1} + 2^{g-1} - 2^{g-2}$ (the overlap is the functions of the middle g − 2 letters), so the quotient has dimension 2^{g-2}; the Δ_v, one for each v, vanish on that subspace and are independent, since each lives on its own four words.

(d) The balance relation at g = 2 gives $f_{ch} = f_{hc}$, and $h = f_{hc} + f_{hh}$, so $f_{cc} = 1 - 2h + p_{hh}$, $f_{ch} = f_{hc} = h - p_{hh}$, $f_{hh} = p_{hh}$. Put these into (a) and multiply out. The last claim is (c) with v empty; two real stackings show the "only if" directly: 4H (hc) and 12R (hhcc) both have h = 1/2, with $p_{hh} = 0$ and 1/4, so their means differ by $\Delta/4$. ∎

The theorem is elementary: (a) is a regrouping, (b) is Kirchhoff's law for factor frequencies, (c) is the statement that a two-by-two table is a sum of a row function and a column function exactly when its interaction term vanishes. The frame is classical in substance: factor frequencies and their balance relations are standard for Sturmian and other words (Dekking [2]; Berthé 1996 [3]; Balková and Pelantová [4]), criterion (c) is the cohomology of locally constant functions on the full shift, and in crystallography the theory of stacking disorder (Hendricks and Teller 1942 [5], Jagodzinski 1949 [1]) and the J_1–J_2 energies of polytype models (the ANNNI model [6]) express quantities through the correlations of a few neighbouring layers. The link to the stacking Hamiltonian is exact. Put s = −1 for c and s = +1 for h and write a pair quantity as $F(x, y) = \alpha + \beta\, s_x + \gamma\, s_y + \delta\, s_x s_y$; then $\Delta = 4\delta$, and (d) reads $\bar{F} = F_0 + (\beta + \gamma)\langle s\rangle + \delta\langle s_i s_{i+1}\rangle$ with $\langle s\rangle = 2h - 1$ and $\langle s_i s_{i+1}\rangle = 1 - 4h + 4p_{hh}$. So the Swap defect is four times the nearest-neighbour coupling J_1 of the stacking Hamiltonian written in c/h spins (in the Hägg spins σ of the ANNNI polytype model, $s_i = -\sigma_i\sigma_{i+1}$, so the same coupling is the J_2 term $\sigma_i\sigma_{i+2}$; Cheng, Needs and Heine 1988 [7]; Price and Yeomans [8]). For general g the Δ_v are the couplings that contain both end spins of the window. The Swap defect is therefore classical as an object; it is new as a name and as the bridge to DICTO's rules. What the theorem adds to DICTO's line of work is the exact form: one law, read one window at a time, with the Swap defect as the coefficient of the one new number at window 2.

3. Corollaries

Corollary 1 (the Hexagonality Law, g = 1). A quantity fixed by one layer has mean F(c) + [F(h) − F(c)]·h: a straight line in the hexagonality, through its values in the pure cubic and the pure hexagonal stacking.

Corollary 2 (the Swap Rule as the test). A quantity fixed by two neighbouring layers is a straight line in h on all stackings if and only if it obeys the Swap Rule, F(ch) + F(hc) = F(cc) + F(hh). If it fails, it is a plane in (h, p_{hh}) with slope Δ in p_{hh}, and no other number enters.

Corollary 3 (defects add). Δ_v is linear in F. If counts F_1, …, F_k by kind add up to a count that is a line in h (or a constant), their defects add up to zero. The counts of gaps of the three classes cc, mixed (ch or hc) and hh have defects +1, −2, +1.

Corollary 4 (same h, different stacking). Two stackings with the same h differ in a window-2 count by exactly $\Delta\cdot(p_{hh} - p'_{hh})$. At a given h the second number runs over $\max(0, 2h - 1) \le p_{hh} < h$ for every mixed periodic stacking (a stacking of period n with both letters has $p_{hh} \le h - 1/n$; the upper edge $p_{hh} = h$ is reached only by the

pure stackings and approached by long blocks of h), so a count with $\Delta \neq 0$ fills a band of width up to $|\Delta|\cdot(h - \max(0, 2h - 1))$ above each h.

Corollary 5 (the lower edge). The lower edge $p_{hh} = \max(0, 2h - 1)$ of that band is reached exactly by the stackings with no hh pair ($h \leq 1/2$) or no cc pair ($h \geq 1/2$). On it every window-2 count is a straight line in h on each side of $h = 1/2$, and the slope jumps there by exactly 2Δ.

Proof. Corollaries 1, 2 and 4 are parts (a) and (d). Corollary 3: Δ is a fixed combination of the values F(cc), F(ch), F(hc), F(hh), so it adds; the count of cc gaps has table (1, 0, 0, 0), the mixed gaps (0, 1, 1, 0), the hh gaps (0, 0, 0, 1). For the band: $f_{cc} = 1 - 2h + p_{hh} \geq 0$ and $f_{ch} = h - p_{hh} \geq 0$. Corollary 5: $p_{hh} = 0$ means $f_{hh} = 0$, and $p_{hh} = 2h - 1$ means $f_{cc} = 0$; the slope in h of $\bar{F}$ changes by $\Delta\cdot 2$ where p_{hh} turns from 0 to $2h - 1$. ∎

Corollary 1 is the Hexagonality Law for every count read from one layer. Corollary 2 marks its exact reach among pair counts: The Focus (#66) proved the direction "Swap Rule ⇒ line" for pieces and showed the two equivalent there; here the statement is for every pair count, with the defect Δ as the exact coefficient of the one number that enters when the rule fails; a pair count that breaks the Swap Rule needs one more number, and only one. Corollary 5 is the door to the approximants (Section 6): the balanced words of one-dimensional quasicrystal theory lie on this lower edge.

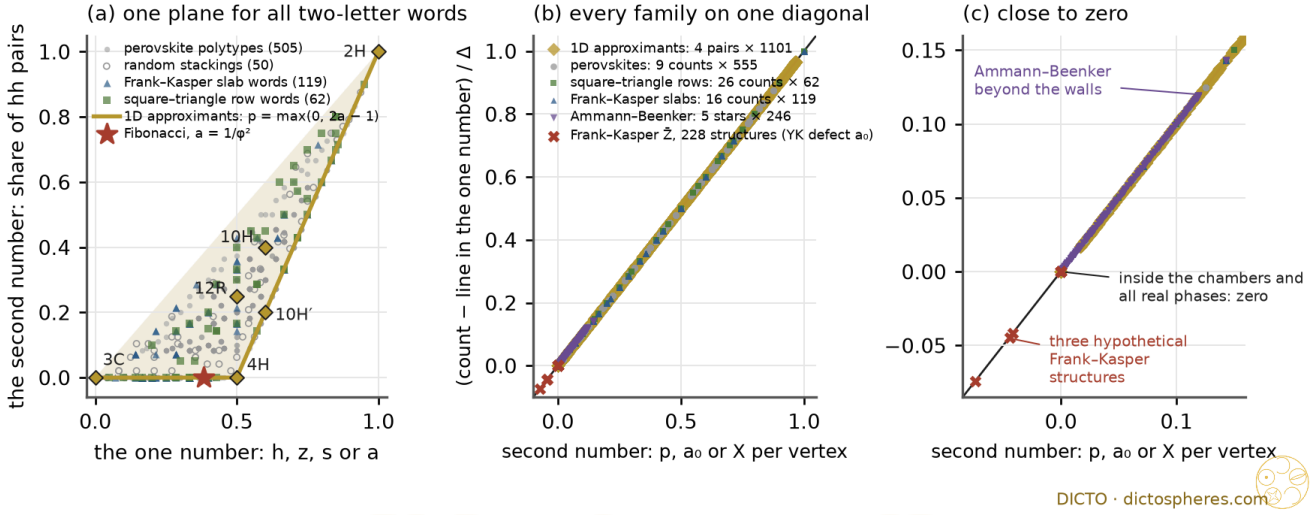


Figure 15.1. The Window Law in one picture. (a) Every two-letter stacking of the three families is a point in one triangle, the one number (h for perovskites; the share z of Z slabs or s of S rows for the Frank–Kasper families; the share a of S tiles in one dimension) against the share of neighbouring pairs of the second letter. Perovskite polytypes fill the triangle; the one-dimensional approximants of the Fibonacci chain lie on its lower edge $\max(0, 2a - 1)$ (gold), the Fibonacci chain at $(1/\varphi^2, 0)$; 4H and 12R, and the two 10H stackings, share an h and differ in p_{hh}. (b) For every count that is not a line in the one number, its distance from the line divided by its fixed coefficient, against the structure's second number: perovskites (9 counts on 555 stackings), the two Frank–Kasper families (26 and 16 window-2 counts per row or slab), the one-dimensional approximants (4 pair counts on 1101 approximants, measured from the chamber line), the Ammann–Beenker square approximants (5 vertex-star counts on 246 approximants, the second number being the forbidden star per vertex) and $\bar{Z}$ of 228 Frank–Kasper rows (the 199 structures computed here and the 29 site-count rows of the Dutour Sikirić–Delgado-Friedrichs–Deza table [9]) (the second number being the Yarmolyuk–Kripyakevich defect a_0). All lie exactly on the diagonal. (c) Close to zero: every approximant inside its chamber and every real phase has zero defect; the three hypothetical Frank–Kasper structures and the Ammann–Beenker approximants beyond the walls do not. Every point is exact and checked in the script before it is drawn.

How DICTO's rules fit together

The Window Law does not compete with DICTO's earlier rules; it is the frame in which each of them has its place. Ordered from the general to the particular:

1. **The Window Law** (the Main Theorem) holds for every quantity of every c/h stacking, with no geometry assumed: a count read from g layers is a line in the shares of the words of length g. It never fails; it only says

how many numbers a count needs.

2. **The Rule of Halves** (#57) is geometry: every piece between two layers is made of a half borrowed from the lower layer's pure stacking and a half from the upper one's. So a count of *halves* is read from one layer (window 1), and a count of whole pieces from two (window 2).
3. **The Swap Rule** (#61) is geometry too: pieces change only by swapping halves, which makes the Swap defect Δ of every piece count zero. By Corollary 2 this is exactly what brings the window-2 piece counts down to window 1.
4. **The Hexagonality Law** (HexOct) is then the result: every count of halves, and every piece count that obeys the Swap Rule, is a straight line in hexagonality (Corollary 1).

So nothing takes precedence over anything else in the sense of overruling it: the Window Law is always true, and the Rule of Halves and the Swap Rule are the geometric facts that decide which window a count lives in. Where they hold, the Hexagonality Law follows; where a count escapes them (the perovskite counts of B with 0, 1 or 2 shared faces, the B–B shells), the Window Law says exactly which extra number it needs. This ordering is our reading of DICTO's rules.

4. Test ground 1: perovskite polytypes

In a hexagonal perovskite ABX_3 the A and X ions form close-packed AX_3 layers and B sits in the octahedral holes made of X only. We model X and A as equal balls (X–X = 1), A on a 2 × 2 sublattice of each layer. In each gap exactly one octahedral hole per cell avoids the A of the lower layer, and the A of the upper layer must avoid it too, which fixes its position: each c/h word gives exactly one perovskite, and the A pattern never doubles the period (checked on every stacking). Coordinates are integers in thirds of the lattice and quarters of the layer spacing, so $72 \cdot r^2 = 8(x^2 + xy + y^2) + 3z^2$ is an integer and every count is exact. The family: all 505 polytypes up to period 12 (every primitive c/h word up to turning over) and 50 random stackings of 48 layers from a two-state chain chosen so that h and p_{hh} vary independently, 555 stackings. Every quantity, per formula unit, was tested exactly over all 555 rows as a line in h, as a plane in (h, p_{hh}), and as a linear form in the shares of longer words.

Result. Of 36 quantities, 16 are lines in h (6 of them constants), 9 are planes in (h, p_{hh}), and 11 need windows of 3 to 6 layers. None needs anything outside the Window Law. Every one of the 6 per-gap quantities that pass the Swap test is a line in h, and every one of the 3 that fail is a plane with coefficient Δ, as Corollary 2 says.

quantity (per B = per ABX_3)	exact on all 555 stackings	window	status
Straight lines in h (Swap Rule holds)			
face-sharing partners; corner-sharing partners	2h; 6(1 – h)	1	known (#72; classical in substance)
X in a shared face (B–X–B = arccos 1/3) / on a straight B–X–B	share h / 1 – h	1	classical in substance
A in a cuboctahedron / anticuboctahedron	1 – h / h	1	classical
A neighbours of B ($r^2 = 3/2$); per-gap values cc 8, ch 7, hc 7, hh 6	8 – 2h	1	new (elementary)
B–B at $r^2 = 2/3, 2, 14/3$	2h, 6 – 6h, 12h	1	first two = face and corner counts; 14/3 new
face-sharing clusters (single octahedra included)	1 – h	1	new
Constants: the same in every stacking			
edge-sharing partners; neighbours of X; of A	0; 8 X + 4 A; 12 X	0	classical for 3C, here for every stacking
empty octahedral holes per gap	3, each A_2X_4 with the two A opposite ($r^2 = 2$)	0	new
tetrahedral holes per gap	8, each AX_3 (6 with A in the base, 2 with A at the tip)	0	new
Planes in (h, p_{hh}) (Swap Rule fails; coefficient of p_{hh} = Δ)			
B with 0 / 1 / 2 shared faces	1 – 2h + p_{hh} / 2h – 2p_{hh} / p_{hh} (Δ = +1, –2, +1)	2	new as stated
strings of two or more octahedra; single octahedra	h – p_{hh}; 1 – 2h + p_{hh}	2	new as stated
B–B at $r^2 = 8/3, 4, 20/3, 8$	2p_{hh}, 12 – 6p_{hh}, 12p_{hh}, 6 – 6p_{hh}	2	new
Longer windows			
B–B at $r^2 = 6, 22/3, 10, 34/3$ / at $r^2 = 32/3, 12$	linear in the shares of 3-layer / 4-layer words	3 / 4	new (window)
clusters of L octahedra (L = 2 … 5); octahedra in strings of 4 or more	share of the word c h^{L-1} c; 4-layer words	L + 1; 4	classical in substance (strings = runs of h)

Clusters (proof). The face-sharing octahedra form chains: one B per gap, and one shared face across each h layer. The pieces of a chain number octahedra minus shared faces, so clusters per B = 1 – h exactly in every stacking except 2H (one endless chain), and the mean cluster holds 1/(1 – h) octahedra: one number, though not a line. 9R (h = 2/3) has trimers, 1/3 per B; 12R (h = 1/2) a trimer and a single octahedron, 1/2 per B.

Holes (proof). The three corners of every hole triangle lie on three different A positions of the 2 × 2 pattern, and each layer has one A per cell. The B hole avoids the A of both its layers, so each of the other three octahedral holes takes exactly one A from each layer, and every tetrahedral hole exactly one A. This holds in every gap of every stacking: in the AX_3 net the kinds and numbers of holes do not see the stacking at all, so at the level of counts the Rule of Halves (#57) has nothing to mix here.

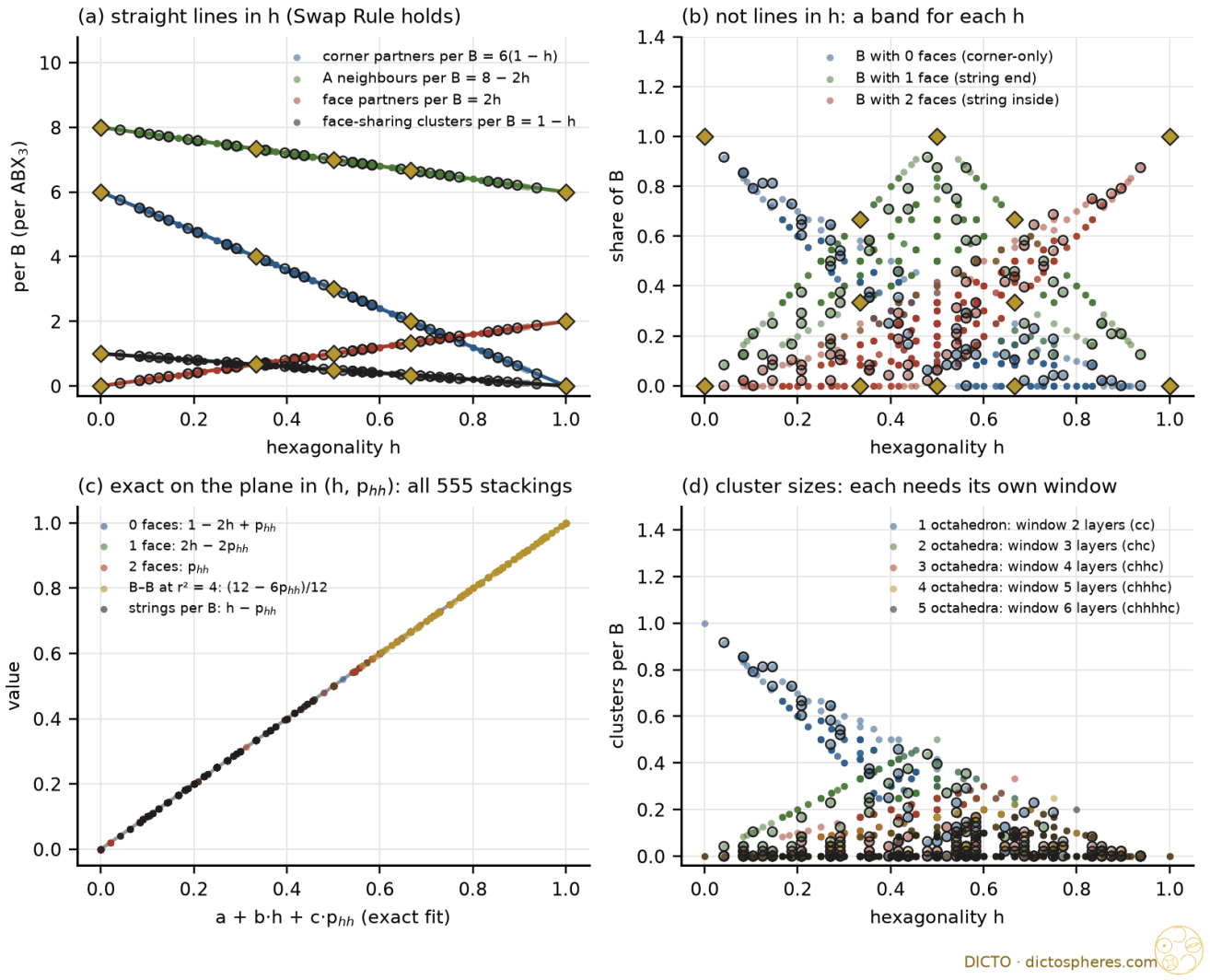


Figure 15.2. The perovskite splash, 555 stackings (dots: all polytypes up to period 12; rings: random stackings of 48 layers; gold diamonds: 3C, 2H, 4H, 6H, 9R). (a) Four counts lie exactly on straight lines in h. (b) The three kinds of B fill bands: h alone does not fix them (Corollary 4). (c) The same three, the strings and the B–B shell at $r^2 = 4$ fall exactly on the diagonal once p_{hh} is added. (d) Clusters of a given size need a window of size + 1 layers.

Real pairs. Corollary 4 at work on known compounds (sequences from Cava et al. [10], Jin et al. [11] and Adkin and Hayward [12]; Ramsdell symbols [13] recomputed from the letters, all eleven matching):

polytype	c/h	example	h	p_{hh}	face partners	B, 0 faces	B, 2 faces	clusters	strings
3C	c	$SrTiO_3$	0	0	0	1	0	1	0
2H	h	$BaNiO_3$	1	1	2	0	1	0	0
6H	hcc	hexagonal $BaTiO_3$	1/3	0	2/3	1/3	0	2/3	1/3
9R	hhc	$BaRuO_3$	2/3	1/3	4/3	0	1/3	1/3	1/3
4H	hc	4H-$BaRuO_3$, $SrMnO_3$	1/2	0	1	0	0	1/2	1/2
12R	hhcc	$Ba_4Fe_3NiO_{12}$	1/2	1/4	1	1/4	1/4	1/2	1/4
10H	hhhcc	$Ba_5Sb_{1-x}Mn_{4+x}O_{15}$	3/5	2/5	6/5	1/5	2/5	2/5	1/5
10H′	hhchc	$Ba_5Fe_4NiO_{13.5}$	3/5	1/5	6/5	0	1/5	2/5	2/5
8H	hhhc	$BaMnO_{3-x}$	3/4	1/2	3/2	0	1/2	1/4	1/4
15R	hhhhc	$BaMnO_{3-x}$	4/5	3/5	8/5	0	3/5	1/5	1/5
5H	hhccc	$Ba_5Ta_4O_{15}$	2/5	1/5	4/5	2/5	1/5	3/5	1/5

4H and 12R share h = 1/2, so they share every window-1 count (face partners 1, clusters 1/2), and differ by p_{hh} = 1/4 in every window-2 count, by exactly Δ/4: B with no shared face 0 against 1/4 (Δ = +1), strings 1/2 against 1/4 (Δ = −1). The two 10H stackings (tetramers; trimers and dimers) share h = 3/5 and differ by 1/5 in p_{hh}, with the same signs. In 2H the octahedra form one endless chain, so its clusters and strings per B are 0. The constant rows of the table for neighbours of X and of A were checked separately on all 505 bracelets, with 0 exceptions. Real compounds with vacancies or two kinds of B are modelled by their stacking only.

5. Test ground 2: Frank–Kasper phases

A Frank–Kasper (tetrahedrally close-packed) structure is cut by its atoms into tetrahedra only. Each atom has Z = 12, 14, 15 or 16 neighbours (its Delaunay neighbours), every bond is shared by 5 or 6 tetrahedra, and Z = 12 + (number of six-fold bonds at the atom). The six-fold bonds form Frank and Kasper's major skeleton [14, 15], Nelson's disclination network [16]. We read the topology of 18 real phases from published coordinates (AFLOW prototype encyclopedia [17] and the Crystallography Open Database [18]) by a periodic Delaunay step, and checked every site: closed triangulated shell with Euler number 2, folds 5 or 6, Z = 12 + #six-fold bonds, no two six-fold neighbours adjacent in a shell. Every count after that is an integer and every test exact. Four candidates fail the check and are not tetrahedrally close-packed: χ (α-Mn) and $Mg_{17}Al_{12}$ (bonds shared by 4 tetrahedra), $Mg_{23}Al_{30}$ and $Mg_{21}Zn_{25}$. Two layered families give the spread: square–triangle rows (Sullivan [19]; Sadoc and Mosseri 2016 [20]; S = A15, T = Z, ST = H, STT = J; all 62 words checked, all primitive words up to length 7 and 30 random words of 20 rows) and kagome slabs (Frank and Kasper 1959 [15]; a step ± is a Laves slab of 6 atoms, a step 0 a Z slab of 7 atoms; + = C15, +− = C14, 0 = Z, 0+ = μ, 0+0− = K_7Cs_6; all 119 words checked, all words up to length 6 and 40 random words of 14 slabs). In all, 199 structures.

> **The identities (classical; exact on all 199).** Per atom, with ρ = six-fold bonds per atom: $\bar{Z} = 12 + 2\rho$, bonds = 6 + ρ, five-fold bonds = 6, tetrahedra = 5 + ρ, triangles = 10 + 2ρ. A tetrahedron has 0, 1 or 2 six-fold edges, and two only when they are opposite.

An atom has 12 five-fold bonds whatever its Z, and its shell has 2Z − 4 triangles, which gives the identities; two six-fold edges at one corner would make adjacent six-fold vertices in that corner's shell. So one number, the disclination density ρ, fixes every topological count per atom, by identity. This is Frank and Kasper's [14, 15] and Nelson's [16] picture, not new.

> **The Yarmolyuk–Kripyakevich line and its defect.** With x_Z the share of sites of coordination Z and $a_0 = x_{12} - x_{14}/3 - 7x_{15}/6 - 2x_{16}$, every structure satisfies
>
> $$\bar{Z} = 68/5 - (2/5)\cdot x_{12} - (6/5)\cdot a_0.$$
>
> On the plane $a_0 = 0$ ($6N_{12} = 2N_{14} + 7N_{15} + 12N_{16}$, Yarmolyuk and Kripyakevich 1974 [21]) this is $\bar{Z} = 66/5 + (2/5)\cdot m$ with $m = 1 - x_{12}$, $\rho = (3 + m)/5$, and $40/3 \le \bar{Z} \le 27/2$.

Proof. Expand both sides with $x_{12} + x_{14} + x_{15} + x_{16} = 1$. The plane $a_0 = 0$ meets the simplex of shares in a triangle with corners A15 ($\bar{Z} = 27/2$), C15 (40/3) and the point Z12 : Z15 = 7 : 6 (174/13), and $\bar{Z}$ is linear, which gives Frank and Kasper's range. ∎

The study plan's candidate for a "hexagonality of tetrahedral packing" (ours, not DICTO's), the non-icosahedral share m, works on every real phase computed (18), on the 13 further known structures tabulated by Deza, Delgado Friedrichs and Dutour Sikirić [9], and in both layered families, where it is a theorem: the units are rows (S = 2 A15 units, T = 1 Z unit) or slabs (± = 2 Laves units, 0 = 1 Z unit). But the line is old: $\bar{Z} = 13.6 - 0.4\cdot x_{12}$ is equivalent to

Yarmolyuk and Kripyakevich's relation of 1974 (see also Shoemaker and Shoemaker 1986 [22]), and it is not a law of all Frank–Kasper structures: the computer search of Dutour Sikirić, Delgado Friedrichs and Deza (2010) [9] found three hypothetical structures off it, 7:4:2:2, 6:6:4:0 and 6:8:4:0 (N_{12}:N_{14}:N_{15}:N_{16}; counts read from the authors' talk "Space fullerenes" [23], the paper being embargoed), with $a_0 = -2/45, -1/24, -2/27$. They leave the line only through a_0, by exactly $-(6/5)a_0$; the third has $\bar{Z} = 122/9 > 13½$. This is the shape of the Window Law, a line in one number plus a fixed coefficient times one defect, with a_0 in the place of the Swap defect term. Here it is an identity of the shares, not an instance of the theorem, since a general Frank–Kasper structure is not a layer sequence.

The Window Law in the layered families (exact on 62 and 119 stackings). Per tile row or per kagome slab, every site and topological count is a line in the share s of S rows or z of Z slabs (window 1): sites S 8, T 7, ± 6, 0 7; Z12 S 2, T 3, ± 4, 0 3; tetrahedra S 46, T 40, ± 34, 0 40; six-fold bonds S 6, T 5, ± 4, 0 5 (the composition matrices of Sadoc and Mosseri and of the Laves/Z intergrowth picture, both known). Every count by kind (bonds by end types, neighbour types, rings of six-fold bonds, tetrahedra by corner types) is fixed by neighbouring pairs of units (window 2), and none needs more. Every window-2 count fails the Swap Rule, and the defects come in the pattern +1, –2, +1 (times a whole number) over the three end-type classes, whose sum, a window-1 count, is a line (Corollary 3):

count (per row or slab)	square–triangle (S/T)	Δ	kagome (Laves L / Z)	Δ
six-fold bonds by end types: S/T Z14–Z14, Z14–Z15, Z15–Z15; L/Z Z14–Z14, Z14–Z16, Z16–Z16	2 + 3s + p, 2s – 2p, 3 – 4s + p (sum 5 + s)	+1, –2, +1	z + p, 2z – 2p, 4 – 5z + p; with Z15–Z15 = 3z (window 1) and Z14–Z15 = 0, sum 4 + z	+1, –2, +1
neighbours of the same pairs (ordered)		+2, –2, –2, +2		+2, –2, –2, +2
tetrahedra 2444, 2445, 2455 (S/T); 2266, 2246, 2244 (L/Z)	36s + 4p, 8s – 8p, …	+4, –8, +4	24 – 30z + 6p, 12z – 12p, 6p	+6, –12, +6
tetrahedra 4444, 4445, 4455 (two opposite six-fold edges)	4s + 2p, …	+2, –4, +2	only 4455: 6z	0

(p: share of neighbouring SS or 00 pairs. Tetrahedron kinds by corners, 2 = Z12, 4 = Z14, 5 = Z15, 6 = Z16.) That the defects sum to zero is Corollary 3; that every window-2 count of both families carries exactly this pattern is observed (catalogue new). **Stacking-blindness:** 22 Z/L patterns of the kagome family are realised by two or more different Laves stackings, and for each all 119 census keys per slab agree (new as stated; elementary). No count of the census sees whether a Laves slab sits cubic or hexagonal: C15, C14, C36, 9R and the long-period Laves phases have one census per atom, and so do μ and K_7Cs_6. Here the two letters that matter are L and Z, and the c/h letter is invisible.

phase	compound (source)	N	Z12:Z14:Z15:Z16	$\bar{Z}$	ρ	YK units A15 + L + Z
A15	Cr_3Si (AFLOW)	8	2:6:0:0	27/2	3/4	2 + 0 + 0
seven Laves: C15, C14, ternary C14, C36, 9R, 9-layer (Ag), long period (Ag)	$MgCu_2$, $MgZn_2$, U_2Cr_3Si, $MgNi_2$ (AFLOW); $Mg(Li,Zn)_2$, $Mg(Zn,Ag)_2$ (COD)	12–126	2:0:0:1	40/3	2/3	0 + N/3 + 0
Z	Zr_4Al_3 (AFLOW)	7	3:2:2:0	94/7	5/7	0 + 0 + 1
σ	CrFe type (AFLOW; COD 2106167)	30	10:16:4:0	202/15	11/15	4 + 0 + 2
μ	W_6Fe_7 (AFLOW)	39	21:6:6:6	174/13	9/13	0 + 6 + 3
δ, P	MoNi (COD 2310246); Mo–Ni–Cr (COD 2311049)	56	24:20:8:4	94/7	5/7	4 + 4 + 4
R	Mo–Co–Cr, $Mn_{85.5}Si_{14.5}$ (COD 2310299, 2107202)	159	81:36:18:24	710/53	37/53	6 + 24 + 9
v	$Mn_{81.5}Si_{18.5}$ (COD 2106034)	186	74:80:20:12	1250/93	67/93	20 + 12 + 10
T (Bergman)	$Mg_{32}(Al,Zn)_{49}$ (AFLOW)	162	98:12:12:40	1082/81	55/81	0 + 40 + 6

All 18 real phases have $a_0 = 0$ and are whole sums of the three YK units (a known structural formula). Z, δ and P share $x_{12} = 3/7$ and so share $\bar{Z} = 94/7$, as the line demands, though their shares of Z14, Z15 and Z16 differ.

One relation observed, not explained. Let σ be the number of six-fold bonds inside an atom's coordination shell. On all 18 real phases and all 181 family members,

$$\Sigma_{Z14\ \mathrm{sites}}\,(\sigma - 4) + 2{\cdot}\Sigma_{Z12\ \mathrm{sites}}\,(6 - \sigma) = 8{\cdot}N_{15}.$$

The identities fix only $\Sigma\sigma = 6\rho N$, so this is an independent relation: one of the four exact relations of the census beyond the identities and YK (the census matrix of all 199 structures has rank 8 where these leave 12; data/relations2.txt), determined only modulo YK. It is stated here as observed: every structure with coordinates lies on the YK plane, so the relation is fixed only up to a multiple of YK, and it cannot be tested on the three hypothetical structures, whose coordinates were not in reach.

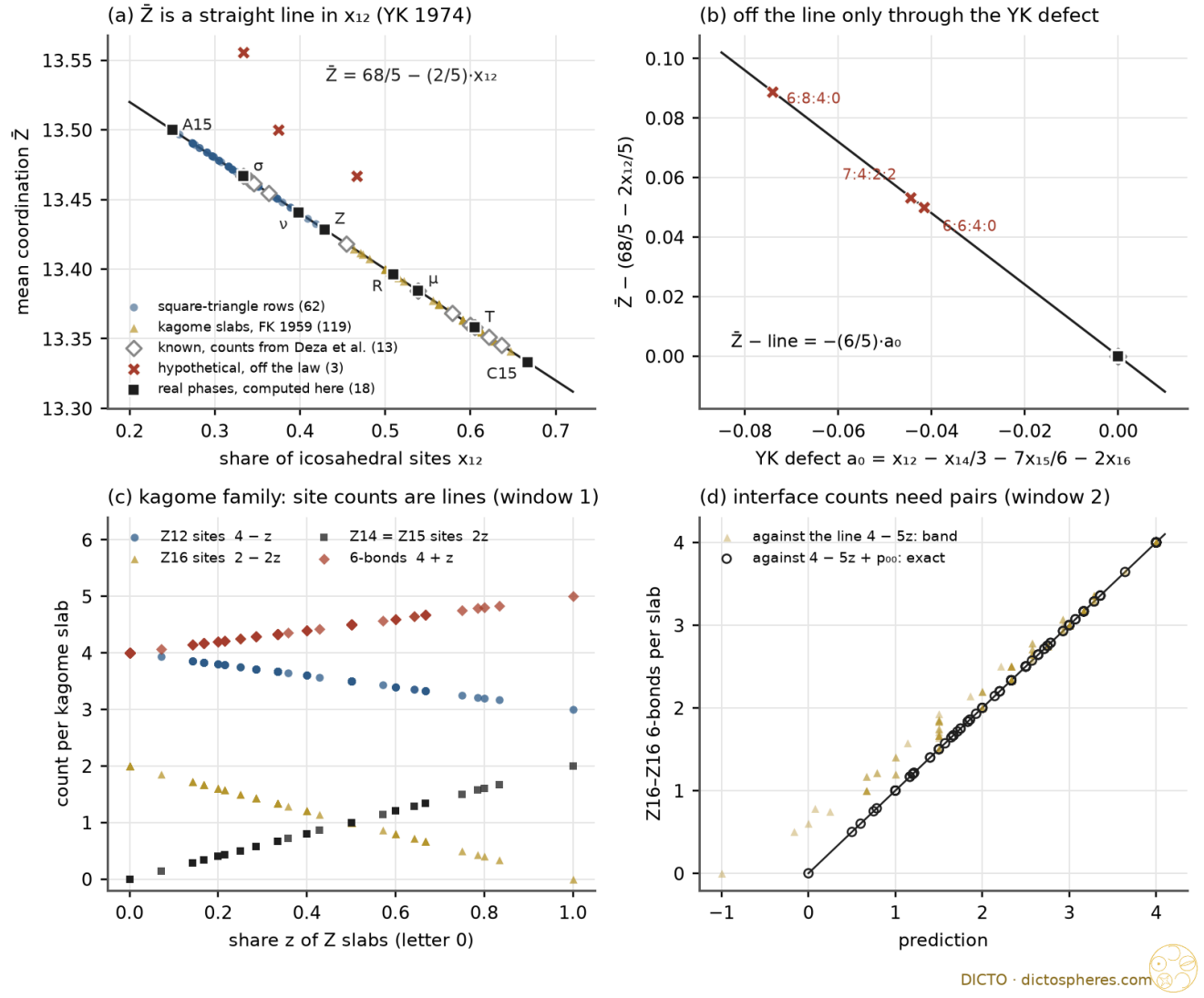


Figure 15.3. The Frank–Kasper splash, 199 tetrahedrally close-packed structures. (a) $\bar{Z}$ against the icosahedral share: every real phase (black squares), every listed known phase (open diamonds) and both families lie on the YK line; three hypothetical structures (red crosses) do not. (b) Their distance from the line is exactly –(6/5) times the YK defect a_0. (c) Kagome family, per slab: site counts and six-fold bonds are lines in the share z of Z slabs. (d) The Z16–Z16 six-fold bonds per slab fill a band against the line 4 – 5z and fall exactly on the diagonal once the share p_{00} of neighbouring Z-slab pairs is added: $4 - 5z + p_{00}$.

6. Test ground 3: quasicrystal approximants

All three families are cut-and-project sets: lattice points of Z^n in a strip around a space E are kept and projected. The quasicrystal takes an irrational E; an approximant replaces the irrational number by a fraction, and the pattern repeats. We used the canonical strip and a generic offset and counted everything per period, exactly: in one dimension the words $s_n = \lfloor (n+1)a + r \rfloor - \lfloor na + r \rfloor$ (S for 1, L for 0, a = share of S) for all 1101 fractions $0 < a = p/q < 1$ with $q \le 60$ and windows g = 1 to 9, and the Fibonacci chain $a = 1/\varphi^2$ in exact arithmetic in $Q(\sqrt{5})$; in two dimensions the Ammann–Beenker (AB) square approximants, $\sqrt{2}/2$ replaced by t = a/b, all 246 fractions $1/2 \le t \le 1$ with $b \le 40$ and the Pell ones up to 99/140, the limit in $Q(\sqrt{2})$; in three dimensions the cubic approximants of the 3D Penrose tiling, φ replaced by F_{k+1}/F_k from 1/1 to 55/34. Two facts hold throughout and were checked on every run: the number of tiles of each kind per period is the absolute value of a minor of a basis of the period lattice (its Plücker coordinate), and vertices equal tiles in number.

One dimension. The S/L words are two-letter stackings, and the Window Law applies as it stands, with S in the place of h. For every window length g and every word w of length g, the share of w in the approximant p/q is a straight line in a on each interval between neighbouring fractions with denominator ≤ g, bending only at those fractions; on the interval containing $1/\varphi^2$ the Fibonacci chain lies on the same line; there are at most three distinct shares per length and g + 1 words of each length g. This is classical (Morse and Hedlund 1940 [24]; Berthé 1996 [3]; the three-distance theorem) and was checked on all 1101 approximants for $g \le 9$. Two readings are ours. First, the approximants lie on the lower edge of the Window-Law plane (Corollary 5): $p_{SS} = \max(0, 2a - 1)$, the smallest share of SS pairs any word with this a can have. Of the 747 primitive periodic words up to 12 letters, 157 lie on this edge (every word with no SS when $a \le 1/2$, or no LL when $a \ge 1/2$) and 590 above it; of the 157, only 47 are balanced, and these are the approximants (Christoffel words). Zero defect at window 2 is therefore necessary, not sufficient: what singles out the approximants is that they stay on the quasicrystal's chamber line window after window, up to a length that grows with the period (below). So every pair count of an approximant is a line in a on each side of 1/2, and its kink at 1/2 is exactly 2Δ: twice the Swap defect. Second, on the Fibonacci side $a < 1/2$ the defect term is zero for every approximant, not only in the limit: SS never occurs, as in the Fibonacci stacking of #69. Longer windows kink with the denominator as factor: at window 3 the vector +3, –3, –3, +3 at 1/3 on LLL, LLS, SLL, SLS. The approximant F_{n-2}/F_n lies on the chain's window-g line exactly when $g < F_{n+2}$ (checked for $g \le 300$), and its share then differs from the chain's by the slope times $p/q - 1/\varphi^2$, the phason strain.

window	chamber of $1/\varphi^2$	shares on it (lines in a) and their values at $a = 1/\varphi^2$	Fibonacci convergents on the line
1	[0, 1]	S: $a = (3 - \sqrt{5})/2$; L: $1 - a$	all
2	[0, 1/2]	LS = SL: a; LL: $1 - 2a = \sqrt{5} - 2$; SS: 0	1/2 and every later one
3	[1/3, 1/2]	LLS = SLL: $1 - 2a = \sqrt{5} - 2$; LSL: a; SLS: $3a - 1 = (7 - 3\sqrt{5})/2$	1/2 and every later one
5	[1/3, 2/5]	adds LLSLL: $2 - 5a = (5\sqrt{5} - 11)/2$	1/3 and every later one
8, 9	[3/8, 2/5]	three values: $2 - 5a$, $3a - 1$, $8a - 3 = 9 - 4\sqrt{5}$	2/5 and every later one

The convergents of $1/\varphi^2$ are F_{n-2}/F_n = 1/2, 1/3, 2/5, 3/8, 5/13, …; F_{n-2}/F_n is on the window-g line exactly when $g < F_{n+2}$. A convergent at an end of the chamber (1/2 at window 2 and 3, 1/3 at window 5, 2/5 at windows 8 and 9) is on the line by continuity.

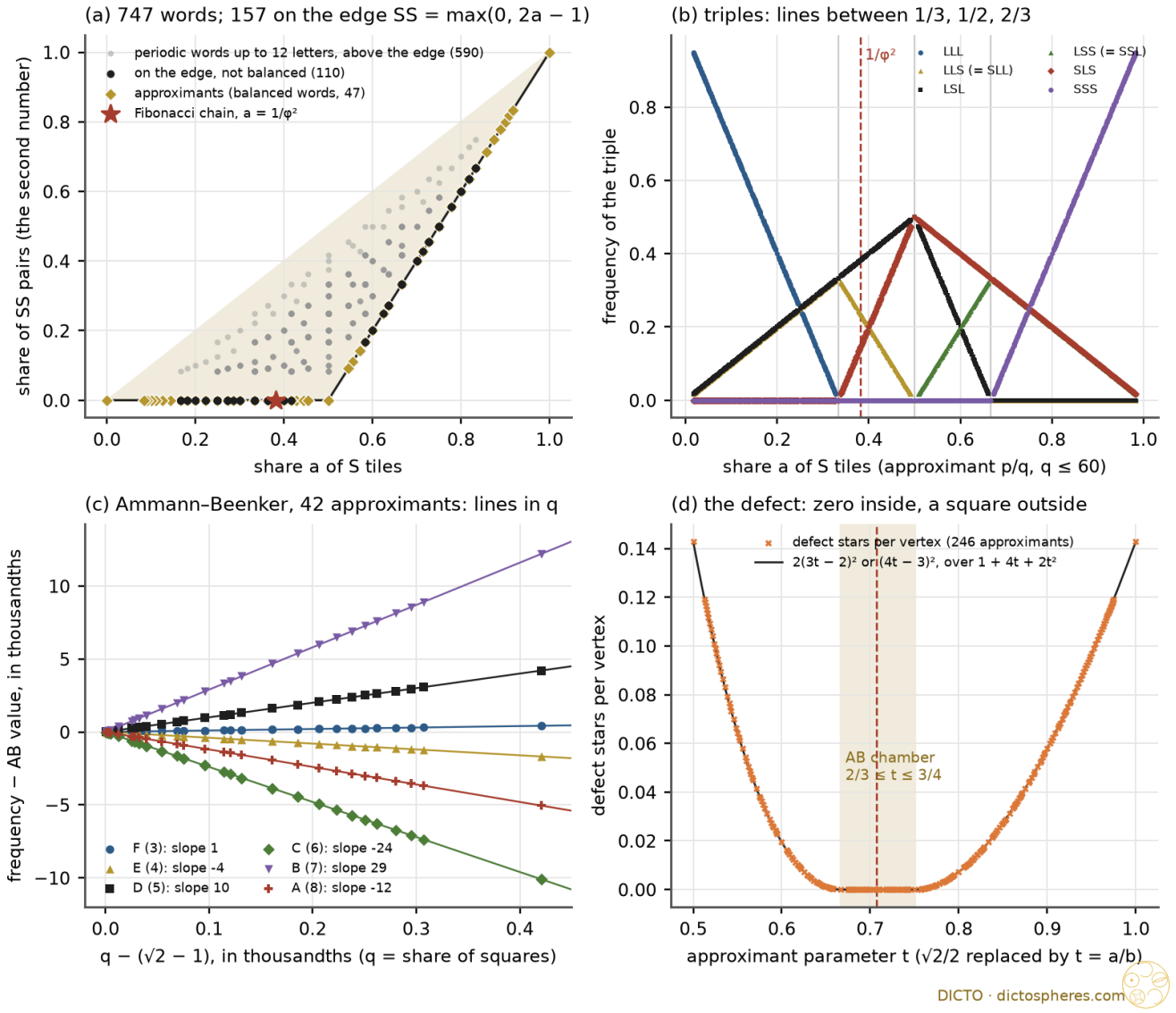


Figure 15.4. The approximants splash. (a) Window 2 in one dimension: share of SS pairs against the share a of S tiles for all 747 primitive periodic words up to 12 letters: 590 above the lower edge max(0, 2a – 1) (grey), 110 on it but not balanced (black), and the 47 approximants (gold, also on the edge); the Fibonacci chain sits at (1/φ², 0). (b) Window 3: the share of every triple in every approximant p/q, q ≤ 60, lies on straight lines that bend only at 1/3, 1/2 and 2/3. (c) Ammann–Beenker square approximants inside the chamber 2/3 ≤ t ≤ 3/4: each vertex-star frequency minus its AB value, against the square share minus (√2 – 1); six straight lines through the origin. (d) The forbidden star: none inside the chamber, a square law outside.

> **The Ammann–Beenker chamber (exact on all 42 square approximants with 2/3 ≤ t ≤ 3/4, b ≤ 40, and on the Pell ones up to 99/140).** With R rhombs and Q squares per period, the six vertex stars occur
>
> F (3 edges) = Q, E (4) = 2R – 2Q, D (5) = –4R + 6Q, C (6) = 10R – 14Q, B (7) = –12R + 17Q, A (8) = 5R – 7Q,
>
> and no other star occurs. Per vertex these are straight lines in the square share q = Q/(R + Q), and at q = √2 – 1 they give exactly the AB frequencies √2 – 1, 6 – 4√2, 10√2 – 14, 34 – 24√2, 29√2 – 41, 17 – 12√2.

The share of squares plays the part of the hexagonality: one number, every count a line, the quasicrystal on every line. The 9 kinds of edge counted by the stars at their two ends are lines on the same chamber (for example F–E edges 4R – 4Q, D–D edges –2R + 3Q). The counts do not depend on the offset of the strip. Example, t = 5/7: 239 vertices per period, R = 140, Q = 99, stars A 7, B 3, C 14, D 34, E 82, F 99. The vertex counts of the Pell approximants are tabulated by Jagannathan and Duneau [25] (2023, Table 1), and our counts reproduce them; the laws on the whole chamber, for all 42 approximants and not only the Pell ones, were not found and are new.

The walls and the defect (exact on all 246 square approximants, 1722 counts). Outside the chamber a seventh star X appears, six edges with two squares side by side at the vertex, which the AB tiling never has. Per normalising cell of $N_u = b^2 + 4ab + 2a^2$ vertices ($R = 4ab$ rhombs and $Q = b^2 + 2a^2$ squares; for even b one period holds half of this cell),

$$X = 2(3a - 2b)^2 \text{ for } t = a/b < 2/3, \quad X = (4a - 3b)^2 \text{ for } t > 3/4, \quad X = 0 \text{ inside,}$$

and every count is its chamber line plus a fixed multiple of X: (F, E, D, C, B, A, X) shift by $\kappa = (0, 0, -1, +2, -3, +1, +1)\cdot X$.

This is the Window-Law shape again: a line in one number plus a defect with a fixed coefficient vector, the two-dimensional counterpart of the pattern +1, –2, +1 (κ also sums to zero, since the vertices still number R + Q). The new feature is the square: in one dimension the forbidden SS pairs grow linearly beyond the wall (2a – 1), in two dimensions the forbidden star grows quadratically, because its acceptance domain is born at a point and grows in two directions. The walls are the Pell approximant 2/3 and its partner 3/4 = 1/(2·2/3). For rational planes without the four-fold symmetry the vertex counts are not fixed by the plane: on 229 of 240 random planes moving the strip changes them (a row of phason flips), which is known in substance; the one-number law belongs to the square approximants, the ones used in practice.

Three dimensions (tile counts). For the cubic approximant $\varphi \approx p/q$, the cubic cell holds prolate = $4(p^3 + 3p^2q + q^3)$ and oblate = $4(p^3 + 3pq^2 - q^3)$ rhombohedra, and as many vertices as rhombohedra; at $p/q = \varphi$ the ratio is φ : 1. The prolate share of the approximant F_{k+1}/F_k is the Fibonacci ratio F_{3k+2}/F_{3k+3} (all nine computed): the 3D approximants carry the 1D approximants of $1/\varphi$ in their tile counts, three steps at a time. The 1/1 and 2/1 counts are known (Elser and Henley 1985 [26]; Henley [27]; Duneau, Mosseri and Oguey 1989 [28]); the closed formula in p and q was not found in the sources checked; the method (minors) is classical. Tile counts are window 1, so they are lines in the prolate share by definition; the vertex types were not computed.

approximant	1/1	2/1	3/2	5/3	8/5	13/8	21/13
prolate + oblate per cubic cell	20 + 12	84 + 52	356 + 220	1508 + 932	6388 + 3948	27060 + 16724	114628 + 70844
prolate share	5/8	21/34	89/144	377/610	1597/2584	6765/10946	28657/46368

7. What is known and what is new

result	exact on	status
The law		
Window Law (a), (b): means linear in word shares, Kirchhoff relations, 2^{g-1} free values	proof	classical in substance (factor frequencies; stacking disorder; J_1–J_2 polytype energies)
Window Law (c), (d): window drops exactly when the Swap defects vanish; Δ is the coefficient of p_{hh}	proof	elementary; criterion classical in substance (cohomology on the shift); $\Delta = 4J_1$ in c/h spins (J_2 in Hägg spins): classical as an object, new as a name and as the bridge to DICTO's rules
Swap Rule ⇔ line in h for pieces fixed by two layers	proof	#66 (DICTO's line of work); count form with coefficient Δ here
Perovskites		
face 2h, corner 6(1 – h) per B	555	known (#72; classical in substance)
clusters per B 1 – h; A per B 8 – 2h; same holes in every gap; B with 0/1/2 faces, strings; B–B shells and their windows	555	new
Frank–Kasper phases		
identities in ρ; tetrahedra with 0, 1, 2 six-fold edges	199	classical (Frank and Kasper [14, 15]; Nelson [16])
YK line; real phases as sums of YK units	18 + 13 + families	known (Yarmolyuk and Kripyakevich 1974 [21]; Gvozdetskyi et al. 2014 [29])
$\bar{Z} = 68/5 - (2/5)x_{12} - (6/5)a_0$; counterexamples off the line through a_0 only	identity; 3	elementary; counterexamples from Dutour Sikirić et al. 2010 [9]; reading as the Window-Law shape ours
window-1 composition of rows and slabs	62 + 119	known (Sadoc and Mosseri [20]; Laves/Z intergrowth)
window-2 catalogue with defects +1, –2, +1; stacking-blindness to the Laves c/h letter	62 + 119	new (catalogue); new as stated, elementary (blindness)
shell relation $\Sigma_{Z14}(\sigma - 4) + 2\Sigma_{Z12}(6 - \sigma) = 8N_{15}$	199	observed, unexplained
Approximants		
1D: Farey lines, three values, complexity g + 1	1101, g ≤ 9	classical (Morse–Hedlund [24]; Berthé 1996 [3])
1D: F_{n-2}/F_n on the window-g line iff $g < F_{n+2}$; offset = slope × phason strain	g ≤ 300	known in substance; this statement not found
1D: approximants on the lower edge (157 of 747 words lie there, 47 balanced = approximants); kink at 1/2 = 2Δ	747 words	elementary; ours as framing
tile counts = Plücker coordinates; AB frequencies; Pell-approximant counts	all	known (canonical projection; Baake–Grimm [30]; Jagannathan–Duneau 2023 [25])
AB: six star laws on the whole chamber 2/3 ≤ t ≤ 3/4; walls; defect star X and its vector κ	42; 246	new (exact, computed)
3D: prolate $4(p^3 + 3p^2q + q^3)$, oblate $4(p^3 + 3pq^2 - q^3)$; prolate share F_{3k+2}/F_{3k+3}	9	1/1, 2/1 counts known (Elser and Henley 1985 [26]); closed formula not found; method classical

8. Conclusion

What the law is. The Window Law is a general frame, and an elementary one: whatever a stacking's layers fix through a window of g layers averages to a linear function of the shares of the words of length g, and the window narrows exactly where the Swap defects vanish. DICTO's Hexagonality Law is its first step, the window of one layer, and DICTO's Swap Rule is the condition for a pair count to stay on it. Nothing in three kinds of crystal needed more than this frame.

What it adds. The Swap-defect reading, carried across three kinds of crystal: the coefficient of the one new number at window 2 is the Swap defect; the same pattern +1, –2, +1 marks the kinds of B in a perovskite and the kinds of bond and tetrahedron in the layered Frank–Kasper phases; the Yarmolyuk–Kripyakevich line is left only

through one defect, in the same shape; and in the approximants the defect is a kink of twice the Swap defect in one dimension and a forbidden star with a fixed transfer vector in two. Alongside, specific new exact counts: clusters per B = 1 − h, A per B = 8 − 2h, the stacking-blind holes and the B–B shells of the perovskites; the window-2 catalogue and the stacking-blindness of the Frank–Kasper families; the Ammann–Beenker chamber laws, walls and defect; the cubic-cell tile formulas of the 3D approximants.

The bridge. DICTO's aim is a bridge between periodic and aperiodic order. The Window Law gives one plank of it: the periodic approximants of a quasicrystal carry zero defect on the chamber that holds the quasicrystal, window after window up to a length that grows with their period (zero defect at window 2 alone is shared by 110 other words up to 12 letters, whose shares part from the approximant's of the same share a at some longer window, since a periodic word is fixed by its factors), so every count of every approximant there lies on one straight line in the share of one tile, and the never-repeating limit lies on the same line. The defect does not shrink towards the limit; inside the chamber it is already zero, and what shrinks is only the distance along the line, the phason strain.

Epilogue. What has been shown

Conclusion

The book set out to show that the endless variety of close-packed stackings is built from a few pieces, and that one rule governs how those pieces change. It has. The space of any stacking divides into finitely many kinds of piece, each borrowed whole or in half from the cubic or the hexagonal stacking (Chapter 1). When a stacking changes, pieces are never created from nothing; they swap, and the two mixed pieces add up to the two pure ones (Chapters 3 and 4). For every piece fixed by its two neighbouring layers, that swap is the same statement as a straight line: every layer keeps what it holds in its own pure stacking, so every count and volume is linear in the hexagonality h (Chapter 5), for random, faulted and never-repeating stackings alike (Chapters 6 and 7). Finite pieces, one rule of exchange, and from them every stacking: endless variety of order from a finite set of quanta.

The second part carried the law into matter. Where the property is a count fixed by geometry, real crystals follow the law exactly: hydrogen sites in titanium and zirconium (2 – h), shared faces in perovskites and battery cathodes (2h), eclipsed bonds in diamond, ice and semiconductors (h/2), kagome planes in magnets (1 + 3(1 – h)). Where the property depends on energy, the law is the first term, and the distance from it is one number per material, the reach of interactions beyond the next layer: tin, germanium, diamond, cobalt and iron lie close to the line; silicon further; silicon carbide, the best-known polytypic material, furthest. In spin ice and water ice the loops that set the frozen-in entropy do not see the stacking at all through every length computed, the law with zero slope (Chapter 13). Geometry fixes what every stacking must contain; the materials choose among the stackings by how far their forces reach.

The third part returned to the pieces. The Rule of Halves gave them; the two HexOct tiles, the c-tile and the h-tile, are what every layer of every stacking is made of, and they fit in any order (Chapter 14). The law gave the pieces, and the pieces prove the law: a string of the two tiles is a stacking, and each of the laws above is a statement about which tile each layer holds.

The fourth part set the laws in one frame. The Window Law (Chapter 15) holds for every count of every c/h stacking, with no geometry assumed: a count read from g neighbouring layers is a linear function of the shares of the words of g layers. The Hexagonality Law is its first case, the window of one layer, and it covers exactly what DICTO's definition names, the counts of pieces, holes, bonds and swaps, read from one layer or from two neighbouring layers in a way that obeys the Swap Rule. At two layers one more number enters, the share p_{hh} of neighbouring hexagonal pairs, and its coefficient is the Swap defect; so the Swap Rule is exactly the condition for a pair count to stay on the straight line, and a pair count beyond it is a plane in (h, p_{hh}). In the physicists' layer model of the second part, the Swap defect is the coupling J_2, and J_3 is the first coupling that needs three layers: the distance of each material from the line, measured in Chapters 8 to 12, is read window by window. The same frame holds in the perovskites (16 of 36 counts on the line, 9 planes, 11 needing longer windows), in the Frank–Kasper phases, which leave the Yarmolyuk–Kripyakevich line only through one defect, and in the approximants of quasicrystals, which carry zero defect on the chamber that holds the quasicrystal, so that the never-repeating limit lies on the same straight lines. That is one plank of the bridge between periodic and aperiodic order that the Fibonacci stacking began (Chapter 7).

How the rules fit together

The Window Law always holds, for every window g: a count read from g neighbouring layers is a linear function of the shares of the c/h words of length g. **The Rule of Halves** and **the Swap Rule** decide which window a count needs. When a count needs only one layer, or two layers and obeys the Swap Rule, the Window Law reduces to **the Hexagonality Law** (g = 1), the straight line in hexagonality. Pair counts that break the Swap Rule are planes in (h, p_{hh}).

rule	place in the hierarchy	proved in
Window Law (#84)	always holds, for every window g	Chapter 15, Main Theorem
Rule of Halves (#57)	decides the window: halves are read from one layer, whole pieces from two	Chapter 1, Theorem (the rule of halves)
Swap Rule (#61, #64)	decides the window: Swap defect Δ = 0 brings two-layer counts down to one	Chapters 3 and 4, Theorem of §3; Chapter 15, Corollary 2
Hexagonality Law	g = 1: a straight line in h	Chapter 15, Corollary 1; Chapter 5, Theorem (Hexagonality Law)
beyond the Swap Rule	Δ ≠ 0: a plane in (h, p_{hh}), coefficient Δ	Chapter 15, Main Theorem (d), Corollary 2

Proved in Chapter 15, Main Theorem and Corollaries 1 and 2; the Rule of Halves in Chapter 1, Theorem (the rule of halves); the Swap Rule in Chapters 3 and 4, Theorem of §3; for pieces fixed by two layers, the Swap Rule and the straight line are equivalent by Chapter 5, Theorem (Hexagonality Law).

The laws and the materials, in one table

Exact on every stacking

count	value	chapter
face-sharing tetrahedral, and octahedral, hole pairs per ball	h, h	10
usable tetrahedral hydrogen sites per metal atom	2 – h	10
faces and corners shared by each octahedron of a hexagonal perovskite	2h, 6(1 – h)	11
lithium–metal face-sharing pairs per lithium site of a layered cathode	2h	11
eclipsed bonds per atom of a tetrahedral network (diamond, ice, zinc blende)	h/2	12
kagome planes through each A atom of an A_3B kagome magnet	1 + 3(1 – h)	13
closed non-backtracking loops of each length on the pyrochlore network (computed to length 34)	independent of h	13
face-sharing clusters, and A neighbours, per octahedron of a hexagonal perovskite	1 – h, 8 – 2h	15
octahedra of a hexagonal perovskite with 0, 1, 2 shared faces (beyond the Swap Rule: planes)	1 – 2h + p_{hh}, 2h – 2p_{hh}, p_{hh}	15

Real materials: distance from the straight line

material, property	result	chapter
grey tin, germanium, diamond, silicon, silicon carbide: polytype energies, $(\|J_2\| + \|J_3\|)/J_1$	0.02, 0.08, 0.15, 0.32, 1.03	12
cobalt, iron: polytype energies, $\|J_2/J_1\|$	0.15, 0.17	10
$BaRuO_3$: volumes of 3C, 4H, 9R	within 0.2%	11
$BaMnO_3$: volume; energy	within 1.3%; far from the line	11
Li_xCoO_2: layer spacing of H1-3	within 0.11%	11
six III-V semiconductors: conduction-band offset; crystal-field splitting	4%; 7% of the full change	12
high-entropy alloy CrCoNi: stacking faults	departs clearly	10

For the values of the other chapters (silicon carbide, silicon, germanium and cerium volumes, hard-sphere entropy, magnesium LPSO compositions) see Chapters 8 to 10; each is stated there with its source.

Open problems

(1) Prove the Loop Equality for all loop lengths and all stackings (Chapter 13). (2) Measure the heat and vapour pressure of stacking-disordered ice against its cubicity (Chapter 12). (3) Measure the 6H cell of $BaRuO_3$ to more figures (Chapter 11). (4) Find the cause of the distance from the law where it is large: silicon carbide, $BaMnO_3$, CrCoNi. (5) Find which HexOct-K tile sets use the stacking itself as the clock, one step per layer, in the interlocking range, where each tile touches four to six tiles above (Chapter 14). (6) Explain the relation $\Sigma_{Z14}(\sigma - 4) + 2\Sigma_{Z12}(6 - \sigma) = 8N_{15}$, observed on all 199 Frank–Kasper structures computed and not yet explained (Chapter 15).

Appendix. Sources by chapter

Each chapter's own list of sources checked, data and code, as published with its paper.

Prologue. From the Octet to Every Stacking: the Web of Bonds (#58, DICTO DOI 10.5281/zenodo.23665944)

Chapter 1. The Rule of Halves: the Holes of Every Close-Packed Stacking (#57, DICTO DOI 10.5281/zenodo.23647898)

[1] DICTO, "Sonsam and Lahdeli: the Delian Bond on the Twinned Octet, with Ian Dury, Blockhead and Keyed Switch Points", paper #56, DICTO DOI: 10.5281/zenodo.23635823 (2026).
[2] DICTO, "The Global Web of Bonds: How Bonds Hand Over Across Crystal Networks", paper #53, DICTO DOI: 10.5281/zenodo.23631140 (2026).
[3] DICTO, "The Delian Bond: Samson, Delilah and the Octet Between Kepler's Rhombic Dodecahedron and the Cube", paper #51, DICTO DOI: 10.5281/zenodo.23606828 (2026).
[4] International Tables for Crystallography, Vol. C, ch. 9.2 (2006), section 9.2.1 on close-packed structures and the h/c notation, https://it.iucr.org/Cb/ch9o2v0001/.
[5] F. Aurenhammer, "Power diagrams: properties, algorithms and applications", SIAM Journal on Computing 16 (1987) 78–96, doi:10.1137/0216006.
[6] T. C. Hales, "An overview of the Kepler conjecture", arXiv:math/9811071 (1998); "Sphere packings I", arXiv:math/9811073; "Sphere packings V", arXiv:math/9811077.
[7] A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer, "Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra", Soft Matter 12 (2016) 7073–7078, doi:10.1039/C6SM01180B; arXiv:1605.07160.
[8] D. Wang, X. An, D. Gou, H. Zhao, L. Wang, F. Huang, "DEM construction of binary hard sphere crystals and radical tessellation", AIP Advances 8 (2018) 105203, doi:10.1063/1.5052478.
[S] DICTO, studies/2026-10-25-stackings (commit b24fd15d): stackings.py, catalogue.py, reach.py, exact_halves.py, exact_tets.py, exact_stretch.py, exact_proof2.py, exact_named.py, appendix.py; outputs exact_*.txt, appendix.json; proof-draft.txt, proof-additions.md, referee-120b.md, prior-art.md.

Chapter 2. DICTO Janus c/h Pyrochlore (#60, DICTO DOI 10.5281/zenodo.23678750)

> **Sources checked** (27 October 2026; full record with queries: study folder, prior-art.md). *Library* (302 references; queries on pyrochlore stackings, weighted division of the holes of close-packed stackings, diamond stackings and bond midpoints): Okamoto, Nilsen, Attfield & Hiroi 2013, breathing pyrochlore lattice; Koca et al. 2015; Al-Siyabi, Koca & Koca 2020; Torquato & Jiao 2012; Betke & Henk 1999; de Graaf, van Roij & Dijkstra 2011; Gray 2002; Damasceno, Engel & Glotzer 2012; Krishnamurthy, Akleman et al. 2022; Quiquandon & Gratias 2014; Katz & Gratias 1993; Sun et al. 2010. *Web* (searches on hexagonal pyrochlore stackings and on stacking variants of the tetrahedral–truncated tetrahedral honeycomb): arXiv 1907.06413, 2404.09023, 1201.0677, 1210.6810, cond-mat/0201144, cond-mat/0412036, 0809.0079 and Phys. Rev. B 74, 172401 (pyrochlore as kagome and triangular layers); arXiv 2608.00138, 2210.17343, 1405.1261 (hexagonal diamond, wurtzite and stacking faults); Wikipedia, Quarter cubic honeycomb and Triakis truncated tetrahedral honeycomb. From #57's record: Hales' Kepler-proof decompositions, the Jagodzinski h/c notation, radical tessellations of fcc and hcp with pore spheres (Wang et al. 2018), Cadotte et al. 2016. Most were seen as titles and abstracts; those credited above were opened. None gives the rule of halves for pyrochlore stackings or their weighted division.

Chapter 3. The Swap Rule (#61, DICTO DOI 10.5281/zenodo.23682596)

> **Sources checked** (27 October 2026; full record with queries: study folder, prior-art.md and proof/prior-art.md). *Library* (1007 references, semantic search, 10 queries): Leppmeier 2019, the dodecahedron as a Voronoi cell; Blatov, Voronoi–Dirichlet polyhedra in crystal chemistry; Garber, Voronoi's conjecture; Hales 1998, overview of the Kepler conjecture; organizing principles for dense packings (arXiv 1206.3716); Hart, Hansen, Kuhs 2021, Markov description of stacking-disordered crystals; Pauling 1929; Torquato and Jiao 2009; Schymura, Seidel and Weltge 2021, lifts for Voronoi cells; Chaidee and Sugihara 2017, spherical Laguerre diagrams; Fedorov 1885, ch. 13. *Web*: IUCr teaching pamphlet 5 on close packing; International Tables Vol. C ch. 9.2 on layer stacking; Bourne, Kok, Roper and Spanjer, Laguerre tessellations and polycrystals (arXiv 1912.07188); Chaidee and Sugihara (arXiv 1705.03911); Voronoi volumes and interstitials in metals (arXiv 2301.04303); VoroTop defect identification. None states the rule.

> **Version 2** (27 October 2026): §6 now proves the Swap Rule for volumes down to s = ¼ (Theorem 2), which version 1 left open.

Data and code. Study 2026-10-27-splash-rules: rule finder (splash.py), map (render_map.py, data/s_*.json), Ganesha check (swap_cut.py, 224/224), proof scripts and logs (proof/; PROOF.md Parts I and II list what each proves; additive.py, additive_proof.py). References: DICTO, The Rule of Halves (#57), DICTO DOI 10.5281/zenodo.23647898; DICTO, DICTO Janus c/h Pyrochlore (#60), DICTO DOI 10.5281/zenodo.23678750.

Chapter 4. The Swap Rule in Crystals (#64, DICTO DOI 10.5281/zenodo.23709753)

Sources checked (28 October 2026). Reference library (ask-library on "Laves phase polytypes C14 C15 C36 stacking; polytypism CdI_2 SiC ZnS Verma Krishna; NiAs rock salt stacking; Voronoi or power diagram of polytypes; Frank–Kasper polyhedra Z12 Z16"): J. Yi and G. S. Canright, "Possible disordered ground states for close-packed polytypes" (1995); E. Abe et al., long-period stacking polytypes in Mg–Zn–Y (2011); A. Haji-Akbari et al., densely packed tetrahedra (2009) and phase diagram of hard tetrahedra (2011); Cadotte et al., hard tetrahedra and octahedra (2016); T. Fujita et al., cluster packing of F-type icosahedral (2013). Web search "Laves phase polytypes C14 C15 C36 stacking sequence Voronoi cell volume Frank–Kasper polyhedra": arXiv 2103.09514 (Laves phase crystal analysis, LaCA); par.nsf.gov 10299033 and 10321376 and RSC Soft Matter d1sm01063h (Laves phases in block polymers, C36 free energy as the average of C14 and C15); APS March Meeting 2022 K16.8 (absence of C36 in diblock melts); open.fau.de 22393; MPI CPfS Laves phase project page; poivs.tsput.ru conference talk. The Verma–Krishna book and Ramsdell's and Jagodzinski's notations are cited from standard knowledge; not read for this paper.

Data and code. Study 2026-10-27-splash-rules (run2/), #57's exact weight-walk engine unchanged. References: DICTO, The Rule of Halves (#57, DOI 10.5281/zenodo.23647898); DICTO Janus c/h Pyrochlore (#60, 10.5281/zenodo.23678750); The Swap Rule (#61, 10.5281/zenodo.23682596).

Chapter 5. The Focus (#66, DICTO DOI 10.5281/zenodo.23717687)

Version 2 (28 October 2026): the map's row on the radial cell projection now says that DICTO made it when he did not yet know the classical reflection construction, an independent rediscovery.

Sources checked (28 October 2026). *Reference library* (ask-library, five searches: polytype stacking energy and ANNNI; loop quantum gravity polyhedra; causal dynamical triangulations; Voronoi volumes in polytypes; Regge calculus): E. Rodriguez-Horta et al., close packed structure with finite range interaction (2017); R. Ganesh and A. K. Nasr, correlations in randomly stacked solids (2023); E. Bianchi, P. Doná and S. Speziale (2011); C. Rovelli and L. Smolin (1995); C. Rovelli (1998); S. Surya (2019); R. Tsuda and T. Fujiwara (2020); A. De Felice and E. Fabri (2001); V. A. Blatov (2004), catalogue entries and first pages. *Web* (search summaries and abstracts, not full texts): arXiv cond-mat/9709191 (SiC polytype energies, citing Cheng, Needs and Heine 1988); arXiv 2306.17756 (Ising-type additivity model for polytype band gaps); University of Pittsburgh, tribute to W. J. Choyke (Choyke, Hamilton and Patrick, Phys. Rev. 133, A1163, 1964); arXiv 1404.3588 (Matsushita and Oshiyama, on the linearity of the band gap in the hexagonality); Phys. Rev. B 49, 7564 (SiC band structures); ECS meeting 236 abstract (lattice constants of metallic polytypes against hexagonality); arXiv 2401.09399, 1111.6938, 1704.00577 (causal dynamical triangulations); arXiv 1508.05937, 2303.01761 (Voronoi topology of fcc and hcp). The 1988 and 1964 papers themselves were not read.

Data and code. Study 2026-10-28-halves-physics (scripts/hexlaw.py, hexlaw_cdi2.py; out/hexlaw.json, hexlaw_cdi2.json), reading the exact weight walks of study 2026-10-27-splash-rules (run2/raw), attached to this record. References: DICTO, The Rule of Halves (#57), DICTO DOI 10.5281/zenodo.23647898; The Swap Rule (#61), 10.5281/zenodo.23682596; The Swap Rule in Crystals (#64), 10.5281/zenodo.23709753; The White Queen 600-cell (#63), 10.5281/zenodo.23690856; The Red Queen 120-cell (#65), 10.5281/zenodo.23716140.

Chapter 6. Random Stackings and Stacking Faults (#67, DICTO DOI 10.5281/zenodo.23718636)

Version 3 (28 October 2026): the three figures are shown in full; in Version 2 they ran over the right edge of the page.

Version 2 (28 October 2026): adds three figures (the pieces of rock salt and nickel arsenide, the straight line for the diamond / ice framework with measured ice, and the six faults as strips of layers).

Sources checked (28 October 2026). *Reference library* (ask-library: "randomly stacked close packed colloidal crystals hexagonality stacking probability"; "stacking fault energy intrinsic extrinsic twin ratio fcc hcp ANNNI"; "hydrogen interstitial octahedral tetrahedral sites fcc hcp"): R. Ganesh and A. K. Nasr, Correlations in randomly stacked solids (2023, arXiv 2306.13569) and A. G. Hart, T. C. Hansen and W. F. Kuhs, A Markov theoretic description of stacking disordered aperiodic crystals including ice and opaline silica (2021, arXiv 2102.08418), catalogue abstracts; Howie and Marks (1984), Marks (1984), Hofmeister (1998) on twinned particles, not relevant. *Web* (search summaries, not full texts): Martelozzo et al., Phys. Rev. E (2002), summarising Pusey et al. (1989); Kegel and Dhont (2000); arXiv 1406.7697; Malkin et al., PNAS (2012), and later cubicities (arXiv 1812.04333; J. Appl. Cryst. 2018; Phys. Chem. Chem. Phys. 2015, c4cp02893g); ANNNI fault energies: arXiv 1704.00313, 2111.03591, 2405.04876, 1505.00431, Phys.

Rev. B 56, 10844, and a density-functional study of Cu, Ag and Au (Leoben). Pusey et al. (1989) and Malkin et al. (2012) were not read in full.

Data and code. Study 2026-10-28-random-stackings-faults (scripts/faults.py, faults.json, fault_volumes_rocksalt.json, scripts/hexlaw_laves8.py and its output), using the exact weight walks of study 2026-10-27-splash-rules (run2) and the law check of study 2026-10-28-halves-physics, attached to this record. References: DICTO, The Focus (#66), DICTO DOI 10.5281/zenodo.23717687; The Swap Rule in Crystals (#64), 10.5281/zenodo.23709753; DICTO Janus c/h Pyrochlore (#60), 10.5281/zenodo.23678750; The Rule of Halves (#57), 10.5281/zenodo.23647898.

Chapter 7. The Hexagonality Law and the Fibonacci Stacking (#69, DICTO DOI 10.5281/zenodo.23721038)

Version 2 (28 October 2026): the 26-layer approximant added (the law exact in all 90 tests); new §5 on hard-core supersymmetric fermions.

Sources checked (28 October 2026). For Version 2: P. Fendley, K. Schoutens and J. de Boer, Lattice models with N = 2 supersymmetry, Phys. Rev. Lett. 90, 120402 (2003), arXiv hep-th/0210161, read (added to the reference library). Reference library, second pass (ask-library: "Fibonacci chain one-dimensional quasicrystal diffraction"; "approximants of quasicrystals Fibonacci periodic approximant phason"; "stacking disorder Markov chain layers hexagonality cubicity"; "Sturmian sequence golden mean substitution frequencies"; "aperiodic order Baake Grimm"): M. Baake, A guide to mathematical quasicrystals (1999), pp. 8–10 read (cut and project, the Fibonacci chain); P. Kramer, Gateways towards quasicrystals (2011), p. 7 (Fibonacci numbers, Lothaire); A. I. Goldman and K. F. Kelton, Quasicrystals and crystalline approximants (1993) and M. Baake and U. Grimm, Aperiodic Order, vol. 1 (2013), catalogue entries; M. Kléman, Defects in quasicrystals, revisited (2013), first page; A. G. Hart, T. C. Hansen and W. F. Kuhs (2021) and R. Ganesh and A. K. Nasr (2023), catalogue abstracts; S. Akiyama et al., Aperiodic tile sets from Sturmian lattices (2026) and M. Baake, D. Damanik and U. Grimm, What is aperiodic order? (2015), catalogue entries. First pass (ask-library "Fibonacci stacking sequence quasiperiodic close-packed layers polytype"): S. Dong, Fibonacci and Lucas sequences in aperiodic monotile supertiles (2024); P. J. Steinhardt and H.-C. Jeong (1996); D. Kamenetsky (2026); C. S. Kaplan (2025); M. Kléman (2013), catalogue entries and pages, none on stackings. Web (search summaries): M. Farkas-Jahnke, IUCr congress abstract E0552 (Fibonacci chains for faulted close-packed lattices); arXiv 1204.5523 and 2307.16052 (Fibonacci magnetic multilayers); arXiv 1012.4880; Int. Tables Vol. C ch. 9.2 (close packing).

Data and code. Study 2026-10-28-fibonacci-stackings (fib_words.py, golden.py, golden_rocksalt.json, hexlaw_fib.py, fig69.py), using the exact weight walks of study 2026-10-27-splash-rules (run2; the new walks lvw_ABACBCACBABCBAC, lvw_ABACBCACBABCACBC, lvw_ABACBCACBABCACBCABACABCBAC) and study 2026-10-28-susy-fermions (count.py, counts.json), attached to this record. References: DICTO, A Fibonacci Spine (#30), DICTO DOI 10.5281/zenodo.23396361; The Focus (#66), 10.5281/zenodo.23717687; Random Stackings and Stacking Faults (#67), 10.5281/zenodo.23718636.

Chapter 8. Real Polytypes and the Hexagonality Law (#68, DICTO DOI 10.5281/zenodo.23719275)

Version 2 (28 October 2026): Figure 1 is shown in full; in Version 1 it ran over the right edge of the page.

Sources checked (28 October 2026). S. Kawanishi and T. Mizoguchi, Effect of van der Waals interactions on the stability of SiC polytypes, arXiv 1512.04626 (Table I, read; its experimental values from H. Schulz and K. H. Thiemann 1979, R. W. G. Wyckoff 1963, M. Stockmeier et al. 2009, Y. M. Tairov and V. F. Tsvetkov, A. H. Gomes de Mesquita 1967, Landolt–Börnstein 1982). M. Keller, A. Belabbes, J. Furthmüller, F. Bechstedt and S. Botti, First-principles insight in structure-property relationships of hexagonal Si and Ge polytypes, arXiv 2302.10822 (Tables I, II and V, read). Wikipedia, Polymorphs of silicon carbide (band gaps to one decimal; two-decimal values from search summaries of the literature); Wikipedia, Cerium (lattice constants at 20 °C). Search summaries, not read in full: Matsushita and Oshiyama (arXiv 1404.3588); Phys. Rev. B 89, 245308 (2014; ZnS, ZnSe, ZnTe polytypes); the RRUFF wurtzite entry; reviews of the γ–β transition of cerium. Zinc sulfide was dropped: no open, consistent table of its polytypes was found. Reference library: ask-library "lattice parameters of silicon carbide polytypes 3C 4H 6H 2H measured" (nothing relevant).

Data and code. Study 2026-10-28-measured-polytypes (sic_linear.py, sige_linear.py, annni_volume.py, gaps_faults_ce.py, fig68.py and their outputs), attached to this record. References: DICTO, The Focus (#66), DICTO DOI 10.5281/zenodo.23717687; Random Stackings and Stacking Faults (#67), 10.5281/zenodo.23718636.

Chapter 9. The Hexagonality Law in Real Materials II (#70, DICTO DOI 10.5281/zenodo.23723565)

Sources checked (28 October 2026). Read: S.-C. Mau and D. A. Huse, Stacking entropy of hard-sphere crystals, Phys. Rev. E 59, 4396 (1999), arXiv cond-mat/9810287; V. Elser, Phonon contribution to the entropy of hard-sphere crystals, arXiv 1403.1784 (2014); S. Kawanishi and T. Mizoguchi, arXiv 1512.04626 (energies and Table I); M. Keller et al., arXiv 2302.10822 (Tables I and II); all in the reference library. Search summaries, not read in full: Y. Sugiyama and M. Togaya (2001, Hokkaido University pdf); the 2018

review of silicon carbide at high pressure and temperature (Daviau and Lee, Yale pdf); Karch et al., Phys. Rev. B (1996); Maitani et al. (2024). Reference library (ask-library: "hard sphere crystal fcc hcp entropy difference stacking"; "Mau Huse stacking entropy hard-sphere crystals"; "Bolhuis Frenkel hard sphere fcc hcp free energy"): Ganesh and Nasr (2023), Hart, Hansen and Kuhs (2021), others not relevant.

Data and code. Studies 2026-10-28-hard-sphere-entropy (README) and 2026-10-28-pressure (pressure.py, pressure.json, fig70.py), attached to this record. References: DICTO, The Focus (#66), DICTO DOI 10.5281/zenodo.23717687; Real Polytypes and the Hexagonality Law (#68), 10.5281/zenodo.23719275.

Chapter 10. The Hexagonality Law in Real Materials III (#71, DICTO DOI 10.5281/zenodo.23726064)

Sources checked (28 October 2026). Read: E. Abe, A. Ono, T. Itoi, M. Yamasaki and Y. Kawamura, Polytypes of long-period stacking structures synchronized with chemical order in a dilute Mg-Zn-Y alloy, arXiv 1106.4944 (pp. 3–5); T. Tsumuraya, I. Watanabe and T. Sawaguchi, Origin of phase stability in Fe with long-period stacking order…, arXiv 2010.07118 v4 (Tables I–II; in the reference library); C. J. Aas, L. Szunyogh, R. F. L. Evans and R. W. Chantrell, Effect of stacking faults on the magnetocrystalline anisotropy of hcp Co, arXiv 1304.1353 (pp. 3, 7); J. Ding, Q. Yu, M. Asta and R. O. Ritchie, Tunable stacking fault energies by tailoring local chemical order in CrCoNi, arXiv 1806.00718 (pp. 9–13). Cited as given in those papers, not read: G. B. Olson and M. Cohen, Metall. Trans. A 7, 1897 (1976); P. J. H. Denteneer and W. van Haeringen, J. Phys. C 20, L883 (1987); G. J. C. Carpenter, J. Nucl. Mater. 48, 264 (1973); A. C. Switendick, Z. Phys. Chem. N.F. 117, 89 (1979). Search summaries, not read in full: arXiv 1811.12442 (δ-hydrides in Zircaloy-4); PMC10532661 (fatigue-induced fcc zirconium); Ikeda et al., Entropy 2018 (PMC7513178); Lizárraga et al., Sci. Rep. 2017 (cobalt hcp–fcc). Web searches: Mg-Zn-Y LPSO stacking sequences; stacking fault energies of CoCrNi, CoCrFeMnNi and Fe-Mn; cobalt fault energies; δ-hydride hcp to fcc mechanism. Library searched for each name above.

Data and code. Study 2026-10-28-law-in-alloys (lpso.py, hydrogen_sites.py, hea_faults.py, cobalt_faults.py, iron_polytypes.py, hydride_bound.py, fig71.py, with outputs), attached to this record. References: DICTO, The Focus (#66), DICTO DOI 10.5281/zenodo.23717687; Random Stackings and Stacking Faults (#67), 10.5281/zenodo.23718636; Real Polytypes and the Hexagonality Law (#68), 10.5281/zenodo.23719275; The Hexagonality Law in Real Materials II (#70), 10.5281/zenodo.23723565.

Chapter 11. The Hexagonality Law in Real Materials IV (#72, DICTO DOI 10.5281/zenodo.23728287)

Sources checked (28 October 2026). Read: C.-Q. Jin et al., High-pressure synthesis of the cubic perovskite $BaRuO_3$ and evolution of ferromagnetism in $ARuO_3$ (A = Ca, Sr, Ba) ruthenates, PNAS 105, 7115 (2008), PMC2438213 (Table 1, synthesis); L. T. Nguyen and R. J. Cava, Hexagonal perovskites as quantum materials, arXiv 2006.05864 (pp. 1–3, notation and $Ba_{1-x}Sr_xRuO_3$; reference list); arXiv 2107.07165 (4H, 9R cells as quoted); T. Motohashi et al., Electronic phase diagram of the layered cobalt oxide system Li_xCoO_2 ($0.0 \le x \le 1.0$), Phys. Rev. B 80, 165114 (2009), arXiv 0909.3556 (Table I and Section III.1); H. Chen and A. J. Millis, Phase diagram of $Sr_{1-x}Ba_xMnO_3$ as a function of chemical doping, epitaxial strain and external pressure, arXiv 1509.06650 (Table II); AFLOW prototype ABC2_hP8_186_b_b_ab-001 (O2-$LiCoO_2$). Abstract only: A. Van der Ven, M. K. Aydinol, G. Ceder, G. Kresse and J. Hafner, Phys. Rev. B 58, 2975 (1998) (repository copy); arXiv 2212.14667. Cited as given in those sources, not read: P. C. Donohue, L. Katz and R. Ward, Inorg. Chem. 5, 335 (1966); the 6H $BaRuO_3$ cell ($c \approx 14$ Å) as quoted second-hand; C. Delmas, C. Fouassier and P. Hagenmuller, Physica B 99, 81 (1980); C. Delmas, J.-J. Braconnier and P. Hagenmuller, Mater. Res. Bull. 17, 117 (1982); D. Carlier et al., On the metastable O2-type $LiCoO_2$, Solid State Ionics 144, 263 (2001); G. G. Amatucci, J.-M. Tarascon and L. C. Klein, J. Electrochem. Soc. 143, 1114 (1996). Web searches: LixCoO2 H1-3 O1 lattice parameters; O2-LiCoO2 Carlier Delmas; BaIrO3 high-pressure polytypes (Cheng et al. 2009, cell volumes not reachable); Katz and Ward hexagonal perovskite stacking (the 1964 paper not reached); Van der Ven 1998 abstract.

Data and code. Study 2026-10-28-perovskites-cathodes (baruo3.py, bamno3.py, cathodes.py, perovskite_counts.py, fig72.py, with outputs), attached to this record. References: DICTO, The Focus (#66), DICTO DOI 10.5281/zenodo.23717687; Real Polytypes and the Hexagonality Law (#68), 10.5281/zenodo.23719275; The Hexagonality Law in Real Materials II (#70), 10.5281/zenodo.23723565; The Hexagonality Law in Real Materials III (#71), 10.5281/zenodo.23726064.

Chapter 12. The Hexagonality Law in Real Materials V (#73, DICTO DOI 10.5281/zenodo.23728921)

Sources checked (28 October 2026). Read: A. Belabbes, C. Panse, J. Furthmüller and F. Bechstedt, Electronic bands of III-V semiconductor polytypes and their alignment, arXiv 1206.2209 (Phys. Rev. B 86, 075208 (2012); Table II); M. Keller, A. Belabbes, J. Furthmüller, F. Bechstedt and S. Botti, First-principles insight in structure-property relationships of hexagonal Si and Ge polytypes, arXiv 2302.10822 (Tables I–II); J. Ziembicki, P. Scharoch, M. P. Polak, M. Wiśniewski and R. Kudrawiec, Electronic and structural properties of group IV materials and their polytypes, arXiv 2404.05718 (Table III); C. M. Tonauer, K. Yamashita, L. del Rosso, M. Celli and T. Loerting, Enthalpy change from pure cubic ice Ic to hexagonal ice Ih, J. Phys. Chem. Lett. 14, 5055 (2023); L. C. O. Dacal and A. Cantarero, An ab initio study of the polytypism in InP (PMC5036193; no polytype energy table). Search summaries,

not read in full: T. L. Malkin et al., Phys. Chem. Chem. Phys. 2015; T. Hondoh, Philos. Mag. 2015; arXiv 2607.03465; C. Panse, D. Kriegner and F. Bechstedt, Phys. Rev. B 84, 075217 (2011) (record only); S. Limpijumnong and W. R. L. Lambrecht, cond-mat/9709191. Cited as given in the reference lists of Belabbes et al., Keller et al. and Tonauer et al., not read: C. Cheng, R. J. Needs and V. Heine, J. Phys. C 21, 1049 (1988); R. E. Algra et al., Nature 456, 369 (2008); P. Caroff et al., Nat. Nanotechnol. 4, 50 (2009); D. Spirkoska et al., Phys. Rev. B 80, 245325 (2009); B. J. Murray, D. A. Knopf and A. K. Bertram, Nature 434, 202 (2005); B. J. Murray and A. K. Bertram, Geophys. Res. Lett. 33, L17801 (2006); T. C. Hansen, M. M. Koza and W. F. Kuhs, J. Phys.: Condens. Matter 20, 285104 (2008); L. del Rosso et al., Nat. Mater. 19, 663 (2020); K. Komatsu et al., Nat. Commun. 11, 464 (2020); L. Lupi et al., Nature 551, 218 (2017); T. L. Malkin et al., PNAS 109, 1041 (2012). Cited as in earlier papers of the series: H. Jagodzinski (1949). Web searches: III-V polytypism ab initio energies and gaps; stacking-disordered ice cubicity and enthalpy; diamond, Si and Ge polytype energies and ANNNI parameters.

Data and code. Study 2026-10-28-ice-diamond-nanowires (tetra_counts.py, iii_v_bands.py, group_iv.py, fig73.py, with outputs), attached to this record. References: DICTO, The Focus (#66), DICTO DOI 10.5281/zenodo.23717687; Real Polytypes and the Hexagonality Law (#68), 10.5281/zenodo.23719275; The Hexagonality Law in Real Materials II (#70), 10.5281/zenodo.23723565; III (#71), 10.5281/zenodo.23726064; IV (#72), 10.5281/zenodo.23728287.

Chapter 13. Spin Ice and Kagome Magnets on Every Stacking (#74, DICTO DOI 10.5281/zenodo.23730421)

Sources checked (28 October 2026). Read: X.-Z. Xu, T.-Y. Lin and G.-M. Zhang, Equivalence of residual entropy of hexagonal and cubic ices from tensor network methods, arXiv 2511.22477 v3 (pp. 1, 7–8). Search summaries, not read in full: C. P. Herrero and R. Ramírez, Configurational entropy of hydrogen-disordered ice polymorphs, arXiv 1406.5929; arXiv 2404.13897 (transfer matrices, Onsager's inequality as reported in Nagle's thesis); arXiv 2009.11591; H. Chen, Q. Niu and A. H. MacDonald, Anomalous Hall effect arising from noncollinear antiferromagnetism, arXiv 1309.4041; J. Kübler and C. Felser, arXiv 1410.5985 (Mn_3Sn, Mn_3Ge); Y. Liu et al., Giant full-space anomalous Hall effect induced by non-coplanar spin state in Mn-rich Mn_3Sn, arXiv 2603.13718 (abstract and reference list read). Cited as given in the reference list of Xu et al., not read: L. Pauling, J. Am. Chem. Soc. 57, 2680 (1935); J. F. Nagle, J. Math. Phys. 7, 1484 (1966); E. H. Lieb, Phys. Rev. Lett. 18, 692 (1967); K. Y. Lin and D. L. Tang, J. Phys. A 9, 1101 (1976); S. V. Isakov, K. S. Raman, R. Moessner and S. L. Sondhi, Phys. Rev. B 70, 104418 (2004); C. P. Herrero and R. Ramírez, Chem. Phys. Lett. 568–569, 70 (2013). Cited as given in the reference list of Liu et al., not read: S. Nakatsuji, N. Kiyohara and T. Higo, Large anomalous Hall effect in a non-collinear antiferromagnet at room temperature, Nature 527, 212 (2015); L. Šmejkal, A. H. MacDonald, J. Sinova, S. Nakatsuji and T. Jungwirth, Anomalous Hall antiferromagnets, Nat. Rev. Mater. 7, 482 (2022). Records checked on arXiv (title, authors, DOI), not read: S. T. Bramwell and M. J. P. Gingras, Spin ice state in frustrated magnetic pyrochlore materials, Science (2001), doi 10.1126/science.1064761 (arXiv cond-mat/0201427); C. Castelnovo, R. Moessner and S. L. Sondhi, Magnetic monopoles in spin ice, Nature (2008), doi 10.1038/nature06433 (arXiv 0710.5515). Web searches: residual entropy of ice Ih and Ic; Ihara zeta function and crystal nets (no result on diamond against lonsdaleite); Mn_3Sn and Mn_3Ir structures.

Data and code. Study 2026-10-28-pyrochlore-kagome (pyro_counts.py, loops2.py, coordseq.py, a3b_counts.py, fig74.py, with outputs; loops.py kept as the superseded small-cell run), attached to this record. References: DICTO, The Rule of Halves (#57), DICTO DOI 10.5281/zenodo.23647898; DICTO Janus c/h Pyrochlore (#60), 10.5281/zenodo.23678750; The Swap Rule (#61), 10.5281/zenodo.23682596; The Swap Rule in Crystals (#64), 10.5281/zenodo.23709753; The Focus (#66), 10.5281/zenodo.23717687; The Hexagonality Law in Real Materials IV (#72), 10.5281/zenodo.23728287; V (#73), 10.5281/zenodo.23728921.

Chapter 14. HexOct: Two Tiles for Every Stacking (#75, DICTO DOI 10.5281/zenodo.23737639)

Computational methods. Python with exact rationals (fractions) and sympy, using the census engine of #53 [2] unchanged; floating point only proposes candidate corners, which are then solved and checked exactly. Sample check (tiles.py): 32 contexts × 10 values of λ, 320 of 320 tiles matched by one motion. Proof (prove.py): 32 of 32 contexts, every sub-interval and breakpoint. Run on the dicto-big-brother node, 29 October 2026.

Sources checked (29 October 2026). Reference library, by meaning search (four searches: unit cell of tetrahedra and octahedra halves matching across cell faces; half-octahedron pieces filling close-packed stackings; polytypes as tilings with layer-to-layer matching rules; periodic unit-cell tilings continuing across cells). Read: R. Gabbrielli, Y. Jiao and S. Torquato 2012 (pp. 1–4); H. Lalvani 1990, NASA TM-102635 (pp. 5–6); Wikipedia, Tetrahedral-octahedral honeycomb. Seen in the hits and judged not relevant (packings and non-periodic monotiles): Y. Kallus, V. Elser and S. Gravel 2009; S. Torquato and Y. Jiao 2009; P. Kramer 1982; D. Smith, J. S. Myers, C. S. Kaplan and C. Goodman-Strauss 2023; I. Tsiokos 2026. Web searches: OD theory and close-packed polytypes (IUCr Acta B 2025, dk5140, read in summary); slab tiles of tetrahedra and octahedra in close packing (arXiv 1407.0692, which splits fcc and hcp between layers into tetrahedra and octahedra, one per triangle); power and radical Voronoi diagrams of interstitial sites (arXiv 2010.01354; Scientific Data 2020, CAVD), which have no layer tiles. Our own papers #57 and the book The Laws of Stacking: no layer tiles.

[1] DICTO, "The Rule of Halves: the Holes of Every Close-Packed Stacking", paper #57, DICTO DOI: 10.5281/zenodo.23647898 (2026).
[2] DICTO, "The Global Web of Bonds: How Bonds Hand Over Across Crystal Networks", paper #53, DICTO DOI: 10.5281/zenodo.23631140 (2026).
[3] DICTO, "Sonsam and Lahdeli: the Delian Bond on the Twinned Octet", paper #56, DICTO DOI: 10.5281/zenodo.23635823 (2026).
[4] International Tables for Crystallography, Vol. C, ch. 9.2 (2006), section 9.2.1, close-packed structures and the h/c notation.
[5] F. Aurenhammer, "Power diagrams: properties, algorithms and applications", SIAM Journal on Computing 16 (1987) 78–96, doi:10.1137/0216006.
[6] K. Dornberger-Schiff and H. Grell-Niemann, "On the theory of order–disorder (OD) structures", Acta Crystallographica 14 (1961) 167–177; and the account of OD theory in Acta Crystallographica B 81 (2025), https://journals.iucr.org/b/issues/2025/06/00/dk5140/.
[7] R. Gabbrielli, Y. Jiao and S. Torquato, "Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings", Physical Review E 86 (2012) 041141, doi:10.1103/PhysRevE.86.041141.
[8] H. Lalvani, "Transformational part-count in layered octahedral-tetrahedral truss configurations", NASA Technical Memorandum 102635 (1990).
[9] DICTO, "The Euclid–Kepler–Pacioli Cell", paper #8, DICTO DOI: 10.5281/zenodo.23284106 (2026).
[10] H. Wang, "Proving theorems by pattern recognition II", Bell System Technical Journal 40 (1961) 1–41.
[11] R. Berger, "The undecidability of the domino problem", Memoirs of the American Mathematical Society 66 (1966).
[12] R. M. Robinson, "Undecidability and nonperiodicity for tilings of the plane", Inventiones Mathematicae 12 (1971) 177–209.
[13] E. Winfree, "Algorithmic self-assembly of DNA", PhD thesis, California Institute of Technology (1998).
[14] M. Cook, Y. Fu and R. Schweller, "Temperature 1 self-assembly: deterministic assembly in 3D and probabilistic assembly in 2D", Proceedings of SODA 2011, arXiv:0912.0027.
[15] D. Šedlbauer, Wang cubes for aperiodic material samples, University of West Bohemia, https://dspace5.zcu.cz/bitstream/11025/30825/1/Sedlbauer.pdf.
[16] L. Kari and others, triangular and hexagonal tile self-assembly, https://www.csd.uwo.ca/~lkari/pdfs/trihexfin.pdf.
[17] D. Smith, J. S. Myers, C. S. Kaplan and C. Goodman-Strauss, "An aperiodic monotile", arXiv:2303.10798 (2023).
[18] R. Akpanya and others, "Topologically interlocking blocks inside the tetroctahedrille", arXiv:2405.01944 (2024).
[S] DICTO, studies/2026-10-29-endless-stack-cell (commit fcb20088): tiles.py, prove.py, render_tiles.py, render_stack.py; outputs tiles.out, prove.out; README.md (prior art and results).
[19] M. Cook, "Universality in elementary cellular automata", Complex Systems 15 (2004) 1–40.
[S2] DICTO, studies/2026-10-29-keyed-hexoct: contacts.py, counter.py, render_counter.py; outputs contacts_0–5.out, counter.out; README.md (prior art and results).
[S3] DICTO, studies/2026-10-29-hexoct-k-clock: clock_contacts.py; outputs ctx_0–31.jsonl, summary.txt; README.md (prior art and results).

Chapter 15. The Window Law (#84, DICTO DOI 10.5281/zenodo.23848832)

Sources checked (31 October 2026; every query and page with its outcome is listed in the three study folders, README.md). *Reference library* (ask-library, 19 queries). Perovskite study, 4 queries (hexagonal perovskite polytypes and face-sharing share; Katz–Ward and Jagodzinski notation; stacking-disorder correlations; face-sharing clusters): Keller et al. 2023 (Si/Ge polytypes, hexagonality, generalized Ising model); Kawanishi and Mizoguchi 2015 (SiC hexagonality); Yi and Canright 1995; Ganesh and Nasr 2023 (correlations between layers n apart through a one-dimensional Ising model); Hart, Hansen and Kuhs 2021 (Markov description of stacking disorder); Rodriguez-Horta et al. 2017; none treats the B network of perovskites. Frank–Kasper study, 6 queries (Frank–Kasper polyhedra Z14–Z16; Yarmolyuk–Kripyakevich mean coordination; disclination network, Nelson; space fullerenes; Shoemaker kagome layers σ μ Laves; Sadoc–Mosseri decoration): no Frank–Kasper paper in the library; hits Kallus–Elser, Haji-Akbari et al. 2009, Papadopolos–Ogievetsky 1999, Koca et al. 2020; Sadoc and Mosseri 2016 catalogue entry only. Approximants study, 7 queries (Sturmian factor frequencies, three distance; periodic approximants, cut and project; AB vertex frequencies; Penrose vertex types, de Bruijn; phason strain; icosahedral cubic approximant 1/1; Christoffel words): read Kléman 2013 (pp. 1–5), Baake 1999 guide (§13), Hann, Socolar and Steinhardt 2016 (p. 3); catalogue only Goldman and Kelton 1993, Yamada et al. 2021, de Bruijn 1981, Fujita et al. 2013, Kramer 2011, Akiyama et al. 2026. For this paper, 2 queries ("frequencies of words of length n satisfy consistency relations, de Bruijn graph circulations": no relevant source; "stacking fault energy ANNNI J1 J2 polytypes hexagonality linear": Hart, Hansen and Kuhs 2021 only). *Web, read:* Cava et al., Hexagonal perovskites as quantum materials, arXiv 2006.05864; Sadoc and Mosseri, arXiv 1609.01624 (pp. 1–7); Deza, Delgado Friedrichs and Dutour Sikirić, talk "Space fullerenes" (table of known structures, the YK conjecture $a_0 = 0$ and its counterexamples); Gvozdetskyi, Gladyshevskii and Hlukhyy, Chem. Met. Alloys 7 (2014) (YK equations, structural formula); Jagannathan and Duneau, Properties of the Ammann–Beenker tiling and its square approximants, arXiv 2308.07701 (§§2.2, 3, 3.1, 3.2; Table 1 = our Pell counts). *Web, abstract or summary only:* Darriet and Subramanian, J. Mater. Chem. 5, 543 (1995); Adkin and Hayward, Chem. Mater. 19, 755 (2007); Jin et al., PNAS 105, 7115 (2008); arXiv 2105.05206; Dutour Sikirić, Delgado Friedrichs and Deza, Acta Cryst. A66, 602 (2010) (PDF

embargoed); Berthé, Theor. Comput. Sci. 165 (1996) 295–309 (bibliographic entry and citing surveys); arXiv 1009.0136, 2212.09406; cond-mat/0402370, 2108.03341, cond-mat/0405298 (two queries on 1/1 tile counts, no source gives them); for this paper, one query on Kirchhoff's law for factor frequencies: Balková and Pelantová, arXiv 0902.0632, and Balková, arXiv 1107.0471 and 1107.5033 (search summaries: Kirchhoff's law for factor frequencies on Rauzy graphs, attributed to Dekking and Berthé). *Not reachable:* Katz and Ward, Inorg. Chem. 3, 205 (1964), cited through Cava et al. *Classical, cited not read:* Cheng, Needs and Heine, J. Phys. C 21, 1049 (1988); Price and Yeomans (ANNNI polytypes); Shoemaker and Shoemaker, Acta Cryst. B42, 3 (1986); Elser and Henley, Phys. Rev. Lett. 55, 2883 (1985); Henley; Hendricks and Teller 1942; Jagodzinski 1949; the ANNNI model of polytypes (J_1, J_2, J_3); Frank and Kasper 1958, 1959; Nelson 1983; Nelson and Spaepen 1989; Yarmolyuk and Kripyakevich 1974; Sinha 1972; Sullivan; Hellner and Pearson; Rivier and Aste 1996; Morse and Hedlund 1940; Lothaire; de Bruijn 1981; Baake and Grimm, Aperiodic Order vol. 1 (2013); Duneau, Mosseri and Oguey 1989; Katz 1988. *In-house:* #57, #61, #64, #66, #69, #72 and the HexOct book, cited for their results.

References. DICTO, The Rule of Halves (#57), DICTO DOI 10.5281/zenodo.23647898; The Swap Rule (#61), 10.5281/zenodo.23682596; The Swap Rule in Crystals (#64), 10.5281/zenodo.23709753; The Focus (#66), 10.5281/zenodo.23717687; The Hexagonality Law and the Fibonacci Stacking (#69), 10.5281/zenodo.23721038; The Hexagonality Law in Real Materials IV (#72), 10.5281/zenodo.23728287; HexOct: The Hexagonality Law (book), 10.5281/zenodo.23731359.

Data and code. Studies 2026-10-31-splash2-perovskite (perov.py exact model, splash.py, data/splash2_perovskite.json: all 555 stackings, every quantity exact), 2026-10-31-splash2-frank-kasper (fk.py census engine, sqtri.py, fk59.py, splash.py, report.py, relations*.py; data/splash2_fk.json, results.md, cif/) and 2026-10-31-splash2-approximants (qf.py, zono.py, sturm.py, ab.py, ablaw.py, abfam.py, ico.py; data/*.json, *.log). Figure 1: papers/window-law/scripts/fig_window_law.py, which re-checks every plotted point exactly against its law; Figure 4: papers/window-law/scripts/fig_approximants.py (the study's figure script, panel (a) redrawn with all 747 words and the 157 on the edge asserted). Exact arithmetic (fractions, Q(√2), Q(√5)) throughout; floats only in the Delaunay step of the Frank–Kasper engine and in drawing. Run on dicto-big-brother.

Outside works cited in the text, numbered in order of first citation; DICTO's own papers are listed above. Every entry was checked against the publisher's or Crossref's record on 1 November 2026, except where the entry says otherwise.

1. H. Jagodzinski, Eindimensionale Fehlordnung in Kristallen und ihr Einfluss auf die Röntgeninterferenzen. I. Berechnung des Fehlordnungsgrades aus den Röntgenintensitäten, *Acta Crystallographica* 2 (1949) 201–207. DOI 10.1107/S0365110X49000552.
2. F. M. Dekking, On the Thue–Morse measure, *Acta Universitatis Carolinae, Mathematica et Physica* 33 (1992) 35–40. No DOI; data as cited in Balková and Pelantová (2009).
3. V. Berthé, Fréquences des facteurs des suites sturmiennes, *Theoretical Computer Science* 165 (1996) 295–309. DOI 10.1016/0304-3975(95)00224-3.
4. L. Balková and E. Pelantová, A note on symmetries in the Rauzy graph and factor frequencies, *Theoretical Computer Science* 410 (2009) 2779–2783. DOI 10.1016/j.tcs.2009.04.002; arXiv 0902.0632.
5. S. Hendricks and E. Teller, X-ray interference in partially ordered layer lattices, *Journal of Chemical Physics* 10 (1942) 147–167. DOI 10.1063/1.1723678.
6. W. Selke, The ANNNI model — theoretical analysis and experimental application, *Physics Reports* 170 (1988) 213–264. DOI 10.1016/0370-1573(88)90140-8.
7. C. Cheng, R. J. Needs and V. Heine, Inter-layer interactions and the origin of SiC polytypes, *Journal of Physics C: Solid State Physics* 21 (1988) 1049–1063. DOI 10.1088/0022-3719/21/6/012.
8. G. D. Price and J. Yeomans, The application of the ANNNI model to polytypic behaviour, *Acta Crystallographica B* 40 (1984) 448–454. DOI 10.1107/S0108768184002469.
9. M. Dutour Sikirić, O. Delgado-Friedrichs and M. Deza, Space fullerenes: a computer search for new Frank–Kasper structures, *Acta Crystallographica A* 66 (2010) 602–615. DOI 10.1107/S0108767310022932.
10. L. T. Nguyen and R. J. Cava, Hexagonal perovskites as quantum materials, *Chemical Reviews* 121 (2021) 2935–2965. DOI 10.1021/acs.chemrev.0c00622; arXiv 2006.05864 (cited in the text as Cava et al.).
11. C.-Q. Jin, J.-S. Zhou, J. B. Goodenough, Q. Q. Liu, J. G. Zhao, L. X. Yang, Y. Yu, R. C. Yu, T. Katsura, A. Shatskiy and E. Ito, High-pressure synthesis of the cubic perovskite $BaRuO_3$ and evolution of ferromagnetism in $ARuO_3$ (A = Ca, Sr, Ba) ruthenates, *Proceedings of the National Academy of Sciences USA* 105 (2008) 7115–7119. DOI 10.1073/pnas.0710928105.
12. J. J. Adkin and M. A. Hayward, $BaMnO_{3-x}$ revisited: a structural and magnetic study, *Chemistry of Materials* 19 (2007) 755–762. DOI 10.1021/cm062055r.
13. L. S. Ramsdell, Studies on silicon carbide, *American Mineralogist* 32 (1947) 64–82. No DOI.
14. F. C. Frank and J. S. Kasper, Complex alloy structures regarded as sphere packings. I. Definitions and basic principles, *Acta Crystallographica* 11 (1958) 184–190. DOI 10.1107/S0365110X58000487.
15. F. C. Frank and J. S. Kasper, Complex alloy structures regarded as sphere packings. II. Analysis and classification of representative structures, *Acta Crystallographica* 12 (1959) 483–499. DOI 10.1107/S0365110X59001499.

16. D. R. Nelson, Order, frustration, and defects in liquids and glasses, *Physical Review B* 28 (1983) 5515–5535. DOI 10.1103/PhysRevB.28.5515.
17. M. J. Mehl, D. Hicks, C. Toher, O. Levy, R. M. Hanson, G. Hart and S. Curtarolo, The AFLOW Library of Crystallographic Prototypes: Part 1, *Computational Materials Science* 136 (2017) S1–S828. DOI 10.1016/j.commatsci.2017.01.017.
18. S. Gražulis, D. Chateigner, R. T. Downs, A. F. T. Yokochi, M. Quirós, L. Lutterotti, E. Manakova, J. Butkus, P. Moeck and A. Le Bail, Crystallography Open Database – an open-access collection of crystal structures, *Journal of Applied Crystallography* 42 (2009) 726–729. DOI 10.1107/S0021889809016690.
19. J. M. Sullivan, The geometry of bubbles and foams, in *Foams and Emulsions* (Springer, Dordrecht, 1999) 379–402. DOI 10.1007/978-94-015-9157-7_23.
20. J.-F. Sadoc and R. Mosseri, Quasiperiodic Frank–Kasper phases derived from the square–triangle dodecagonal tiling, *Structural Chemistry* 28 (2017) 63–73 (online 2016). DOI 10.1007/s11224-016-0846-2; arXiv 1609.01624.
21. Ya. P. Yarmolyuk and P. I. Kripyakevich, *Soviet Physics – Crystallography* 19 (1974) 334–337. No DOI; data as cited by Gvozdetskyi, Gladyshevskii and Hlukhyy (2014), ref. 4; the paper itself was not reached.
22. D. P. Shoemaker and C. B. Shoemaker, Concerning the relative numbers of atomic coordination types in tetrahedrally close packed metal structures, *Acta Crystallographica B* 42 (1986) 3–11. DOI 10.1107/S0108768186098671.
23. O. Delgado Friedrichs, M. Deza and M. Dutour Sikirić, Space fullerenes, talk slides, https://poivs.tsput.ru/conf/international/XVI/files/talks/newSpaceFull.pdf (no date or venue in the record found).
24. M. Morse and G. A. Hedlund, Symbolic dynamics II. Sturmian trajectories, *American Journal of Mathematics* 62 (1940) 1–42. DOI 10.2307/2371431.
25. A. Jagannathan and M. Duneau, Properties of the Ammann–Beenker tiling and its square periodic approximants, *Israel Journal of Chemistry* 64 (2024) e202300119. DOI 10.1002/ijch.202300119; arXiv 2308.07701 (2023).
26. V. Elser and C. L. Henley, Crystal and quasicrystal structures in Al–Mn–Si alloys, *Physical Review Letters* 55 (1985) 2883–2886. DOI 10.1103/PhysRevLett.55.2883.
27. C. L. Henley, Crystals and quasicrystals in the aluminum–transition metal system, *Journal of Non-Crystalline Solids* 75 (1985) 91–96. DOI 10.1016/0022-3093(85)90208-X.
28. M. Duneau, R. Mosseri and C. Oguey, Approximants of quasiperiodic structures generated by the inflation mapping, *Journal of Physics A: Mathematical and General* 22 (1989) 4549–4564. DOI 10.1088/0305-4470/22/21/017.
29. V. Gvozdetskyi, R. Gladyshevskii and V. Hlukhyy, Structure of a new multicomponent stannide and comments on the Yarmolyuk–Kripyakevich criterion for Frank–Kasper phases, *Chemistry of Metals and Alloys* 7 (2014) 169–180.
30. M. Baake and U. Grimm, *Aperiodic Order. Volume 1: A Mathematical Invitation* (Cambridge University Press, Cambridge, 2013). DOI 10.1017/CBO9781139025256.

Papers referred to, not in this book

#8	The Euclid–Kepler–Pacioli Cell	10.5281/zenodo.23284106
#30	A Fibonacci Spine	10.5281/zenodo.23396361
#40	The DICTO OG Family on Every Stacking	10.5281/zenodo.23490114
#42	The DICTO Sunstar	10.5281/zenodo.23497702
#47	Every Mix of the Platonic Solids Fills Space	10.5281/zenodo.23551326
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#53	The Global Web of Bonds: How Bonds Hand Over Across Crystal Networks	10.5281/zenodo.23631140
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About the author

DICTO is an artist in Osaka, Japan. His background is architectural sculpture, stonecarving and Buddhist statuary. He trained as an architectural stonecarver at the City and Guilds of London Art School (1998–2001), worked at St James's Palace, and recarved the Roman lettering of the Civil Service Rifles War Memorial at Somerset House, London, for its rededication in 2002. Since 2009 he has worked in Japan in the sculpture and restoration of Buddhist statues, among them a statue at Takaosan Jingo-ji, Kyoto, traditionally attributed to Kōbō Daishi Kūkai.

He builds geometric sculpture from recycled PET bottles with two modular systems of his own, B1NR and DICTOspheres, joined without glue. He holds six Japanese design registrations; five rest on spherical projections of the Platonic solids, the five forms behind his name: Dodecahedron, Icosahedron, Cube, Tetrahedron, Octahedron. His sculptures have been shown in Kyoto at the Kyoto Municipal Museum of Art (Interdependent Origination III, 2025), the Insho Domoto Museum of Fine Arts, the Kyoto Botanical Garden, the Kansho Exhibition and the Kyoto Takasegawa Art Festival.

The geometry behind the sculptures led to seventy-five papers, gathered in *DICTO: Collected Geometric Works*, and to four books: this one, *The Golden Cycle*, *Alicia in Wonderland: The Red & White Queens*, and *Quanta of Space: A Manifesto* (all 2026). Every paper and book is published openly with its own DICTO DOI at Zenodo. More at https://dictospheres.com.